

Chapter 24

Probability

Solutions

SECTION - A

Objective Type Questions (One option is correct)

1. If n coins are thrown then number of elements in sample space is

(1) ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n$ (2) ${}^nC_0 + {}^nC_2 + {}^nC_4 + \dots$
(3) ${}^nC_1 + {}^nC_3 + {}^nC_5 + \dots$ (4) 2^{n+1}

Sol. Answer (1)

Sample space contains, no head, one head, two head and so on upto n heads therefore total number of elements $= {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n$

2. Two dice are thrown simultaneously and three events A , B and C are defined as

A : The sum of the numbers is 10

B : The sum of the numbers is 9

C : The sum of the numbers is 8

If $P(E)$ is the probability of event E , then which of the followings is true ?

(1) $P(A) = \frac{1}{11}$ (2) $P(B) = \frac{1}{8}$ (3) $P(C) = \frac{5}{36}$ (4) $P(C) = \frac{1}{9}$

Sol. Answer (3)

$$A = \{(4,6), (5,5), (6,4)\}$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

$$B = \{(3,6), (4,5), (5,4), (6,3)\}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

$$C = \{(2,6), (3,5), (4,4), (5,3), (6,2)\}$$

$$P(C) = \frac{n(C)}{n(S)} = \frac{5}{36}$$

3. If a coin is tossed four times then the probability that head appears at least once is

- (1) $\frac{1}{16}$ (2) $\frac{15}{16}$ (3) $\frac{13}{16}$ (4) $\frac{11}{16}$

Sol. Answer (2)

$$\text{Probability that head appears atleast once} = 1 - \frac{1}{16} = \frac{15}{16}$$

4. A dice is thrown the probability of the event that either an even number appears or odd number appears is

- (1) 1 (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) $\frac{1}{6}$

Sol. Answer (1)

Let event A is occurrence of even number

Event B is occurrence of odd number

$$\text{Then } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{6} + \frac{3}{6} - 0 = 1$$

5. Two dice are thrown simultaneously and the sum of the numbers is noted. If the sum of the numbers is 7, then the probability is

- (1) $\frac{1}{5}$ (2) $\frac{1}{4}$ (3) $\frac{1}{6}$ (4) $\frac{1}{7}$

Sol. Answer (3)

Let event A is getting the sum of number = 7

$$A = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

6. A die is thrown four times and the sum of the numbers is noted. If the sum of the numbers is 23 then the probability is

- (1) $\frac{1}{324}$ (2) $\frac{1}{342}$ (3) $\frac{1}{243}$ (4) $\frac{1}{322}$

Sol. Answer (1)

Total number of ways = $6 \times 6 \times 6 \times 6$.

Total number of ways getting sum of number 23 = 4

$$\text{Therefore probability} = \frac{4}{6 \times 6 \times 6 \times 6} = \frac{4}{36 \times 9} = \frac{1}{324}$$

7. One card is drawn from 52 playing card. The probability that this will be red card or a King is

- (1) $\frac{7}{13}$ (2) $\frac{29}{52}$ (3) $\frac{25}{52}$ (4) $\frac{23}{52}$

Sol. Answer (1)

$$\text{Probability } P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{26}{52} + \frac{4}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13}$$

8. Four cards are drawn at random from a pack of 52 playing cards, one of them is King, other is Queen, third is Jack and fourth is Ace. If the probability is $\frac{K}{52C_4}$ then K is

- (1) 64 (2) 4 (3) 16 (4) 256

Sol. Answer (4)

$$K = {}^4C_1 \times {}^4C_1 \times {}^4C_1 \times {}^4C_1 = 256$$

9. A number k is selected from the set $\{1, 2, 3, 4, \dots, 10\}$. If $k^2 - 4k + 3 < 0$ then the probability is

- (1) $\frac{1}{5}$ (2) $\frac{2}{5}$ (3) $\frac{3}{10}$ (4) $\frac{1}{10}$

Sol. Answer (4)

$$k^2 - 4k + 3 < 0$$

$$k^2 - 3k - k + 3 < 0$$

$$k(k - 3) - 1(k - 3) < 0$$

$$(k - 3)(k - 1) < 0$$

$$k \in (1, 3)$$

$$k = 2$$

$$\text{Probability} = \frac{1}{10}$$

10. Numbers of three digits with repetition using digits $\{1, 2, 3, \dots, 9\}$ are formed. The probability that a selected number is divisible by 5 is

- (1) $\frac{2}{9}$ (2) $\frac{3}{9}$ (3) $\frac{1}{9}$ (4) $\frac{1}{8}$

Sol. Answer (3)

The number is divisible by 5 if number has last digit 5 or 0. But number is formed by using digits 1 to 9.

Therefore total number of ways = $9 \times 9 \times 9$

Favourable ways = $9 \times 9 \times 1$

$$\text{Probability} = \frac{1}{9}$$

11. Four digit numbers are formed by using the digits 1, 2, 3, 6 without repetition. The probability that a selected number is divisible by 3 is

- (1) $\frac{1}{4}$ (2) $\frac{1}{3}$ (3) $\frac{1}{2}$ (4) 1

Sol. Answer (4)

It is clear that all the numbers formed by 1, 2, 3, 6 are divisible by 3.

12. In a game of six digits, two digits are already fixed for prize. If any one will find these two fixed digits then he will win the prize. Mr. A selects two numbers. The probability that Mr. A will win the prize is

(1) $\frac{1}{15}$

(2) $\frac{1}{6}$

(3) $\frac{1}{5}$

(4) $\frac{1}{1}$

Sol. Answer (1)

$$\text{No. of ways of selecting 2 digit from 6 digit} = {}^6C_2 = \frac{6 \times 5}{2} = 15$$

$$\text{Probability} = \frac{1}{15}$$

13. A number of four digit is selected. If the probability that number has non-repeated digits is P then the value of $125P$ is

(1) 63

(2) 64

(3) 65

(4) 66

Sol. Answer (1)

$$\text{Probability } P = \frac{9 \times 9 \times 8 \times 7}{9 \times 10 \times 10 \times 10} = \frac{63}{125}$$

$$\text{Then } 125P = 63$$

14. Three dice are thrown simultaneously. If the probability that numbers on them are same is P , then the value of $36P$ is

(1) 3

(2) 2

(3) 1

(4) 4

Sol. Answer (3)

$$\text{Probability of getting same number } P = \frac{6}{216} = \frac{1}{36}$$

$$36P = 1$$

15. Two cards are drawn at random from a pack of 52 playing cards. If one is red and other is black then probability is

(1) $\frac{26}{51}$

(2) $\frac{1}{2}$

(3) $\frac{25}{51}$

(4) $\frac{29}{51}$

Sol. Answer (1)

$$\text{Probability} = \frac{{}^{16}C_1 \times {}^{26}C_1}{{}^{52}C_2} = \frac{26 \times 26}{\frac{52 \times 51}{2}} = \frac{2 \times 26 \times 26}{52 \times 51} = \frac{26}{51}$$

16. A bag contains 5 black and 4 white balls. Two balls are drawn. The probability that the balls are of same colour is

(1) $\frac{4}{9}$

(2) $\frac{8}{9}$

(3) $\frac{2}{9}$

(4) $\frac{1}{9}$

Sol. Answer (1)

$$\text{Required probability} = \frac{{}^5C_2 + {}^4C_2}{{}^9C_2} = \frac{10 + 6}{36} = \frac{4}{9}$$

17. A bag contains 3 red and 5 black balls. Two balls are drawn at random. The probability that one ball is red and other is black is P then the value of $28P - 15$ is

- (1) 1 (2) 0 (3) $\frac{1}{2}$ (4) 2

Sol. Answer (2)

$$\text{Required probability } P = \frac{3 \times 5}{{}^8C_2} = \frac{3 \times 5}{\frac{8 \times 7}{2}} = \frac{2 \times 3 \times 5}{8 \times 7}$$

$$P = \frac{15}{28}$$

$$28P - 15 = 0$$

18. If n is a positive integer then the probability that 3^n has 3 at unit place is

- (1) $\frac{1}{4}$ (2) $\frac{1}{5}$ (3) $\frac{3}{10}$ (4) $\frac{1}{8}$

Sol. Answer (1)

$$3^1 = 3, 3^2 = 9, 3^3 = 27, 3^4 = 81, 3^5 = 3$$

$$\text{Probability} = \frac{1}{4}$$

19. If $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}, P(C) = \frac{1}{2}$ and

$$P(A \cap B) = P(B \cap C) = P(C \cap A) = \frac{1}{4} \text{ and}$$

$$P(A \cap B \cap C) = \frac{1}{8} \text{ then the probability that at least one of them appears is}$$

- (1) $\frac{17}{24}$ (2) $\frac{19}{24}$ (3) $\frac{13}{24}$ (4) $\frac{11}{24}$

Sol. Answer (1)

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$= \frac{1}{2} + \frac{1}{3} + \frac{1}{2} - \frac{3}{4} + \frac{1}{8} = \frac{17}{24}$$

20. If $P(A) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{3}$ then the probability that A happens but B does not happen is

- (1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{5}{6}$ (4) $\frac{1}{6}$

Sol. Answer (4)

$$P(A) = \frac{1}{2} \quad P(A \cap B) = \frac{1}{3}$$

$$P(A \cap B') = P(A) - P(A \cap B)$$

$$\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

21. Three students Ram, Ali and John appear at a competitive examinations. The probability of Ram comes first is thrice that of Ali and the probability of Ali coming first is thrice of John. The probability of John coming first is

- (1) $\frac{1}{13}$ (2) $\frac{3}{13}$ (3) $\frac{12}{13}$ (4) $\frac{9}{13}$

Sol. Answer (1)

Let $P(\text{John}) = p$, and $P(\text{Ali}) = 3p$ and $P(\text{Ram}) = 9p$.

But $P(\text{John}) + P(\text{Ali}) + P(\text{Ram}) = 1$

$$\Rightarrow 13p = 1 \Rightarrow p = \frac{1}{13}$$

22. A number is selected from first 100 natural numbers. The probability that number is perfect square or a perfect cube is

- (1) $\frac{7}{50}$ (2) $\frac{3}{25}$ (3) $\frac{2}{25}$ (4) $\frac{1}{25}$

Sol. Answer (2)

A = getting a number which is perfect square $\{1, 4, 9, 16, 25, 36, 49, 64, 81, 100\}$

B = getting a number which is perfect cube $\{1, 8, 27, 64\}$

$A \cap B = \{1, 64\}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= \frac{10}{100} + \frac{4}{100} - \frac{2}{100} = \frac{12}{100} = \frac{3}{25}$$

Option (2) is correct.

23. Two numbers a and b are chosen at random from the set of integers 1, 2, 3, 15. The probability that the equation $2a - 3b = 0$ is satisfied, is

- (1) $\frac{1}{21}$ (2) $\frac{2}{15}$ (3) $\frac{2}{21}$ (4) $\frac{3}{31}$

Sol. Answer (1)

$a, b \in \{1, 2, 3, \dots, 15\}$

$n(S) = {}^{15}C_2 = 15 \times 7$

$2a = 3b$

$$a = \frac{3}{2} \cdot b \quad \Rightarrow \quad b \text{ must be even number}$$

b is 2, 4, 6, 8, 10

$$n(E) = \frac{5}{15 \times 7} = \frac{1}{21}$$

24. 5 cards are drawn from a pack of 52 cards. The probability that these 5 will contain just one king is

(1) $\frac{{}^{48}C_4}{{}^{52}C_5}$ (2) $\frac{{}^{48}C_4}{{}^{51}C_4}$ (3) $\frac{{}^{48}C_{44} \times 4}{{}^{52}C_5}$ (4) $\frac{{}^{48}C_{44} \times 4}{{}^{51}C_4}$

Sol. Answer (3)

$$\text{We have } n(S) = {}^{52}C_5$$

$$n(E) = {}^4C_1 \times {}^{48}C_4, \quad P(E) = \frac{4 \times {}^{48}C_4}{{}^{52}C_5} = \frac{4 \times {}^{48}C_{44}}{{}^{52}C_5}$$

25. Let $n = 2^3 \cdot 4^5 \cdot 6^8 \cdot 5^4$. A positive factor is taken at random from the possible positive factors of n . Then the probability that the selected factor is a perfect square and divisible by 100 is

(1) $\frac{5}{99}$ (2) $\frac{10}{99}$ (3) $\frac{1}{6}$ (4) $\frac{71}{99}$

Sol. Answer (2)

$$\text{We have, } n = 2^3 \times 4^5 \times 6^8 \times 5^4 = 2^{21} \times 3^8 \times 5^4$$

$$n(S) = \text{The number of possible positive factors} = 22 \times 9 \times 5.$$

$$n(E) = \text{The number of factors which is a perfect square and divisible by 100.}$$

$$= (2^2, 2^4, 2^6, \dots, 2^{20}) \times (5^2, 5^4) \times (3^0, 3^2, 3^4, 3^6, 3^8)$$

$$= 10 \times 2 \times 5$$

$$P(E) = \frac{10 \times 2 \times 5}{22 \times 9 \times 5} = \frac{10}{99}$$

26. 12 members of a committee are to sit down at random round a table. Probability that there are 3 members between the two particular members A and B , is

(1) $\frac{2}{11}$ (2) $\frac{7}{11}$ (3) $\frac{4}{11}$ (4) $\frac{5}{11}$

Sol. Answer (1)

$$\text{Number of ways of arrangement of 12 members round a table} = (12 - 1)! = 11!$$

$$'3' \text{ members is selected from 10 member (excluding } A \text{ and } B) \text{ by } {}^{10}C_3 \text{ ways.}$$

$$\text{Assuming 3 selected members and } A \text{ and } B \text{ as single unit}$$

$$n(E) = {}^{10}C_3 \times (8 - 1)! \times 3! \times 2!$$

$$P(E) = \frac{{}^{10}C_3 \times 7! \times 3! \times 2!}{11!} = \frac{2}{11}$$

27. Two integers x and y are chosen with replacement out of the set $\{0, 1, 2, \dots, 10\}$. The probability that $|x - y|$ doesn't exceed 5 is

(1) $\frac{5}{11}$

(2) $\frac{81}{121}$

(3) $\frac{3}{11}$

(4) $\frac{91}{121}$

Sol. Answer (4)

We have, $-5 \leq x - y \leq 5$

$$n(S) = 11 \times 11 = 121$$

x	y	No. of ways
0	0 to 5	6
1	0 to 6	7
2	0 to 7	8
3	0 to 8	9
4	0 to 9	10
5	0 to 10	11
6	1 to 10	9
7	2 to 10	8
8	3 to 10	7
10	4 to 10	6
		91

$$n(E) = \frac{91}{121}$$

28. Dialing a telephone number, an old man forgets the last two digits remembering only that they are different, dialled at random. The probability that the number is dialled correctly

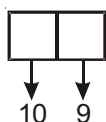
(1) $\frac{1}{90}$

(2) $\frac{1}{45}$

(3) $\frac{2}{45}$

(4) $\frac{2}{100}$

Sol. Answer (1)



total number of ways to dial last two digits = $10 \times 9 = 90$

$$n(E) = 1$$

$$P(E) = \frac{1}{90}$$

29. There are four balls of different colours and four boxes of colours same as of the balls the number of ways in which the balls, one in each box, could be placed such that a ball does not go to a box of its own colour is

(1) $\frac{5}{8}$

(2) $\frac{3}{8}$

(3) $\frac{1}{8}$

(4) $\frac{7}{8}$

Sol. Answer (2)

$$n(S) = 4 \times 3 \times 2 \times 1 = 24$$

$$n(E) = 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right) = 9$$

$$P(E) = \frac{9}{24} = \frac{3}{8}$$

30. Four digit number is formed from all possible ways. The probability that a number $xyzt$ chosen from the number satisfy $x < y = z > t$ is

(1) $\frac{2}{75}$

(2) $\frac{3}{75}$

(3) $\frac{2}{25}$

(4) $\frac{3}{25}$

Sol. Answer (1)

$$\text{We have } n(S) = 9 \times 10 \times 10 \times 10$$

For fixed y and z as n , x can be selected in $n - 1$ ways and t can be selected in n ways.

$$\therefore n(E) = \sum_{n=1}^9 n(n-1) = 240$$

$$P(E) = \frac{240}{9 \times 10 \times 10 \times 10} = \frac{2}{75}$$

31. A coin is tossed twice. Events E and F are defined as follows A = heads on first toss, B = heads on second toss. The probability of the occurrence of the event $A \cup B$ is

(1) $\frac{1}{2}$

(2) $\frac{3}{4}$

(3) $\frac{1}{4}$

(4) $\frac{1}{3}$

Sol. Answer (2)

$$S = \{HH, HT, TH, TH\}$$

$$n(S) = 4, n(A) = 2, n(B) = 2, n(A \cap B) = 1$$

$$\therefore P(A \cup B) = \frac{2}{4} + \frac{2}{4} - \frac{1}{4} = \frac{3}{4}$$

32. A coin is tossed 10 times. The probability of getting at most 6 heads is

(1) $\frac{51}{64}$

(2) $\frac{53}{64}$

(3) $\frac{105}{512}$

(4) $\frac{193}{512}$

Sol. Answer (2)

Probability of getting at most 6 heads

$$= {}^{10}C_0 \cdot p^0 q^{10} + {}^{10}C_1 \cdot p^1 q^9 + {}^{10}C_2 \cdot p^2 q^8 + {}^{10}C_3 \cdot p^3 q^7 + {}^{10}C_4 \cdot p^4 q^6 + {}^{10}C_5 \cdot p^5 q^5 + {}^{10}C_6 \cdot p^6 q^4$$

$$\text{where } p = \frac{1}{2}, q = \frac{1}{2}$$

$$= \left(\frac{1}{2}\right)^{10} \left[{}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3 + {}^{10}C_4 + {}^{10}C_5 + {}^{10}C_6 \right]$$

$$= \frac{848}{2^{10}} = \frac{53}{64}$$

33. A pair of dice is rolled together till a sum of either 5 or 7 is obtained. The probability that 5 comes before 7 is

(1) $\frac{1}{5}$

(2) $\frac{2}{5}$

(3) $\frac{3}{5}$

(4) $\frac{2}{3}$

Sol. Answer (2)Let E_1 = the event of getting 5 in a roll of two dice

$$= \{(1, 4), (2, 3), (3, 2), (4, 1)\}$$

$$\therefore P(E_1) = \frac{4}{36} = \frac{1}{9}$$

Let E_2 = the event of getting either 5 or 7

$$= \{(1, 4), (2, 3), (3, 2), (4, 1), (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

$$\therefore P(E_2) = \frac{10}{36} = \frac{5}{18}$$

$$\therefore P(E_2) = \frac{13}{18}$$

Now, the probability of getting 5 before 7

$$= P(E_1) + P(\bar{E}_2 E_1) + P(\bar{E}_2 \bar{E}_2 E_1) + \dots$$

$$= \frac{1}{9} + \frac{13}{18} \cdot \frac{1}{9} + \frac{13}{18} \cdot \frac{13}{18} \cdot \frac{1}{9} + \dots$$

$$= \frac{1}{9} \left[1 + \frac{13}{18} + \left(\frac{13}{18}\right)^2 + \dots \text{to } \infty \right]$$

$$= \frac{1}{9} \left[\frac{1}{1 - \frac{13}{18}} \right] = \frac{1}{9} \cdot \frac{18}{5} = \frac{2}{5}$$

34. Given two events A and B . If odds against A are 2 : 1 and those in favour of $A \cup B$ are as 3 : 1, then

(1) $\frac{1}{2} \leq P(B) \leq \frac{3}{9}$

(2) $\frac{5}{12} \leq P(B) \leq \frac{3}{4}$

(3) $\frac{1}{2} \leq P(B) \leq \frac{3}{5}$

(4) $\frac{1}{3} \leq P(B) \leq \frac{3}{4}$

Sol. Answer (2)

Odd against A are 2 : 1

$$\therefore P(A) = \frac{1}{3}$$

Odd in favour of $A \cup B$ is 3 : 1

$$\therefore P(A \cup B) = \frac{3}{4}$$

$$\text{Now, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) \leq P(A) + P(B)$$

$$\Rightarrow P(B) \geq \frac{5}{12}$$

$$\text{Also } P(B) \leq P(A \cup B)$$

$$\therefore P(B) \leq \frac{3}{4}$$

35. A bag contains 5 brown and 4 white socks. A man pulls out two socks. The probability that these are of the same colour, is

(1) $\frac{5}{108}$

(2) $\frac{18}{108}$

(3) $\frac{30}{108}$

(4) $\frac{48}{108}$

Sol. Answer (4)

Probability of getting 2 socks either brown or white

$$= \frac{{}^5C_2 + {}^4C_2}{{}^9C_2} = \frac{10+6}{36} = \frac{16}{36}$$

36. The rod of length 10 cm is broken into three parts, so that each part is having a length as an integral multiple of 1 cm. The probability that the parts are forming a triangle, is

(1) $\frac{1}{4}$

(2) $\frac{1}{2}$

(3) $\frac{3}{4}$

(4) $\frac{1}{3}$

Sol. Answer (1)

Let x, y and z be the parts and $x \leq y \leq z$

Then $(x, y, z) \in (1, 1, 8), (1, 2, 7), (1, 3, 6), (1, 4, 5), (2, 3, 5), (2, 4, 4), (2, 2, 6), (3, 3, 4)$

(x, y, z) formed a triangle when $(2, 4, 4), (3, 3, 4)$

$$\therefore \text{Required probability} = \frac{2}{8} = \frac{1}{4}$$

37. Two numbers are selected randomly from the set $S = \{1, 2, 3, 4, 5, 6\}$ without replacement one by one. The probability that minimum of the two number is less than 4, is

(1) $\frac{1}{15}$

(2) $\frac{14}{15}$

(3) $\frac{1}{5}$

(4) $\frac{4}{5}$

Sol. Answer (4)

Number of elements in sample space = ${}^6C_2 = 15$

Favourable events = The minimum of the two numbers is less than 4

= (one is less than 4) or (both are less than 4)

$$= {}^3C_1 \times {}^3C_1 + {}^3C_2 = 9 + 3 = 12$$

$$\therefore \text{Required probability} = \frac{12}{15} = \frac{4}{5}$$

38. If two balls are drawn from a bag containing 2 white, 4 red and 5 black balls, the chance that both are red is

(1) $\frac{1}{55}$

(2) $\frac{2}{55}$

(3) $\frac{4}{55}$

(4) $\frac{6}{55}$

Sol. Answer (4)

$$n(E) = {}^4C_2$$

$$n(S) = {}^{11}C_2$$

$$\Rightarrow P(E) = \frac{{}^4C_2}{{}^{11}C_2} = \frac{6}{55}$$

39. In the previous question the probability that one is red and the other is black is

(1) $\frac{1}{11}$

(2) $\frac{2}{11}$

(3) $\frac{3}{11}$

(4) $\frac{4}{11}$

Sol. Answer (4)

$$n(E) = {}^4C_1 \cdot {}^5C_1$$

$$n(S) = {}^{11}C_2$$

$$\Rightarrow P(E) = \frac{{}^5C_1 \cdot {}^4C_1}{{}^{11}C_2} = \frac{4}{11}$$

40. Out of 20 consecutive numbers, two are chosen at random, the probability that their sum is odd is

(1) $\frac{9}{10}$

(2) $\frac{11}{20}$

(3) $\frac{10}{19}$

(4) $\frac{10}{21}$

Sol. Answer (3)

$$n(S) = {}^{20}C_2$$

The sum of the numbers will be odd only if one of them is odd and the other is even

$$\therefore n(E) = {}^{10}C_1 \cdot {}^{10}C_1$$

$$P(E) = \frac{100}{{}^{20}C_2} = \frac{100 \times 2}{20 \times 19} = \frac{10}{19}$$

41. If the letters of "Attempt" are written down at random, then the chance that no two t's are together is

(1) $\frac{1}{7}$

(2) $\frac{2}{7}$

(3) $\frac{3}{7}$

(4) $\frac{4}{7}$

Sol. Answer (2)

$$x_1 \text{ } T \text{ } x_2 \text{ } T \text{ } x_3 \text{ } T \text{ } x_4$$

$$x_1 + x_2 + x_3 + x_4 = 4$$

$$x_1 \geq 0; x_2 \geq 1; x_3 \geq 1; x_4 \geq 0$$

$$\text{So, the total number of solutions} = {}^{2+4-1}C_{4-1} = {}^5C_3 = 10$$

So, total number of ways in which they can be arranged is $10 \times 4!$

$$n(S) = \frac{7!}{3!}$$

$$P(E) = \frac{10 \times 4!}{7!} \times 3! = \frac{10 \times 3 \times 2}{7 \times 6 \times 5} = \frac{2}{7}$$

42. n men sit in a row. The probability that not more than m men ($n > m$) sit between two particular men A and B is

(1) $\frac{(m+1)(2n-m-2)}{n(n-1)}$

(2) $\frac{m(2n-m-2)}{n(n-1)}$

(3) $\frac{(2n-m-2)}{n(n-1)}$

(4) $\frac{(2n-m)(m+1)}{n(n-1)}$

Sol. Answer (1)

Let m people be seated in between A and B



$$\begin{aligned} \text{So, total number of ways of arrangement} &= {}^{n-2}C_m \times m! \cdot (n-m-1)! \times 2! \\ &= 2(n-m-1) \cdot (n-2)! \end{aligned}$$

Similarly when there are $(m-1)$ people in between total number of arrangements

$$= {}^{n-2}C_{m-1} \times (m-1)! \cdot (n-m)! \times 2! = 2(n-m) \cdot (n-2)!$$

$$\text{So, the required probability} = \frac{2}{n(n-1)} \{n-m-1 + n-m + n-m+1 + \dots + n-1\}$$

$$= \frac{(m+1)(2n-m-2)}{n(n-1)}$$

43. From a bag containing a balls, m balls are drawn simultaneously and replaced; then n balls are drawn. The chance that exactly r balls are common to the two drawings is

(1) $\frac{{}^m C_r}{{}^a C_n}$

(2) $\frac{{}^{a-m} C_{n-r}}{{}^a C_n}$

(3) $\frac{{}^m C_r \cdot {}^{a-m} C_{n-r}}{{}^a C_n}$

(4) $\frac{{}^m C_r \cdot {}^{n-r} C_{a-m}}{{}^a C_n}$

Sol. Answer (3)

$$n(S) = {}^a C_n$$

Now out of these n things r must be from the first group.

$$\text{So, total number of ways} = {}^m C_r \cdot {}^{(a-r)-(m-r)} C_{n-r} = {}^m C_r \cdot {}^{a-m} C_{n-r}$$

$$\text{Hence, the required probability is} = \frac{{}^m C_r \cdot {}^{a-m} C_{n-r}}{{}^a C_n}$$

44. One number is selected from the set $\{1, 2, 3, \dots, 100\}$. The probability that the number is even or a multiple of 5 is

(1) $\frac{7}{10}$

(2) $\frac{3}{5}$

(3) $\frac{4}{5}$

(4) $\frac{2}{5}$

Sol. Answer (2)

A : An even number appears $\Rightarrow n(A) = 50$

B : A multiple of 5 occurs $\Rightarrow n(B) = 20$

$$\therefore n(A \cap B) = 10$$

$$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{50}{100} + \frac{20}{100} - \frac{10}{100}$$

$$= \frac{3}{5}$$

45. Out of n persons sitting at a round table, three, A, B, C are chosen at random. The chance that no two of these three are sitting next to one another is

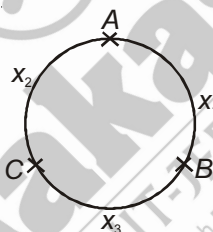
(1) $\frac{(n-4)(n-5)}{2 \times (n-1)}$

(2) $\frac{(n-4)(n-5)}{2 \times (n-1)(n-2)(n-3)}$

(3) $\frac{(n-4)(n-5)}{(n-1)(n-2)}$

(4) $\frac{n(n-4)}{(n-1)(n-2)}$

Sol. Answer (3)



Let three persons be seated around a table such that there x_1, x_2, x_3 number of people between them.

$$x_1 + x_2 + x_3 = n - 3$$

$$x_1 \geq 1; x_2 \geq 1; x_3 \geq 1$$

So, total number of integral solutions = ${}^{n-4}C_2$

So, total number of ways of arrangement = ${}^{(n-4)}C_2 \times 2! \times (n-3)!$

$$\text{So, required probability} = \frac{{}^{(n-4)}C_2 \times (n-3)! \times 2!}{(n-1)!}$$

$$= \frac{(n-4)(n-5)}{(n-1)(n-2)}$$

46. A and B play with two dice on the condition that A wins if he throws 6 before B throw 7, then the probability that A wins is

(1) $\frac{36}{1141}$

(2) $\frac{180}{1141}$

(3) $\frac{172}{1141}$

(4) $\frac{191}{1141}$

Sol. Answer (2)

The probability that sum of numbers will be 6 is $\frac{5}{36}$

The probability that sum of numbers is 7 is $\frac{1}{6}$

So, probability that A will win is

$$\left(\frac{5}{36}\right) + \left(\frac{31}{36}\right) \cdot \frac{5}{6} \cdot \left(\frac{5}{36}\right) + \left(\frac{31}{36}\right) \cdot \frac{5}{6} \cdot \left(\frac{31}{36}\right) \cdot \frac{5}{6} \cdot \frac{5}{36} + \dots$$

$$\left(\frac{5}{36}\right) \left\{ 1 + \left(\frac{31 \times 5}{36^2}\right) + \left(\frac{31 \times 5}{36^2}\right)^2 + \dots \infty \right\}$$

$$\frac{5}{36} \left\{ \frac{1}{1 - \frac{155}{36 \times 36}} \right\} = \frac{5}{36} \times \frac{36 \times 36}{1141} = \frac{180}{1141}$$

47. The decimal part of logarithm of two numbers taken at random are found to seven places. The chance that second can be subtracted from the first

(1) $(0.5)^7$ (2) $(0.2)^7$ (3) $(0.55)^7$ (4) $(0.66)^7$

Sol. Answer (3)

$$n(S) = 10^7 \times 10^7 = 10^{14}$$

Now, suppose if the digit selected is 9, we have 10 options, if its 8, we have 9 options and so on.

So, for each digit we have $(10 + 9 + 8 + \dots + 1) = 55$ options

$$P(E) = \frac{55^7}{10^{14}} = (0.55)^7$$

48. A draws a card from a pack of n cards marked 1, 2, 3, ..., n ; the card is replaced in the pack and then B draws a card. The chance that A draws a higher card is

(1) $\frac{1}{n} \left(\frac{1}{n} + \frac{2}{n} + \dots + \frac{n-1}{n} \right)$

(2) $\left(\frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \dots + \frac{n-1}{n} \right)$

(3) $\frac{1}{n} \left(\frac{1}{n} + \frac{2}{n} + \dots + \frac{n}{n} \right)$

(4) $\left(\frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \dots + \frac{n}{n} \right)$

Sol. Answer (1)

The probability that A draws a card of particular number = $\frac{1}{n}$

Now the favourable cases for A that A draws higher card than B $\Rightarrow$ B draws a smaller card than A are 0, 1, 2, ..., $n-1$ as A draws 1, 2, ..., n respectively.

$$\text{The required probability} = \frac{1}{n} \left[\frac{1}{n} + \frac{2}{n} + \dots + \frac{n-1}{n} \right]$$

49. In a singing competition a group of 10 people, participate, each person has exactly one friend and rest are strangers, the probability that both the singers of the duet are complete strangers is

- (1) $\frac{1}{15}$ (2) $\frac{1}{45}$ (3) $\frac{8}{9}$ (4) $\frac{22}{45}$

Sol. Answer (3)

$$n(S) = {}^{10}C_2 = 45$$

Out of 5 pairs 2 pairs can be selected in 5C_2 ways

$$\therefore n(E) = {}^5C_2 \cdot 2^2$$

$$\therefore P(E) = \frac{{}^5C_2 \cdot 2^2}{{}^{10}C_2} = \frac{40}{45} = \frac{8}{9}$$

50. If A and B are two events such that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$ and $P(A \cap B) = \frac{1}{5}$, then $P(\bar{B} | \bar{A}) =$

- (1) $\frac{37}{40}$ (2) $\frac{37}{45}$ (3) $\frac{23}{40}$ (4) $\frac{23}{45}$

Sol. Answer (1)

$$\text{Here, } P(A \cup B) = \frac{1}{3} + \frac{1}{4} - \frac{1}{5} = \frac{23}{60}$$

$$P\left(\frac{\bar{B}}{\bar{A}}\right) = \frac{1 - P(A \cup B)}{1 - P(A)} = \frac{1 - \frac{23}{60}}{1 - \frac{1}{3}} = \frac{\frac{37}{60}}{\frac{2}{3}} = \frac{37}{40}$$

51. If E and F are independent events such that $0 < P(E) < 1$ and $0 < P(F) < 1$, then

- (1) E and F^c (the complement of the event F) are independent
 (2) E^c and F^c are independent

(3) $P\left(\frac{E}{F}\right) + P\left(\frac{E^c}{F^c}\right) = 1$

- (4) All of these

Sol. Answer (4)

$$P(E \cap F) = P(E) \cdot P(F)$$

$$\begin{aligned} \text{Now, } P(E \cap F^c) &= P(E) - P(E \cap F) \\ &= P(E) [1 - P(F)] = P(E) P(F^c) \end{aligned}$$

$$\begin{aligned} \text{and } P(E^c \cap F^c) &= 1 - P(E \cup F) \\ &= 1 - [P(E) + P(F) - P(E \cap F)] \\ &= [1 - P(E)] [1 - P(F)] \\ &= P(E^c) \cdot P(F^c) \end{aligned}$$

52. If A and B are two independent events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{5}$, then

(1) $P(A|B) = \frac{1}{2}$

(2) $P(A|(A \cup B)) = \frac{5}{6}$

(3) $P((A \cap B) | (A' \cup B')) = 0$

(4) All of these

Sol. Answer (4)

$$P(A|B) = P(A) \text{ as independent event} = \frac{1}{2}$$

$$P(A|A \cup B) = \frac{P[A \cap (A \cup B)]}{P(A \cup B)}$$

$$\{\text{Since } A \cap (A \cup B) = A\}$$

$$\Rightarrow P\left(\frac{A}{A \cup B}\right) = \frac{P(A)}{P(A \cup B)} = \frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{5} - \frac{1}{10}} = \frac{5}{6}$$

$$\text{Similarly } P\left(\frac{A \cap B}{A' \cup B'}\right) = 0$$

53. Let $0 < P(A) < 1$, $0 < P(B) < 1$ and $P(A \cup B) = P(A) + P(B) - P(A)P(B)$. Then

(1) $P(B|A) = P(B) - P(A)$

(2) $P(A^c \cup B^c) = P(A^c) + P(B^c)$

(3) $P(A \cup B)^c = P(A^c) \cdot P(B^c)$

(4) $P(A|B) = P(A) | P(B)$

Sol. Answer (3)

$$\text{Since } P(A \cap B) = P(A) \cdot P(B)$$

It means that A and B are independent events so A^c and B^c will also be independent. Hence

$$P(A \cup B)^c = P(A^c \cap B^c) = P(A^c) P(B^c)$$

($\because$ De-Morgan's Law)

As A is independent of B , hence

$$P(A|B) = P(A) \quad (\because P(A \cap B) = P(B) P(A|B))$$

54. A coin is tossed three times in succession. If E is the event that there are at least two heads and F is the event in which first throw is a head, then $P(E|F)$ is equal to

(1) $\frac{3}{4}$

(2) $\frac{3}{8}$

(3) $\frac{1}{2}$

(4) $\frac{1}{8}$

Sol. Answer (1)

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

$$n(E) = 4, n(F) = 4 \text{ and } n(E \cap F) = 3$$

$$\therefore P\left(\frac{E}{F}\right) = \frac{P(E \cap F)}{P(F)} = \frac{\frac{3}{8}}{\frac{4}{8}} = \frac{3}{4}$$

55. One dice is thrown three times and the sum of the thrown numbers is 15. The probability for which number 4 appears in first throw is

- (1) $\frac{1}{18}$ (2) $\frac{1}{36}$ (3) $\frac{1}{9}$ (4) $\frac{1}{3}$

Sol. Answer (1)

We have to find the bounded probability to get sum 15 when 4 appears first. Let the event of getting sum

15 of three thrown number is A and the event of appearing 4 is B . So we have to find $P\left(\frac{A}{B}\right)$. But

$P\left(\frac{A}{B}\right) = \frac{n(A \cap B)}{n(B)}$ when $n(A \cap B)$ and $n(B)$ respectively denote the number of digits in $A \cap B$ and B . Now $n(B) = 36$, because first throw is of 4. So another two throws stop by $6 \times 6 = 36$ types. Three dices have only two throws, which starts from 4 and given sum 15. i.e., (4, 5, 6) and (4, 6, 5)

So, $n(A \cap B) = 2$, $n(B) = 36$

$$\therefore P\left(\frac{A}{B}\right) = \frac{2}{36} = \frac{1}{18}$$

56. Bag A contains 4 green and 3 red balls and bag B contains 4 red and 3 green balls. One bag is taken at random and a ball is drawn and noted it is green. The probability that it comes from bag B.

- (1) $\frac{2}{7}$ (2) $\frac{2}{3}$ (3) $\frac{3}{7}$ (4) $\frac{1}{3}$

Sol. Answer (3)

It is based on Bayes' theorem

Probability of picked bag A, $P(A) = \frac{1}{2}$

Probability of picked bag B, $P(B) = \frac{1}{2}$

Probability of green ball picked from bag A = $P(A) \cdot P\left(\frac{G}{A}\right) = \left(\frac{1}{2}\right)\left(\frac{4}{7}\right) = \frac{2}{7}$

Probability of green ball picked from bag B = $P(B) \cdot P\left(\frac{G}{B}\right) = \frac{1}{2} \times \frac{3}{7} = \frac{3}{14}$

Total probability of green ball = $\frac{2}{7} + \frac{3}{14} = \frac{1}{2}$

Probability of fact that green ball is drawn from bag B

$$P\left(\frac{G}{B}\right) = \frac{P(B) P\left(\frac{G}{B}\right)}{P(A) P\left(\frac{G}{A}\right) + P(B) P\left(\frac{G}{B}\right)}$$

$$= \frac{\left(\frac{1}{2}\right)\left(\frac{3}{7}\right)}{\frac{1}{2} \times \frac{4}{7} + \frac{1}{2} \times \frac{3}{7}} = \frac{3}{7}$$

57. Two dice are thrown. The probability that the sum of the numbers appearing on the two dice is 11, if 5 appears on the first is

- (1) $\frac{1}{36}$ (2) $\frac{1}{6}$ (3) $\frac{5}{6}$ (4) $\frac{5}{36}$

Sol. Answer (2)

Since we are given that 5 appears on first die so to get sum 11, six must be on the second and hence, the required probability = $\frac{1}{6}$

58. If A and B are two events such that $P(A) = \frac{3}{8}$, $P(B) = \frac{5}{8}$ and $P(A \cup B) = \frac{3}{4}$, then $P(A|B)$ is equal to

- (1) $\frac{2}{5}$ (2) $\frac{2}{3}$ (3) $\frac{3}{5}$ (4) $\frac{3}{4}$

Sol. Answer (1)

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\left(\frac{3}{8}\right) + \left(\frac{5}{8}\right) - \left(\frac{3}{4}\right)}{\left(\frac{5}{8}\right)} = \frac{2}{5}$$

59. If $4P(A) = 6P(B) = 10P(A \cap B) = 1$, then $P\left(\frac{B}{A}\right)$

- (1) $\frac{2}{5}$ (2) $\frac{3}{5}$ (3) $\frac{7}{10}$ (4) $\frac{19}{60}$

Sol. Answer (1)

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{10}}{\frac{1}{4}} = \frac{2}{5}$$

60. One Indian and four American men and their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian man is seated adjacent to his wife given that each American man is seated adjacent to his wife is

- (1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{2}{5}$ (4) $\frac{1}{5}$

Sol. Answer (3)

$A \rightarrow$ The event that the Indian man is seated adjacent to his wife.

$B \rightarrow$ The event that the American man is seated adjacent to his wife.

$$P(A) = \frac{2 \times 8!}{9!}; \quad P(B) = \frac{2^4 \times 5!}{9!}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{2^5 \times 4!}{9!}}{\frac{2^4 \times 5!}{9!}} = \frac{2}{5}$$

61. For a biased die the probabilities for different faces to turn up are given below

Face	1	2	3	4	5	6
Probability	0.1	0.32	0.21	0.15	0.05	0.17

The die is tossed and you are told that either face 1 or 2 has turned up. Then the probability that it is face 1, is

- (1) $\frac{5}{21}$ (2) $\frac{5}{22}$ (3) $\frac{4}{21}$ (4) $\frac{2}{11}$

Sol. Answer (1)

$$\text{Required probability} = \frac{0.1}{0.1+0.32} = \frac{0.1}{0.42} = \frac{5}{21}$$

62. A and B are two events such that $P(A) = 0.8$, $P(B) = 0.6$ and $P(A \cap B) = 0.5$, then the value of $P(A|B)$ is

- (1) $\frac{5}{6}$ (2) $\frac{5}{8}$ (3) $\frac{9}{10}$ (4) $\frac{5}{7}$

Sol. Answer (1)

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{0.5}{0.6} = \frac{5}{6}$$

63. One ticket is selected at random from 50 tickets numbered 00, 01, 02, 03, ..., 49. Then the probability that the sum of the digits on the selected ticket is 8, given that the product of these digits is zero, equals

- (1) $\frac{1}{14}$ (2) $\frac{1}{7}$ (3) $\frac{5}{14}$ (4) $\frac{1}{50}$

Sol. Answer (1)

Y be the event that product of digits on lottery = 0

' X ' the event that sum of digit is 8

$$P(Y = 0) = \frac{14}{50}$$

$$(X = 8) \cap (Y = 0) = \{08\}$$

$$P(E) = P\{(X = 8) \cap (Y = 0)\} = \frac{1}{50}$$

$$\frac{P(E)}{P(Y = 0)} = \frac{\frac{1}{50}}{\frac{14}{50}} = \frac{1}{14}$$

64. It is given that the events A and B are such that $P(A) = \frac{1}{4}$, $P(A|B) = \frac{1}{2}$ and $P(B|A) = \frac{2}{3}$. Then $P(B)$ is

- (1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) $\frac{1}{2}$ (4) $\frac{1}{6}$

Sol. Answer (1)

$$P(A) = \frac{P(A \cap B)}{P(B|A)} \Rightarrow P(A \cap B) = \frac{1}{4} \times \frac{2}{3} = \frac{1}{6}$$

$$\therefore P(B) = \frac{P(A \cap B)}{P(A|B)} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}$$

65. If the events A and B are mutually exclusive, then $P(A|B) =$

- (1) 0 (2) 1 (3) $\frac{1}{2}$ (4) $\frac{1}{3}$

Sol. Answer (1)

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

Since A and B are mutually exclusive

$$\text{So, } P(A \cap B) = 0$$

$$\text{Hence, } P\left(\frac{A}{B}\right) = \frac{0}{P(B)} = 0$$

66. Two aeroplanes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2 respectively. The second plane will bomb only if the first misses the target. The probability that the target is hit by the second plane is

- (1) 0.06 (2) 0.14 (3) 0.32 (4) 0.7

Sol. Answer (3)

$$P(E_1) = 0.3; P(E_2) = 0.2$$

$$P(\bar{E}_1) = 0.7; P(\bar{E}_2) = 0.8$$

$$\text{Required probability} = (0.7)(0.2) + (0.7)(0.8)(0.7)(0.2) + (0.7)(0.8)(0.7)(0.8)(0.7)(0.2) + \dots$$

$$= \frac{(0.7)(0.2)}{1 - (0.7)(0.8)} = \frac{14}{44} = 0.32$$

67. If two events A and B are such that $P(A^c) = 0.3$ and $P(B) = 0.4$ and $P(AB^c) = 0.5$, then $P[B|(A \cup B^c)]$ is equal to

- (1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{1}{4}$ (4) $\frac{1}{5}$

Sol. Answer (3)

$$\begin{aligned} P[B|(A \cup B^c)] &= \frac{P(B \cap (A \cup B^c))}{P(A \cup B^c)} \\ &= \frac{P(A \cap B)}{P(A) + P(B^c) - P(A \cap B^c)} \\ &= \frac{P(A) - P(A \cap B^c)}{P(A) + P(B^c) - P(A \cap B^c)} \\ &= \frac{0.7 - 0.5}{0.8} = \frac{1}{4} \end{aligned}$$

68. In a certain town, 40% of the people have brown hair, 25% have brown eyes and 15% have both brown hair and brown eyes. If a person selected at random from the town, has brown hair, the probability that he also has brown eyes, is

- (1) $\frac{1}{5}$ (2) $\frac{3}{8}$ (3) $\frac{1}{3}$ (4) $\frac{2}{3}$

Sol. Answer (2)

$$P(A) = \frac{40}{100}, \quad P(B) = \frac{25}{100} \text{ and } P(A \cap B) = \frac{15}{100}$$

$$\text{So, } P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{15}{100}}{\frac{40}{100}} = \frac{3}{8}$$

69. If $\bar{E}$ and $\bar{F}$ are the complementary events of events E and F respectively and if $0 < P(F) < 1$, then

- (1) $P(E|F) + P(\bar{E}|F) = 1$ (2) $P(E|F) + P(E|\bar{F}) = 1$
 (3) $P(\bar{E}|F) + P(E|\bar{F}) = 1$ (4) $P(E|\bar{F}) + P(\bar{E}|\bar{F}) = 0$

Sol. Answer (1)

$$\begin{aligned} P(E|F) + P(\bar{E}|F) &= \frac{P(E \cap F) + P(\bar{E} \cap F)}{P(F)} \\ &= \frac{P\{(E \cap F) \cup (\bar{E} \cap F)\}}{P(F)} \quad (\because E \cap F \text{ and } \bar{E} \cap F \text{ are disjoint}) \\ &= \frac{P\{(E \cup \bar{E}) \cap F\}}{P(F)} = 1 \end{aligned}$$

Similarly we can show that (2) and (3) are not true

$$P\left(\frac{E}{F}\right) + P\left(\frac{\bar{E}}{\bar{F}}\right) = \frac{P(E \cap \bar{F})}{P(\bar{F})} + \frac{P(\bar{E} \cap \bar{F})}{P(\bar{F})} = \frac{P(\bar{F})}{P(\bar{F})} = 1$$

70. A pair of dice is thrown simultaneously. The probability that exactly one 3 turns up, is

- (1) $\frac{5}{18}$ (2) $\frac{5}{9}$ (3) $\frac{1}{4}$ (4) $\frac{1}{3}$

Sol. Answer (1)

$$P(3 \text{ comes exactly once}) = 2 \times \frac{5}{6} \times \frac{1}{6} = \frac{5}{18}$$

71. If A and B are two events such that $P(A) \neq 0$ and $P(B) \neq 1$, then $P\left(\frac{\bar{A}}{\bar{B}}\right) =$

- (1) $1 - P\left(\frac{A}{B}\right)$ (2) $1 - P\left(\frac{\bar{A}}{B}\right)$ (3) $\frac{1 - P(A \cup B)}{P(\bar{B})}$ (4) $\frac{P(\bar{A})}{P(\bar{B})}$

Sol. Answer (3)

$$P\left(\frac{\bar{A}}{\bar{B}}\right) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{P(\overline{A \cup B})}{P(\bar{B})} = \frac{1 - P(A \cup B)}{P(\bar{B})}$$

72. If A and B are two events such that $A \subseteq B$, then $P(B|A)$ is equal to

- (1) 0 (2) 1 (3) $\frac{1}{2}$ (4) $\frac{1}{3}$

Sol. Answer (2)

$$\text{Since } A \subseteq B \Rightarrow A \cap B = B \cap A = A$$

$$\text{Hence, } P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$$

73. A family has two children. If one of them is boy, then the probability that other is also a boy, is

- (1) $\frac{1}{2}$ (2) $\frac{1}{4}$ (3) $\frac{1}{3}$ (4) $\frac{1}{5}$

Sol. Answer (3)

$$\text{Let } P(A) = \text{probability of a boy in two children} = \frac{3}{4}$$

Because cases are $BB, BG, GB, GG = 4$

Favourable cases are $BB, BG, GB = 3$

$$\text{Probability that the second child is also boy is } P(A \cap B) = \frac{1}{4}$$

$$\text{We have to find } P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

74. A divisor of $N = 2^5 3^4 5^3$ is selected at random. The probability that selected number is divisible by 200 if it is known that it is divisible by 100 is

- (1) $\frac{3}{4}$ (2) $\frac{1}{2}$ (3) $\frac{1}{4}$ (4) $\frac{4}{5}$

Sol. Answer (1)

$$\text{Required probability} = \frac{3 \times 5 \times 2}{4 \times 2 \times 5} = \frac{3}{4}$$

75. For two events A and B , if $P(A) = P(A|B) = \frac{1}{4}$ and $P(B|A) = \frac{1}{2}$, then

(1) A and B are independent

(2) $P(A'|B) = \frac{3}{4}$

(3) $P(B'|A') = \frac{1}{2}$

(4) All of these

Sol. Answer (4)

$$P\left(\frac{B}{A}\right) = \frac{1}{2} \Rightarrow \frac{P(B \cap A)}{P(A)} = \frac{1}{2} \Rightarrow P(B \cap A) = \frac{1}{8}$$

$$P\left(\frac{A}{B}\right) = \frac{1}{4} \Rightarrow \frac{P(A \cap B)}{P(B)} = \frac{1}{4} \Rightarrow P(B) = \frac{1}{2}$$

$$P(A \cap B) = \frac{1}{8} = P(A) \cdot P(B)$$

$\therefore$ Events A and B are independent

$$\text{Now, } P\left(\frac{A'}{B}\right) = \frac{P(A' \cap B)}{P(B)} = \frac{P(A')P(B)}{P(B)} = \frac{3}{4}$$

$$\text{and } P\left(\frac{B'}{A'}\right) = \frac{P(B' \cap A')}{P(A')} = \frac{P(B') \cdot P(A')}{P(A')} = \frac{1}{2}$$

76. A bag A contains 2 white and 3 red balls and bag B contains 4 white and 5 red balls. One ball is drawn at random from a randomly chosen bag and is found to be red. The probability that it was drawn from bag B was

(1) $\frac{5}{14}$

(2) $\frac{5}{16}$

(3) $\frac{5}{18}$

(4) $\frac{25}{52}$

Sol. Answer (4)

Let E_1 be the event that the ball is drawn from bag A , E_2 the event that it is drawn from bag B and E that the ball is red. We have to find $P(E_2|E)$.

Since both the bags are equally likely to be selected, we have $P(E_1) = P(E_2) = \frac{1}{2}$

$$\text{Also, } P(E|E_1) = \frac{3}{5} \text{ and } P(E|E_2) = \frac{5}{9}$$

$$P(E_2|E) = \frac{P(E_2) \cdot P(E|E_2)}{P(E_1) \cdot P(E|E_1) + P(E_2) \cdot P(E|E_2)}$$

$$= \frac{\frac{1}{2} \cdot \frac{5}{9}}{\frac{1}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{5}{9}} = \frac{25}{52}$$

77. In a single throw of two dice what is the probability of obtaining a number greater than 7, if 4 appears on the first dice

(1) $\frac{1}{3}$

(2) $\frac{1}{2}$

(3) $\frac{1}{12}$

(4) $\frac{1}{6}$

Sol. Answer (2)

Since 4 has appeared on the first, so we are required 4 or 5 or 6 on second dice.

Hence required probability = $\frac{3}{6} = \frac{1}{2}$

78. Three coins are tossed. If one of them shows tail, then the probability that all three coins show tail, is

(1) $\frac{1}{7}$

(2) $\frac{1}{8}$

(3) $\frac{2}{7}$

(4) $\frac{1}{6}$

Sol. Answer (1)

Let E be the event in which all three coins shows tail and F be the event in which a coin shows tail.

$\therefore F = \{HHT, HTH, THH, HTT, THT, TTH, TTT\}$ and $E = \{TTT\}$

Required probability = $P(E | F) = \frac{P(E \cap F)}{P(F)} = \frac{1}{7}$

79. For $k = 1, 2, 3$, the box B_k contains k red balls and $(k + 1)$ white balls. Let $P(B_1) = \frac{1}{2}$, $P(B_2) = \frac{1}{3}$ and $P(B_3) = \frac{1}{6}$. A box is selected at random and a ball is drawn from it. If a red ball is drawn, then the probability that it has come from box B_2 , is

(1) $\frac{35}{78}$

(2) $\frac{14}{39}$

(3) $\frac{10}{13}$

(4) $\frac{12}{13}$

Sol. Answer (2)

In a box, $B_1 = 1R, 2W$; $B_2 = 2R, 3W$, and $B_3 = 3R, 4W$

Also, given that, $P(B_1) = \frac{1}{2}$, $P(B_2) = \frac{1}{3}$ and $P(B_3) = \frac{1}{6}$

$$P\left(\frac{B_2}{R}\right) = \frac{P(B_2) \cdot P\left(\frac{R}{B_2}\right)}{P(B_1) \cdot P\left(\frac{R}{B_1}\right) + P(B_2) \cdot P\left(\frac{R}{B_2}\right) + P(B_3) \cdot P\left(\frac{R}{B_3}\right)}$$

$$= \frac{\left(\frac{1}{3}\right)\left(\frac{2}{5}\right)}{\frac{1}{2} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{5} + \frac{1}{6} \times \frac{3}{7}} = \frac{14}{39}$$

80. Two cards are drawn one by one from a pack of cards. The probability of getting first card an ace and second honoured one is (before drawing second card first card is not placed again in the pack)

(1) $\frac{1}{52}$ (2) $\frac{5}{52}$ (3) $\frac{5}{221}$ (4) $\frac{4}{13}$

Sol. Answer (3)

$$P(E_1) = \frac{4}{52} = \frac{1}{13}, \quad P\left(\frac{E_2}{E_1}\right) = \frac{15}{51} = \frac{5}{17}$$

$$P(E_1 \cap E_2) = P(E_1) \cdot P\left(\frac{E_2}{E_1}\right) = \frac{1}{13} \cdot \frac{5}{17} = \frac{5}{221}$$

81. 8 coins are tossed simultaneously. The probability of getting at least 6 heads is

(1) $\frac{57}{64}$ (2) $\frac{229}{256}$ (3) $\frac{7}{64}$ (4) $\frac{37}{256}$

Sol. Answer (4)

The required probability

$$= {}^8C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^2 + {}^8C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right) + {}^8C_8 \left(\frac{1}{2}\right)^8$$

$$= \frac{37}{256}$$

82. If three dice are thrown together, then the probability of getting 5 on at least one of them is

(1) $\frac{125}{216}$ (2) $\frac{215}{216}$ (3) $\frac{1}{216}$ (4) $\frac{91}{216}$

Sol. Answer (4)

$$\text{Required probability} = {}^3C_1 \cdot \left(\frac{1}{6}\right) \left(\frac{5}{6}\right)^2 + {}^3C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right) + {}^3C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^0$$

$$= \frac{91}{216}$$

83. In binomial probability distribution, mean is 3 and standard deviation is $\frac{3}{2}$. Then the value of p is

(1) $\frac{1}{4}$ (2) $\frac{3}{4}$ (3) $\frac{1}{2}$ (4) $\frac{1}{3}$

Sol. Answer (1)

$$\text{Mean} = np = 3$$

$$\text{S.D.} = \sqrt{npq} = \frac{3}{2}$$

$$q = \frac{npq}{np} = \frac{9}{4 \times 3} = \frac{3}{4}$$

$$p = 1 - \frac{3}{4} = \frac{1}{4}$$

84. The probability of a bomb hitting a bridge is $\frac{1}{2}$ and two direct hits are needed to destroy it. The least number of bombs required so that the probability of the bridge being destroyed is greater than 0.9, is
- (1) 8 (2) 7 (3) 6 (4) 9

Sol. Answer (2)

Let n be the least number of bombs required and X the number of bombs that hit the bridge. Then X follows a binomial distribution with parameter n and $P = \frac{1}{2}$.

$$\text{Now, } P(X \geq 2) > 0.9 \Rightarrow 1 - P(X < 2) > 0.9$$

$$\Rightarrow P(X = 0) + P(X = 1) < 0.1$$

$$\Rightarrow {}^nC_0 \left(\frac{1}{2}\right)^n + {}^nC_1 \left(\frac{1}{2}\right)^{n-1} \left(\frac{1}{2}\right) < 0.1$$

$$\Rightarrow 10(n + 1) < 2^n$$

85. In a simultaneous toss of four coins, the probability of getting exactly three heads is

- (1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{1}{4}$ (4) $\frac{1}{6}$

Sol. Answer (3)

$$\text{Required probability} = {}^4C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right) = \frac{1}{4}$$

86. A die is tossed twice. Getting a number greater than 4 is considered a success. Then the variance of the probability distribution of the number of success is

- (1) $\frac{2}{9}$ (2) $\frac{4}{9}$ (3) $\frac{1}{3}$ (4) $\frac{2}{3}$

Sol. Answer (2)

$$p = \frac{2}{6} = \frac{1}{3} \Rightarrow q = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\text{Also, } n = 2$$

$$\therefore \text{Variance} = npq = 2 \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{9}$$

87. A biased coin with probability p , $1 > p > 0$, of heads is tossed until a head appears for the first time. If the probability that the number of tosses required is even is $\frac{2}{5}$, then p equals to

- (1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{1}{4}$ (4) $\frac{1}{5}$

Sol. Answer (2)

Let X denote the number of tosses required.

$$\text{Then, } P(X = r) = (1 - p)^{r-1} \cdot p \text{ for } r = 1, 2, 3, \dots$$

Let E denote the event that the number of tosses is even

Then, $P(E) = P(X = 2) \cup (X = 4) \cup (X = 6) \dots$

$$P(E) = P(X = 2) + P(X = 4) + P(X = 6) + \dots$$

$$P(E) = (1 - p)p + (1 - p)^3 p + (1 - p)^5 p + \dots = \frac{1 - p}{2 - p}$$

$$\therefore P(E) = \frac{2}{5} \Rightarrow p = \frac{1}{3}$$

88. If two coins are tossed 5 times, then the probability of getting 5 heads and 5 tails is

- (1) $\frac{63}{256}$ (2) $\frac{1}{1024}$ (3) $\frac{2}{205}$ (4) $\frac{9}{64}$

Sol. Answer (1)

Probability for a head = $\frac{1}{2}$ i.e., $p = \frac{1}{2}$, $q = \frac{1}{2}$

$$\therefore \text{Probability} = {}^{10}C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^5 = \frac{63}{256}$$

89. A dice is thrown 100 times and getting an even number is considered a success. The variance of the number of success will be

- (1) 10 (2) 25 (3) 18 (4) 15

Sol. Answer (2)

Let E = event of getting an even number from $\{2, 4, 6\} \Rightarrow n(E) = 3$

$\therefore$ Probability of success $p = \frac{3}{6} = \frac{1}{2}$ and probability of failure, $q = \frac{1}{2}$

$$\text{Now, variance} = npq = 100 \times \frac{1}{2} \times \frac{1}{2} = 25$$

90. A contest consist of predicting the result win, draw or defeat of 7 football matches. A sent his entry predicting at random. The probability that his entry will contain exactly 4 correct predictions is

- (1) $\frac{8}{3^7}$ (2) $\frac{16}{3^7}$ (3) $\frac{280}{3^7}$ (4) $\frac{560}{3^7}$

Sol. Answer (3)

$$P(\text{correct prediction}) = \frac{1}{3}; P(\text{wrong prediction}) = \frac{2}{3}$$

For exactly 4 right predictions

$$\text{Probability} = {}^7C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^3 = \frac{280}{3^7}$$

91. In a bag there are three tickets numbered 1, 2, 3. A ticket is drawn at random and put back and this is done four times. The probability that the sum of the numbers is even, is

- (1) $\frac{41}{81}$ (2) $\frac{39}{81}$ (3) $\frac{40}{81}$ (4) $\frac{42}{81}$

Sol. Answer (1)

The total number of ways of selecting 4 tickets = $3^4 = 81$.

The favourable number of ways = Sum of coefficients of $x^2, x^4, \dots$ in $(x + x^2 + x^3)^4$
 = Sum of coefficients of $x^2, x^4, \dots$ in $x^4 (1 + x + x^2)^4$

Let $(1 + x + x^2)^4 = 1 + a_1x + a_2x^2 + \dots + a_8x^8$

Then, $3^4 = 1 + a_1 + a_2 + a_3 + \dots + a_8$ (Put $x = 1$)

$1 = 1 - a_1 + a_2 - a_3 + \dots + a_8$ (Put $x = -1$)

$\therefore 3^4 + 1 = 2(1 + a_2 + a_4 + a_6 + a_8)$

$\Rightarrow a_2 + a_4 + a_6 + a_8 = 41$

Thus, sum of the coefficients of $x^2, x^4, \dots = 41$

Hence probability = $\frac{41}{81}$.

92. Assuming that for a husband-wife couple the chance of their child being a boy or a girl are the same, the probability of their two children being a boy and a girl is

(1) $\frac{1}{4}$

(2) 1

(3) $\frac{1}{2}$

(4) $\frac{1}{8}$

Sol. Answer (3)

The chance of being a boy or girl

$$p = \frac{1}{2}, q = \frac{1}{2}$$

Total child = 2, Number of boys = 1

$$P(1 \text{ boy, } 1 \text{ girl}) = {}^2C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{2-1} = \frac{1}{2}$$

93. A bag contains 2 white and 4 black balls. A ball is drawn 5 times with replacement. The probability that at least 4 of the balls are white is

(1) $\frac{8}{141}$

(2) $\frac{10}{243}$

(3) $\frac{11}{243}$

(4) $\frac{8}{41}$

Sol. Answer (3)

$$\text{Probability for white ball} = \frac{2}{6} = \frac{1}{3}$$

$$\text{Probability for black ball} = \frac{4}{6} = \frac{2}{3}$$

$$\therefore \text{Probability} = {}^5C_5 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^0 + {}^5C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right) = \left(\frac{1}{3}\right)^4 \left[\frac{1}{3} + 5 \cdot \frac{2}{3}\right] = \frac{11}{3^5} = \frac{11}{243}$$

94. In a binomial distribution the probability of getting a success is $\frac{1}{4}$ and standard deviation is 3, then its mean is

(1) 6

(2) 8

(3) 12

(4) 10

Sol. Answer (3)

$$\text{Probability of success } p = \frac{1}{4}$$

Probability of unsuccess $q = \frac{3}{4}$

Mean = np

Standard deviation = $\sqrt{\text{Variance}}$

$\therefore$ Variance = 9

$$npq = 9 \Rightarrow n \cdot \frac{1}{4} \cdot \frac{3}{4} = 9$$

$$n = 48; \text{Mean} = np = \frac{1}{4} \times 48 = 12$$

95. If X follows a binomial distribution with parameter $n = 100$ and $p = \frac{1}{3}$, then $P(X = r)$ is maximum when r is equal to

- (1) 16 (2) 32 (3) 33 (4) 50

Sol. Answer (3)

Since $(n+1)p = \frac{101}{3}$ is not an integer,

therefore, $P(X = r)$ is maximum when $r = \left[\frac{101}{3} \right] \equiv 33$

96. If X has binomial distribution with mean np and variance npq , then $\frac{P(X = k)}{P(X = k-1)}$ is equal to

- (1) $\frac{n-k}{k-1} \frac{p}{q}$ (2) $\frac{n-k+1}{k} \frac{p}{q}$ (3) $\frac{n+1}{k} \frac{q}{p}$ (4) $\frac{n-1}{k+1} \frac{q}{p}$

Sol. Answer (2)

Here, Mean = np and variance = npq

$$\therefore \frac{P(X = k)}{P(X = k-1)} = \frac{{}^nC_k p^k q^{n-k}}{{}^nC_{k-1} p^{k-1} q^{n-k+1}} = \frac{n-k+1}{k} \cdot \frac{p}{q}$$

97. In a box containing 100 eggs, 10 eggs are rotten. The probability that out of a sample of 5 eggs none is rotten if the sampling is with replacement is

- (1) $\left(\frac{1}{10}\right)^5$ (2) $\left(\frac{1}{5}\right)^5$ (3) $\left(\frac{9}{5}\right)^5$ (4) $\left(\frac{9}{10}\right)^5$

Sol. Answer (4)

$$\text{Let, } P(\text{fresh egg}) = \frac{90}{100} = \frac{9}{10} = p$$

$$P(\text{rotten egg}) = \frac{10}{100} = \frac{1}{10} = q; n = 5, r = 5$$

$$\text{So the probability that none egg is rotten} = {}^5C_5 \left(\frac{9}{10}\right)^5 \left(\frac{1}{10}\right)^0 = \left(\frac{9}{10}\right)^5$$

98. If a dice is thrown 7 times, then the probability of obtaining 5 exactly 4 times is

- (1) ${}^7C_4 \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^3$ (2) ${}^7C_4 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^4$ (3) $\left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^3$ (4) $\left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^4$

Sol. Answer (1)

$$\text{Required probability} = {}^7C_4 \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^3$$

99. A dice is thrown ten times. If getting even number is considered as a success, then the probability of four success is

- (1) ${}^{10}C_4 \left(\frac{1}{2}\right)^4$ (2) ${}^{10}C_4 \left(\frac{1}{2}\right)^6$ (3) ${}^{10}C_4 \left(\frac{1}{2}\right)^8$ (4) ${}^{10}C_6 \left(\frac{1}{2}\right)^{10}$

Sol. Answer (4)

$$\text{Required probability} = {}^{10}C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^6 = {}^{10}C_4 \left(\frac{1}{2}\right)^{10} = {}^{10}C_6 \left(\frac{1}{2}\right)^{10}$$

100. In a binomial distribution, the mean is 4 and variance is 3. Then its mode is

- (1) 6 (2) 5 (3) 4 (4) 7

Sol. Answer (3)

$$\text{Given that } np = 4 \text{ and } npq = 3$$

$$\Rightarrow q = \frac{3}{4}, p = 1 - q, 1 - \frac{3}{4} = \frac{1}{4}$$

Since mode is an integer x such that

$$np + p > x > np - q$$

$$\Rightarrow 4 + \frac{1}{4} > x > 4 - \frac{3}{4}$$

$$\Rightarrow \frac{17}{4} > x > \frac{13}{4}$$

$$\Rightarrow 3.25 > x > 4.25$$

$$\Rightarrow x = 4$$

101. A coin is tossed successively three times. The probability of getting exactly one head or 2 heads, is

- (1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) $\frac{3}{4}$ (4) $\frac{1}{3}$

Sol. Answer (3)

$$\text{Required probability} = 3 \left(\frac{1}{2}\right) \left(\frac{1}{2}\right)^2 + {}^3C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right) = \frac{3}{4}$$

102. If there are n independent trials, p and q are the probability of success and failure respectively, then probability of exactly r success

- (1) ${}^nC_{r+1} p^r q^{n-r}$ (2) ${}^nC_r p^{r-1} q^{r+1}$ (3) ${}^nC_r q^{n-r} p^r$ (4) ${}^nC_r p^{r+1} q^{r-1}$

Sol. Answer (3)

It is a fundamental concept.

103. In a trial the probability of success is twice the probability of failure. In six trials the probability of at least four successes will be

- (1) $\frac{496}{729}$ (2) $\frac{400}{729}$ (3) $\frac{500}{729}$ (4) $\frac{600}{729}$

Sol. Answer (1)

Let the probability of success and failure be p and q respectively

$$\therefore p = 2q \text{ and } p + q = 1$$

$$3q = 1; q = \frac{1}{3}; p = \frac{2}{3}$$

$$\therefore \text{Required probability} = {}^6C_4 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^2 + {}^6C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right) + {}^6C_6 \left(\frac{2}{3}\right)^6 = \frac{496}{729}$$

104. The probability that a student is not a swimmer is $\frac{1}{5}$. What is the probability that out of 5 students, 4 are swimmers?

- (1) ${}^5C_4 \left(\frac{4}{5}\right)^4 \left(\frac{1}{5}\right)$ (2) $\left(\frac{4}{5}\right)^4 \left(\frac{1}{5}\right)$ (3) ${}^5C_1 \left(\frac{1}{5}\right) \left(\frac{4}{5}\right)^4 \times {}^5C_4$ (4) ${}^5C_4 \left(\frac{4}{5}\right) \left(\frac{1}{5}\right)^4$

Sol. Answer (1)

The probability that student is not swimmer $p = \frac{1}{5}$ and probability that student is swimmer $q = \frac{4}{5}$

$$\therefore \text{Probability that out of 5 students 4 are swimmer} = {}^5C_4 \left(\frac{4}{5}\right)^4 \left(\frac{1}{5}\right)^1$$

105. A coin is tossed $2n$ times. The chance that the number of times one gets head is not equal to the number of times one gets tail is

- (1) $\frac{2n!}{(n!)^2} \cdot \left(\frac{1}{2}\right)^{2n}$ (2) $1 - \frac{(2n)!}{(n!)^2}$ (3) $1 - \frac{(2n)!}{(n!)^2} \cdot \frac{1}{4^n}$ (4) $\frac{(2n)!}{(n!)^2}$

Sol. Answer (3)

The required probability

$$= 1 - \text{probability of equal number of heads and tails}$$

$$= 1 - {}^{2n}C_n \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n = 1 - \frac{(2n)!}{n!n!} \left(\frac{1}{4}\right)^n$$

106. A coin is tossed 10 times. The probability of getting exactly six heads is

(1) $\frac{512}{513}$

(2) $\frac{105}{512}$

(3) $\frac{100}{153}$

(4) $\frac{100}{512}$

Sol. Answer (2)

$$p = P(\text{getting a head}) = \frac{1}{2}, \quad q = \frac{1}{2}$$

$\therefore$ Required probability = $P(\text{six success})$

$$= {}^{10}C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^4 = \frac{105}{512}$$

107. One coin is thrown 100 times. The probability of coming tails in odd number

(1) $\frac{1}{2}$

(2) $\frac{1}{8}$

(3) $\frac{3}{8}$

(4) $\frac{3}{4}$

Sol. Answer (1)

$$\text{Let } p = \text{probability of getting tail} = \frac{1}{2}$$

$$q = \text{probability of getting head} = \frac{1}{2}$$

Also, $p + q = 1$ and $n = 100$

$$\begin{aligned} \therefore \text{Required probability} &= P(X = 1) + P(X = 3) + \dots + P(X = 99) \\ &= {}^{100}C_1 p q^{99} + {}^{100}C_3 p^3 q^{97} + \dots + {}^{100}C_{99} p^{99} q \\ &= \left(\frac{1}{2}\right)^{100} [{}^{100}C_1 + {}^{100}C_3 + \dots + {}^{100}C_{99}] \\ &= \left(\frac{1}{2}\right)^{100} [2^{100} - {}^{100}C_0] \\ &= \left(\frac{1}{2}\right)^{100} (2^{100} - 1) \end{aligned}$$

108. Two cards are drawn successively with replacement from a well shuffled deck of 52 cards then the mean of the number of aces is

(1) $\frac{1}{13}$

(2) $\frac{3}{13}$

(3) $\frac{2}{13}$

(4) $\frac{4}{13}$

Sol. Answer (3)

Let X denote a random variable which is the number of aces. Clearly, X takes values 1, 2

$$\therefore p = \frac{4}{52} = \frac{1}{13}; \quad q = 1 - \frac{1}{13} = \frac{12}{13}$$

$$P(X = 1) = 2 \times \left(\frac{1}{13}\right) \times \left(\frac{12}{13}\right) = \frac{24}{169}$$

$$P(X = 2) = 2 \left(\frac{1}{13}\right)^2 \left(\frac{12}{13}\right) = \frac{1}{169}$$

$$\text{Mean} = \sum p_i x_i = \frac{24}{169} + \frac{2}{169} = \frac{2}{13}$$

109. If X denotes the number of sixes in four consecutive throws of a dice, thus $P(X = 4)$ is

- (1) $\frac{1}{1296}$ (2) $\frac{4}{6}$ (3) 1 (4) $\frac{1295}{1296}$

Sol. Answer (1)

Probability of coming 'six' in one throw is $\frac{1}{6}$

Hence required probability is given by

$$= {}^4C_4 \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^0 = \frac{1}{1296}$$

110. A pair of dice is thrown independently three times. The probability of getting a score of 9 exactly twice is

- (1) $\frac{1}{729}$ (2) $\frac{8}{9}$ (3) $\frac{8}{729}$ (4) $\frac{8}{243}$

Sol. Answer (4)

Probability of getting score 9 in a single throw = $\frac{4}{36} = \frac{1}{9}$

Probability of getting score 9 exactly twice = ${}^3C_2 \cdot \left(\frac{1}{9}\right)^2 \left(\frac{8}{9}\right) = \frac{8}{243}$

111. A die is tossed thrice. If getting a four is considered a success, then the mean and variance of the probability distribution of the number of successes are

- (1) $\frac{1}{2}, \frac{1}{12}$ (2) $\frac{1}{6}, \frac{5}{12}$ (3) $\frac{5}{6}, \frac{1}{2}$ (4) $\frac{1}{2}, \frac{5}{12}$

Sol. Answer (4)

We are given that $n = 3$, $p = \frac{1}{6}$, $q = \frac{5}{6}$

$$\text{Mean} = np = 3 \times \frac{1}{6} = \frac{1}{2}$$

$$\text{Variance} = npq = 3 \times \frac{1}{6} \times \frac{5}{6} = \frac{5}{12}$$

112. The items produced by a firm are supposed to contain 5% defective items. The probability that a sample of 8 items will contain less than 2 defective items, is

- (1) $\frac{27}{20} \left(\frac{19}{20}\right)^7$ (2) $\frac{533}{400} \left(\frac{19}{20}\right)^6$ (3) $\frac{153}{20} \left(\frac{1}{20}\right)^7$ (4) $\frac{35}{16} \left(\frac{1}{20}\right)^6$

Sol. Answer (1)

$${}^8C_1 \left(\frac{1}{20}\right)^1 \left(\frac{19}{20}\right)^7 + {}^8C_0 \left(\frac{1}{20}\right)^0 \left(\frac{19}{20}\right)^8 = \left(\frac{27}{20}\right) \left(\frac{19}{20}\right)^7$$

113. A die is tossed thrice. A success is getting 1 or 6 on a toss. The mean and the variance of number of successes

- (1) $\mu = 1, \sigma^2 = \frac{2}{3}$ (2) $\mu = \frac{2}{3}, \sigma^2 = 1$ (3) $\mu = 2, \sigma^2 = \frac{2}{3}$ (4) $\mu = 1, \sigma^2 = 1$

Sol. Answer (1)

For binomial distribution,

Mean = np and Variance = npq

$$n = 3, p = \frac{2}{6} = \frac{1}{3}, q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\text{So, mean } (\mu) = 3 \times \frac{1}{3} = 1$$

$$\text{Variance } (\sigma^2) = 3 \times \frac{1}{3} \times \frac{2}{3} = \frac{2}{3}$$

114. The probability that a bulb produced by a factory will fuse after 150 days of use is 0.05. What is the probability that out of 5 such bulbs none will fuse after 150 days of use?

- (1) $1 - \left(\frac{19}{20}\right)^5$ (2) $\left(\frac{19}{20}\right)^5$ (3) $\left(\frac{3}{4}\right)^5$ (4) $90\left(\frac{1}{4}\right)^5$

Sol. Answer (2)

$$\text{Here } p = \frac{19}{20}, q = \frac{1}{20}, n = 5, r = 5$$

$$\text{The required probability} = {}^5C_5 \left(\frac{19}{20}\right)^5 \left(\frac{1}{20}\right)^0 = \left(\frac{19}{20}\right)^5$$

115. If A and B are two independent events such that $P(A') = \frac{7}{10}$, $P(B') = \alpha$, $P(A \cup B) = \frac{8}{10}$, then $\alpha =$

- (1) $\frac{2}{7}$ (2) $\frac{5}{7}$ (3) 1 (4) $\frac{3}{7}$

Sol. Answer (1)

$$\text{We have } P(A) = \frac{3}{10}, P(B) = 1 - \alpha, P(A \cap B) = P(A)P(B) = \frac{3}{10}(1 - \alpha)$$

$$\text{Now } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{8}{10} = \frac{3}{10} + 1 - \alpha - \frac{3}{10}(1 - \alpha)$$

$$\Rightarrow \alpha = \frac{2}{7}$$

116. If $P(A \cup B) = P(A \cap B)$ if and only if the relation between $P(A)$ and $P(B)$ is

$$(1) P(A) + P(B) = 2P(A)P\left(\frac{B}{A}\right)$$

$$(2) P(A) + P(B) = 2P(A)P\left(\frac{A}{B}\right)$$

$$(3) P(A) + P(B) = 2P(A)P(B)$$

$$(4) P(A) + P(B) = P(A)P(B)$$

Sol. Answer (1)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$2P(A \cup B) = P(A) + P(B)$$

$$\Rightarrow 2P(A \cap B) = P(A) + P(B)$$

$$2P(A).P\left(\frac{B}{A}\right) = P(A) + P(B)$$

117. The probability that a man aged 50 years will die in a year is p . The probability that out of n men $A_1, A_2, A_3, \dots, A_n$ each aged 50 years, A_1 will die and first to die is

$$(1) \frac{1 - (1 - p)^n}{n}$$

$$(2) \frac{1 - (1 - p)^n}{n^2}$$

$$(3) 1 - (1 - p)^n$$

$$(4) \frac{[1 - (1 - p)^n]^2}{n^2}$$

Sol. Answer (1)

The probability that the person not die in 50 years = $1 - p$

The probability that no one will die is $(1 - p)^n$

The probability that at least one will die is $1 - (1 - p)^n$

The probability that A_1 will die first is $\frac{1 - (1 - p)^n}{n}$

118. The probabilities that a student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects the student has 75% chance of passing in at least one subject, 50% chance of passing in at least two and 40% chance of passing in exactly two. Which is true?

$$(1) p + m + c = \frac{27}{20}, pmc = \frac{1}{10}$$

$$(2) p + m + c = \frac{19}{20}, pmc = \frac{3}{10}$$

$$(3) p + m + c = \frac{17}{20}, pmc = \frac{7}{10}$$

$$(4) p + m + c = \frac{13}{20}, pmc = \frac{1}{4}$$

Sol. Answer (1)

Let A, B, C denote the events of the student passing in mathematics, physics and chemistry respectively.

According to question, probability of not passing in the three subjects i.e. $(1 - m)(1 - p)(1 - c) = \frac{25}{100}$

$$[1 - (m + p) + mp](1 - c) = \frac{1}{4}$$

$$\text{or, } 1 - (m + p + c) + mp + mc + pc - mpc = \frac{1}{4} \quad \dots (1)$$

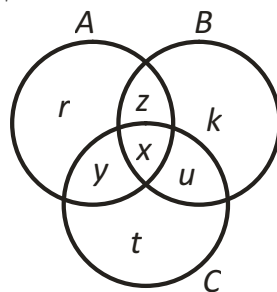
It is given that, $mp(1-c) + mc(1-p) + pc(1-m) + mpc = \frac{50}{100} = \frac{1}{2}$

or, $mp + mc + pc - 2mpc = \frac{1}{2}$... (2)

Also, $mp(1-c) + mc(1-p) + pc(1-m) = \frac{40}{100} = \frac{2}{5}$... (3)

From (1), (2), (3)

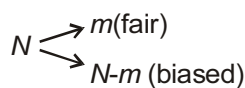
$$m + p + c = \frac{27}{20}, mpc = \frac{1}{10}$$



119. A box contains N coins, m of which are fair and the rest are biased. The probability of getting a head when a fair coin is tossed is $\frac{1}{2}$, while it is $\frac{2}{3}$ when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows heads and the second time it shows tails. The probability that the coin drawn is fair is

- (1) $\frac{9m}{8N+m}$ (2) $\frac{9m}{N+8m}$ (3) $\frac{4m}{8N+m}$ (4) $\frac{4m}{N+8m}$

Sol. Answer (1)



'A' be the event that selected coin is fair $P(A) = \frac{m}{N}$

$$P(A') = \frac{N-m}{N}$$

'E' be the event that first time coin shows heads and second time it shows tails.

$$P\left(\frac{E}{A}\right) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$P\left(\frac{E}{A'}\right) = \frac{2}{3} \times \frac{1}{3} = \frac{2}{9}$$

$$\therefore P\left(\frac{A}{E}\right) = \frac{P(A) \cdot P\left(\frac{E}{A}\right)}{P(A) \cdot P\left(\frac{E}{A}\right) + P(A') \cdot P\left(\frac{E}{A'}\right)}$$

$$= \frac{\frac{m}{N} \times \frac{1}{4}}{\frac{m}{N} \times \frac{1}{4} + \frac{N-m}{N} \times \frac{2}{9}}$$

$$= \frac{9m}{9m + 8(N-m)}$$

$$= \frac{9m}{8N+m}$$

120. If A, B, C are events such that $P(A) = 0.3, P(B) = 0.4, P(C) = 0.8, P(AB) = 0.09, P(AC) = 0.28, P(ABC) = 0.08$. If $P(A \cup B \cup C) \geq 0.75$ then $P(BC)$ lies in the interval

- (1) $[0.12, 0.67]$ (2) $[0.12, 0.48]$ (3) $[0.21, 0.46]$ (4) $[0.23, 0.67]$

Sol. Answer (3)

$$\text{We have, } P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$\Rightarrow P(A \cup B \cup C) = 0.4 + 0.3 + 0.8 - 0.08 - 0.28 - P(BC) + 0.09$$

$$\Rightarrow P(A \cup B \cup C) = 1.59 - 0.36 - P(BC) = 1.23 - P(BC)$$

$$\Rightarrow P(BC) = 1.23 - P(A \cup B \cup C) \quad (0.75 \leq P(A \cup B \cup C) \leq 1)$$

$$\Rightarrow P(BC) \in [0.23, 0.48]$$

121. Three dice, each numbered 1 to 6 are rolled. One die is fair and others are biased so that for each of them a six is twice as likely to occur as any other score. One of the dice is chosen at random and on two throws it shows a six on each occasion. The probability that the die chosen was biased is

- (1) $\frac{423}{674}$ (2) $\frac{288}{337}$ (3) $\frac{118}{337}$ (4) $\frac{323}{674}$

Sol. Answer (2)

A = event that dice is fair.

B = event that dice is biased.

E = event that outcome is '6'

$$P\left(\frac{E}{A}\right) = \frac{1}{6}, P(A) = \frac{1}{3}$$

$$P\left(\frac{E}{B}\right) = \frac{2}{7}, P(B) = \frac{2}{3}$$

$$P\left(\frac{B}{E}\right) = \frac{P(B) \cdot P\left(\frac{E}{B}\right)}{P(B) \cdot P\left(\frac{E}{B}\right) + P(A) \cdot P\left(\frac{E}{A}\right)}$$

$$= \frac{\frac{2}{3} \times \frac{2}{7} \times \frac{2}{7}}{\frac{2}{3} \times \frac{2}{7} \times \frac{2}{7} + \frac{1}{3} \times \frac{1}{6} \times \frac{1}{6}}$$

$$= \frac{8}{3 \times 49} \div \frac{1}{3 \times 6 \times 6}$$

$$= \frac{8 \times 6 \times 6}{8 \times 6 \times 6 + 49} = \frac{288}{337}$$

122. Bag A contains 2 white and 3 red marbles and bag B contains 4 white and 5 red marbles. One marble is drawn at random from one of the bags and is found to be red. The probability that it was drawn from the bag B is

- (1) $\frac{4}{13}$ (2) $\frac{25}{52}$ (3) $\frac{36}{65}$ (4) $\frac{41}{78}$

Sol. Answer (2)

'R' be the event that selected marble is red

$$P\left(\frac{R}{A}\right) = \frac{3}{5}, P\left(\frac{R}{B}\right) = \frac{5}{9}$$

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{2}$$

$$P\left(\frac{B}{R}\right) = \frac{P(B) \cdot P\left(\frac{R}{B}\right)}{P(B) \cdot P\left(\frac{R}{B}\right) + P(A) \cdot P\left(\frac{R}{A}\right)}$$

$$= \frac{\frac{1}{2} \times \frac{5}{9}}{\frac{1}{2} \times \frac{5}{9} + \frac{1}{2} \times \frac{3}{5}}$$

$$= \frac{\frac{5}{18}}{\frac{5}{18} + \frac{3}{10}}$$

$$= \frac{5 \times 10}{50 + 54}$$

$$= \frac{50}{104} = \frac{25}{52}$$

123. 5 different games are to be distributed among 4 children randomly. The probability that each child gets at least one game is

- (1) $\frac{1}{4}$ (2) $\frac{15}{64}$ (3) $\frac{13}{64}$ (4) $\frac{17}{64}$

Sol. Answer (2)

Total number of ways = 4^5

$n(S) = 4^5$

Total ways of distributing so that each child get at least one game = $4^5 - {}^4C_3 \cdot 3^5 + {}^4C_2 \cdot 2^5 - {}^4C_1 \cdot 1^5$
 $= 240$

Required probability = $\frac{240}{4^5} = \frac{15}{64}$

124. Let A and B be two independent events with $P(A) = p$ and $P(B) = q$. If p and q are the roots of the equation $ax^2 + bx + c = 0$, then the probability of the occurrence of at least one of the two events is

- (1) $\frac{b+c}{a}$ (2) $\frac{b-c}{a}$ (3) $\frac{-(b+c)}{a}$ (4) $\frac{-b+c}{a}$

Sol. Answer (3)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= P(A) + P(B) - P(A) \cdot P(B), \text{ as } A \text{ and } B \text{ are independent events}$$

$$= p + q - pq$$

$$= -\frac{b}{a} - \frac{c}{a}$$

$$= -\left(\frac{b+c}{a}\right)$$

125. A circle of maximum area is inscribed in an ellipse. If p is the probability that a point within the ellipse chosen at random lies outside the circle, then the eccentricity of the ellipse is

- (1) $\sqrt{1-p}$ (2) $\sqrt{1-(1-p)^2}$ (3) $\sqrt{1-p^2}$ (4) $\sqrt{(1+p)^2-1}$

Sol. Answer (2)

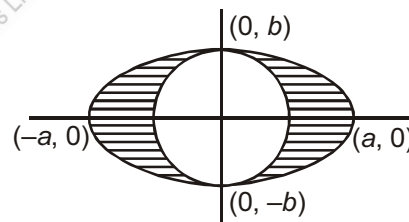
The probability that the point lies outside the circle and inside the ellipse

$$p = \frac{\pi ab - \pi b^2}{\pi ab} = \frac{ab - b^2}{ab}$$

$$p = \frac{a-b}{a} \Rightarrow p = 1 - \frac{b}{a}$$

$$\Rightarrow \frac{b}{a} = 1 - p$$

$$\Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - (1-p)^2}$$



126. The probability of a candidate passing an examination at any one attempt is $\frac{3}{5}$. He carries on entering until he passes and each entry costs him ₹ 1. The expected cost of his passing the exam is (in ₹)

- (1) $\frac{25}{9}$ (2) $\frac{5}{3}$ (3) $\frac{125}{27}$ (4) $\frac{625}{81}$

Sol. Answer (2)

We have,

$$S = \frac{3}{5} + \frac{2}{5} \times \frac{3}{5} \times 2 + \frac{2}{5} \times \frac{2}{5} \times \frac{3}{5} \times 3 + \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{3}{5} \times 4 + \dots$$

$$\Rightarrow \frac{2}{5} pS = \frac{3}{5} \times \frac{2}{5} + \frac{2}{5} \times \frac{2}{5} \times \frac{3}{5} \times 2 + \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{3}{5} \times 3 + \dots$$

On subtraction, $\left(1 - \frac{2}{5}\right)S = \frac{3}{5} + \frac{2}{5} \times \frac{3}{5} + \frac{2}{5} \times \frac{2}{5} \times \frac{3}{5} + \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{3}{5} + \dots$

$$= S = \frac{5}{3}$$

$\Rightarrow$ The expected cost of passing the exam = $\frac{5}{3}$

127. Hari and Krishna throw with one die for prize of ₹ 121, which will be won by a player who throws 4 first. If Hari starts, then the mathematical expectation for Krishna is

- (1) ₹ 57 (2) ₹ 104 (3) ₹ 55 (4) ₹ 98

Sol. Answer (3)

Probability of winning Hari

$$= \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \dots \text{up to } \infty$$

$$= \frac{1}{6} \left(1 + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^4 + \dots \infty \right)$$

$$= \frac{1}{6} \left(\frac{1}{1 - \frac{25}{36}} \right)$$

$$= \frac{1}{6} \times \left(\frac{36}{11} \right) = \frac{6}{11}$$

Probability of winning Krishna = $\frac{5}{11}$

Mathematical expectation for Krishna = $\frac{5}{11} \times 121 = ₹ 55$

128. Three players A, B, C toss a coin in turn till head shows. Let p be the probability that the coin shows a head. Let α, β, γ be respectively the probabilities that A, B and C get first head then

(1) $\alpha = \frac{p}{1 - q^3}$, where $q = 1 - p$

(2) $\beta = \frac{p(1 - p)}{1 - q^3}$

(3) $\gamma = \frac{p - 2p^2 + p^3}{1 - q^3}$

(4) All of these

Sol. Answer (4)

Let A throws first then B and then C, P be the probability that the coin shows head and q the probability that coin shows tail

$$\alpha = p + qqp + qqqqp + \dots$$

$$\alpha = p \left(\frac{1}{1-q^3} \right) = qp + q.q.q.p + qqqqqqp + \dots$$

$$\beta = \frac{qp}{1-q^3} = \frac{p(1-p)}{1-q^3}$$

$$\gamma = q.q.p + qqqqqqp + \dots \infty.$$

$$\gamma = \frac{qp}{1-q^3} = \frac{p(1-p)(1-p)}{1-q^3}$$

129. Let $S = \{1, 2, \dots, 100\}$. The probability of choosing an integer k , $1 \leq k \leq 100$ is proportional to $\log k$. The conditional probability of choosing the integer 2, given that an even integer is chosen is

(1) $\frac{\log 2}{50 \log 2 + \log(50)!}$ (2) $\frac{\log 2}{100 \log 2 + \log(50)!}$ (3) $\frac{\log 2}{50 \log 2 + \log(100)!}$ (4) $\frac{\log 2}{100 \log 2 + (100)!}$

Sol. Answer (1)

P_i = probability of selecting i

Now, $P_1 = r \log 1$

$P_2 = r \log 2$

$P_3 = r \log 3$

.....

.....

$P_{100} = r \log 100$

A be the event that '2' is selected, B be the event that even number is selected.

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{r \log 2}{r \log 2 + r \log 4 + \dots + r \log 100}$$

$$= \frac{\log 2}{\log(2^{50}(1.2.3 \dots 50))} = \frac{\log 2}{50 \log 2 + \log(50)!}$$

130. Digits 1, 2, 3, ..., 9 are written in a random order to form a 9 digit number. If the number happens to be divisible by 4. Then probability that it will be divisible by 36 is given by

(1) $\frac{2}{9}$ (2) $\frac{3}{4}$ (3) $\frac{7}{9}$ (4) 1

Sol. Answer (4)

Clearly, we are given the number is divisible by 4 and number is divisible by 36 if it is also divisible by 9

$$\text{and } 1 + 2 + 3 + \dots + 9 = \frac{9 \times 10}{2} = 45$$

Clearly, every number formed by 1, 2, 3, ..., 9 is always divisible by 9. Thus the number formed by 1, 2, 3 ... 9 is divisible by 4, then it is sure that the number is divisible by 36 also.

$\therefore$ Required probability = 1

131. If m things are distributed among a men and b women. Then the probability that number of things received by men is odd is

$$(1) \frac{1}{2} \left[\frac{(b+a)^m + (b-a)^m}{(b+a)^m} \right]$$

$$(2) \frac{1}{2} \left[\frac{(a+b)^m}{(a-b)^m} \right]$$

$$(3) \frac{1}{2} \left[\frac{(b+a)^m - (b-a)^m}{(b+a)^m} \right]$$

(4) Data inadequate

Sol. Answer (3)

m things can be distributed among $(a+b)$ persons in $(a+b)^m$ ways

i.e. $n(S) = (a+b)^m$

Man will get odd no. of thing

$$\therefore n(E) = {}^m C_1 a \cdot b^{m-1} + {}^m C_3 a^3 b^{m-3} + {}^m C_5 a^5 b^{m-5} + \dots$$

$$= \frac{1}{2} [(a+b)^m - (b-a)^m]$$

$$P(E) = \frac{1}{2} \frac{[(a+b)^m - (b-a)^m]}{(a+b)^m}$$

132. If n biscuits be distributed at random amongst N beggars. The chance that a particular beggar receives r ($< n$) biscuits, is

$$(1) \frac{{}^n C_r (N-1)^{n-r}}{N^n}$$

$$(2) \frac{{}^n C_r \cdot (N-1)^{n-r+1}}{N^n}$$

$$(3) \frac{{}^n C_r \cdot (N-2)^{n-r}}{N^{n+1}}$$

$$(4) \frac{{}^n C_r \cdot (N-1)^{n-r}}{N^{n+1}}$$

Sol. Answer (1)

The number of ways in which n biscuits can be distributed amongst N beggars

$$= N \times N \times N \times \underbrace{\dots}_{n \text{ times}} \times N = N^n$$

If a specified beggar gets r biscuits then the remaining $(n-r)$ biscuits can be distributed amongst $(N-1)$ beggars in $(N-1)^{n-r}$ ways. Also specified beggar can take r biscuits out of n is ${}^n C_r$ ways.

$$\therefore \text{Required probability} = \frac{{}^n C_r \cdot (N-1)^{n-r}}{N^n}$$

133. There are 5 duplicate and 10 original items in an automobile shop and 3 items are brought at random by a customer. The probability that none of items is duplicate, is

$$(1) \frac{20}{91}$$

$$(2) \frac{22}{91}$$

$$(3) \frac{24}{91}$$

$$(4) \frac{89}{91}$$

Sol. Answer (3)

Duplicate = 5, original = 10

Taking 3 items

Probability that none of is duplicate i.e., all three are original,

$$= \frac{{}^{10} C_3}{{}^{15} C_3} = \frac{24}{91}$$

134. A consignment of 15 record players contains 4 defectives. The record players are selected at random, one by one, and examined. The one is examined, are not put back. Then the probability that 9th one examined is the last defective, is

- (1) $\frac{7}{195}$ (2) $\frac{11}{195}$ (3) $\frac{8}{195}$ (4) $\frac{13}{195}$

Sol. Answer (3)

Let A and B be two events such that

A = Getting exactly 3 defectives in examination of 8 record players.

B = 9th record player is defective.

Then,

Required probability is

$$P(A \cap B) = P(A) \cdot P\left(\frac{B}{A}\right)$$

$$\text{Now, } P(A) = \frac{{}^4C_3 \times {}^{11}C_5}{{}^{15}C_8}$$

$$\text{and } P\left(\frac{B}{A}\right) = \frac{1}{7}$$

$$\therefore P(A \cap B) = \frac{{}^4C_3 \times {}^{11}C_5}{{}^{15}C_8} \times \frac{1}{7} = \frac{8}{195}$$

135. A coin is tossed 15 times, then find the probability that exactly 9 consecutive heads appears, is

- (1) $\frac{7}{2^{11}}$ (2) $\frac{9}{2^{11}}$ (3) $\frac{11}{2^{11}}$ (4) $\frac{13}{2^{11}}$

Sol. Answer (2)

The sequence of exactly 9 consecutive heads will starts from 1st toss, 2nd toss, 3rd toss; so on and last toss will be 7th toss.

Let A_i is the event that sequence of exactly 9 consecutive heads will starts from i^{th} toss, then required probability is

$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_7) = P(A_1) + P(A_2) + \dots + P(A_7)$$

$$\text{Now, } P(A_1) = \underbrace{\frac{1}{2} \times \frac{1}{2} \times \dots \times \frac{1}{2} \times \frac{1}{2}}_{10 \text{ times}} \times 1 \times 1 \times \dots \times 1 = \frac{1}{2^{10}}$$

$$P(A_2) = \underbrace{\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \dots \times \frac{1}{2}}_{11 \text{ times}} \times 1 \times \dots = \frac{1}{2^{11}}$$

Similarly,

$$P(A_3) = P(A_4) = P(A_5) = P(A_6) = \frac{1}{2^{11}}$$

$$\text{and } P(A_7) = \frac{1}{2^{10}}$$

Thus required probability

$$\begin{aligned}
 &= \frac{1}{2^{10}} + \frac{1}{2^{11}} + \frac{1}{2^{11}} + \frac{1}{2^{11}} + \frac{1}{2^{11}} + \frac{1}{2^{11}} + \frac{1}{2^{10}} \\
 &= \frac{2}{2^{10}} + \frac{5}{2^{11}} \\
 &= \frac{4+5}{2^{11}} = \frac{9}{2^{11}}
 \end{aligned}$$

136. A team of 8 couples (husband and wife) attend a lucky draw in which 4 persons picked for a prize. Then the probability that there is at least one couple is

(1) $\frac{11}{39}$ (2) $\frac{12}{39}$ (3) $\frac{14}{39}$ (4) $\frac{15}{39}$

Sol. Answer (4)

Required probability

= 1 – Probability at not selecting any couple for the prize

$$= 1 - \frac{{}^{16}C_1 \times {}^{14}C_1 \times {}^{12}C_1 \times {}^{10}C_1}{{}^{16}C_4 \times 4!}$$

$$= \frac{15}{39}$$

137. Three persons A, B and C are to speak at a function along with five others. If they all speak in random order, then probability that A speaks before B and B speaks before C is

(1) $\frac{3}{8}$ (2) $\frac{1}{6}$ (3) $\frac{3}{5}$ (4) $\frac{5}{6}$

Sol. Answer (2)

Total number of ways in which 8 person can speak is ${}^8P_8 = 8!$

The number of ways which can be arranged in the specified speaking order is 8C_3 . There are 5! ways in which the other five can speak. So, favourable number of ways is ${}^8C_3 \times 5!$

$$\text{Required probability} = \frac{{}^8C_3 \times 5!}{8!} = \frac{8! \times 5!}{3! \times 5! \times 8!} = \frac{1}{6}$$

138. Three numbers are selected at random without replacement from the set of numbers $\{1, 2, 3, \dots, n\}$. The conditional probability that the 3rd number lies between the 1st two. If the 1st number is known to be smaller than the 2nd, is

(1) $\frac{1}{6}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) $\frac{5}{6}$

Sol. Answer (3)

Consider the following events

A = 1st number is less than the 2nd

B = The 3rd number lies between the 1st and the 2nd.

We have to find $P\left(\frac{B}{A}\right)$

Number of ways of choosing three numbers out of n is nC_3 . Let the selected numbers be a_1, a_2, a_3 . Then they satisfy exactly one of

$$\begin{array}{lll} a_1 < a_2 < a_3, & a_1 < a_3 < a_2, & a_2 < a_1 < a_3, \\ a_2 < a_3 < a_1, & a_3 < a_1 < a_2, & a_3 < a_2 < a_1 \end{array}$$

The total number of ways of selecting 3 numbers, then arranging them in ${}^nC_3 \times 3!$

$$\therefore P(A) = \frac{{}^nC_3 \times 3}{{}^nC_3 \times 3!} \text{ and } P(A \cap B) = \frac{{}^nC_3}{{}^nC_3 \times 3!}$$

$$\therefore \text{Required probability} = \frac{P(A \cap B)}{P(A)} = \frac{1}{3}$$

139. One hundred identical coins, each with probability ' p ' of showing heads are tossed once. If $0 < p < 1$ and the probability of heads showing on 50 coins is equal to that of head showing on 51 coins, then the value of p is

(1) $\frac{1}{2}$ (2) $\frac{51}{101}$ (3) $\frac{49}{101}$ (4) $\frac{47}{101}$

Sol. Answer (2)

Let X be the number of coins showing heads, then X is a binomial variate with parameters $n = 100$ and P . Now, we have

$$P(X = 50) = P(X = 51)$$

$$\Rightarrow {}^{100}C_{50} P^{50} \cdot (1 - P)^{50} = {}^{100}C_{51} \cdot P^{51} (1 - P)^{49}$$

$$\Rightarrow \frac{(1 - P)}{P} = \frac{{}^{100}C_{51}}{{}^{100}C_{50}} = \frac{50}{51} \Rightarrow P = \frac{51}{101}$$

140. If mean and variance of a binomial distribution are 4 and 2 then $P(|x - 4| \leq 2)$ where x is binomial variate is given by

(1) $\frac{119}{128}$ (2) $\frac{117}{128}$ (3) $\frac{113}{128}$ (4) $\frac{111}{128}$

Sol. Answer (1)

Let n and p be binomial parameter. Then we have

$$np = 4 \text{ and } npq = 2 \Rightarrow q = \frac{1}{2}, p = \frac{1}{2} \text{ and } n = 8$$

Required probability is

$$\begin{aligned} P(|X - 4| \leq 2) &= P(-2 \leq X - 4 \leq 2) \\ &= P(2 \leq X \leq 6) \\ &= P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) \\ &= {}^8C_2 \left(\frac{1}{2}\right)^8 + {}^8C_3 \cdot \left(\frac{1}{2}\right)^8 + {}^8C_4 \left(\frac{1}{2}\right)^8 + {}^8C_5 \left(\frac{1}{2}\right)^8 + {}^8C_6 \left(\frac{1}{2}\right)^8 \\ &= \left(\frac{1}{2}\right)^8 \cdot ({}^8C_2 + {}^8C_3 + \dots + {}^8C_6) = \frac{238}{2^8} = \frac{119}{128} \end{aligned}$$

141. A biased die is tossed and the respective probabilities for various faces to turn up are

Face	1	2	3	4	5	6
Probability	0.1	0.24	0.19	0.18	0.15	0.14

If an even face has turned up, then the probability that it is face 2 or face 4 is

(1) 0.25 (2) 0.42 (3) 0.75 (4) 0.9

Sol. Answer (3)

Let A be the event that an even face turns up and B be the event it is 2 or 4.

$$\text{Then, } P(A) = 0.24 + 0.18 + 0.14 = 0.56$$

$$P(B) = 0.24 + 0.18 = 0.42$$

$$P(A \cap B) = 0.32$$

$$\therefore \text{ Required probability} = P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{0.42}{0.56} = \frac{3}{4} = 0.75$$

142. If three squares of dimension 1×1 are selected at random from the chess board, then the probability that they form the letter L is

(1) $\frac{196}{64C_3}$

(2) $\frac{49}{64C_3}$

(3) $\frac{36}{64C_3}$

(4) $\frac{98}{64C_3}$

Sol. Answer (1)

$$n(S) = {}^{64}C_3$$

Let E be the event of selecting 3 squares which form letter L .

Number of ways of selecting squares consisting of 4 unit squares ${}^7C_1 \times {}^7C_1 = 49$.

Each square with 4 unit square forms L -shapes consisting 3 unit square

$$\therefore n(E) = 7 \times 7 \times 4 = 196$$

$$\therefore \text{ Required probability} = \frac{196}{64C_3}$$

143. 2 cards are selected randomly from a pack of 52 cards. The probability that the first card is a club and second is not a queen, is

(1) $\frac{82}{221}$

(2) $\frac{41}{221}$

(3) $\frac{1}{221}$

(4) $\frac{4}{17}$

Sol. Answer (2)

$$\text{Favourable ways} = {}^{12}C_1 \times {}^{37}C_1 + 1 \times {}^{48}C_1$$

$$\text{Required probability} = \frac{12 \times 41}{52 \times 51} = \frac{41}{221}$$

144. If x and y coordinates of any point P are chosen randomly from intervals $[0, 2]$ and $[0, 1]$ respectively, then the probability $y \leq x^2$ is

(1) $\frac{1}{2}$

(2) $\frac{2}{3}$

(3) $\frac{3}{4}$

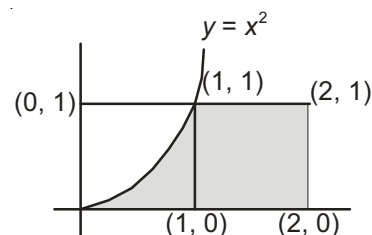
(4) $\frac{1}{4}$

Sol. Answer (2)

We take as our sample space.

Now the condition $y \leq x^2$ satisfied if point (x, y) lie in the shaded region.

$$\therefore \text{ Required probability} = \frac{\int_0^1 x^2 dx + \text{Area of square}}{2} = \frac{\frac{4}{3 \times 2}}{2} = \frac{2}{3}$$



145. One Indian and four American men and their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian man is seated adjacent to the wife given that each American man is seated adjacent to his wife is **[IIT-JEE 2007]**

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{2}{5}$ (4) $\frac{1}{5}$

Sol. Answer (3)

$$\text{Required probability} = \frac{4! \times 2^5}{5! \times 2^4} = \frac{2}{5}$$

146. Let E^c denote the complement of an event E . Let E, F, G be pairwise independent events with $P(G) > 0$ and $P(E \cap F \cap G) = 0$. Then $P(E^c \cap F^c | G)$ equals **[IIT-JEE 2007]**

(1) $P(E^c) + P(F^c)$ (2) $P(E^c) - P(F^c)$ (3) $P(E^c) - P(F)$ (4) $P(E) - P(F^c)$

Sol. Answer (3)

Using the properties of set theory.

147. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is **[IIT-JEE 2008]**

(1) 2, 4 or 8 (2) 3, 6 or 9 (3) 4 or 8 (4) 5 or 10

Sol. Answer (4)

$$P(A) = \frac{2}{5}$$

For independent events

$$P(A \cap B) = P(A) P(B)$$

$$P(A \cap B) \leq \frac{2}{5}$$

$$\Rightarrow P(A \cap B) = \frac{1}{10}, \frac{2}{10}, \frac{3}{10}, \frac{4}{10} \quad (\text{Maximum 4 outcomes may be in } A \cap B)$$

$$(i) \quad P(A \cap B) = \frac{1}{10}$$

$$\Rightarrow P(A) \cdot P(B) = \frac{1}{10}$$

$$P(B) = \frac{1}{10} \times \frac{5}{2} = \frac{1}{4}, \text{ not possible}$$

$$(ii) \quad P(A \cap B) = \frac{2}{10}$$

$$\Rightarrow \frac{2}{5} \times P(B) = \frac{2}{10}$$

$$\Rightarrow P(B) = \frac{5}{10}, \text{ outcomes of } B = 5$$

$$(iii) P(A \cap B) = \frac{3}{10}$$

$$P(A)P(B) = \frac{3}{10}$$

$$\frac{2}{5} \times P(B) = \frac{3}{10}$$

$$P(B) = \frac{3}{4}, \text{ not possible}$$

$$(iv) P(A \cap B) = \frac{4}{10}$$

$$\Rightarrow P(A) \cdot P(B) = \frac{4}{10}$$

$$\Rightarrow P(B) = 1, \text{ outcomes of } B = 10$$

148. A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively, is received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is $\frac{3}{4}$. If the signal received at station B is green, then the probability that the original signal was green is [IIT-JEE 2010]

$$(1) \frac{3}{5}$$

$$(2) \frac{6}{7}$$

$$(3) \frac{20}{23}$$

$$(4) \frac{9}{20}$$

Sol. Answer (3)

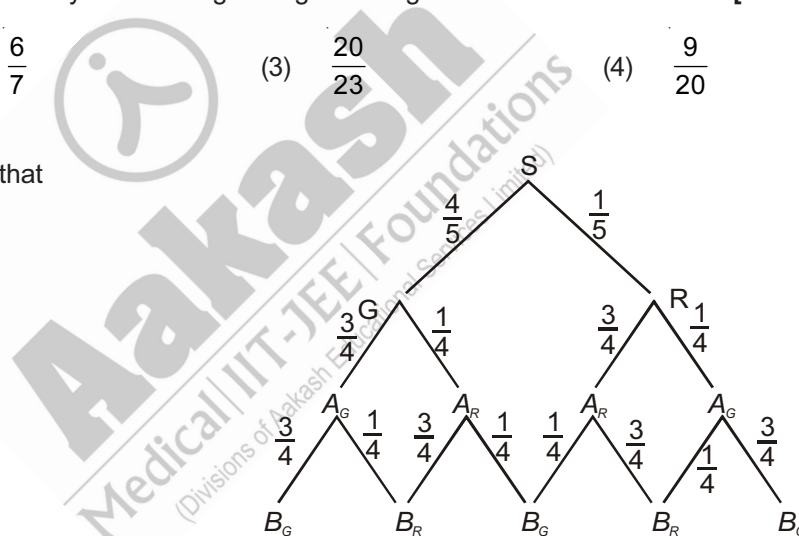
From the tree-diagram it follows that

$$P(B_G) = \frac{46}{80}$$

$$P(B_G | G) = \frac{10}{16} = \frac{5}{8}$$

$$\therefore P(B_G \cap G) = \frac{5}{8} \times \frac{4}{5} = \frac{1}{2}$$

$$P(G | B_G) = \frac{\frac{1}{2}}{\frac{46}{80}} = \frac{1}{2} \times \frac{80}{46} = \frac{20}{23}$$



149. Let ω be a complex cube root of unity with $\omega \neq 1$. A fair die is thrown three times. If r_1 , r_2 and r_3 are the numbers obtained on the die, then the probability that $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$ is [IIT-JEE 2010]

$$(1) \frac{1}{18}$$

$$(2) \frac{1}{9}$$

$$(3) \frac{2}{9}$$

$$(4) \frac{1}{36}$$

Sol. Answer (3)

ω is a complex cube root of unity with $\omega \neq 1$

$\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$ where r_1, r_2, r_3 can be successively selected from the sets $\{1, 4\}$, $\{2, 5\}$ and $\{3, 6\}$.

$$n(S) = 6 \times 6 \times 6 = 216$$

$$n(E) = {}^2C_1 \times {}^2C_1 \times {}^2C_1 \times {}^3C_1 \times {}^2C_1 = 6 \times 8$$

$$\text{Required probability} = \frac{6 \times 8}{216} = \frac{2}{9}$$

150. Four fair dice D_1, D_2, D_3 and D_4 , each having six faces numbered 1, 2, 3, 4, 5 and 6, are rolled simultaneously. The probability that D_4 shows a number appearing on one of D_1, D_2 and D_3 is [IIT-JEE 2012]

- (1) $\frac{91}{216}$ (2) $\frac{108}{216}$ (3) $\frac{125}{216}$ (4) $\frac{127}{216}$

Sol. Answer (1)

Required probability

$$\begin{aligned}
 &= \text{Getting equal number on two dice} + \text{equal number on three dice} + \text{equal number on four dice} \\
 &= \frac{{}^6C_1 \times 3 \times 5 \times 5}{6^4} + \frac{{}^6C_1 \times 3 \times 5}{6^4} + \frac{{}^6C_1}{6^4} \\
 &= \frac{91}{216}
 \end{aligned}$$

151. Four persons independently solve a certain problem correctly with probabilities $\frac{1}{2}, \frac{3}{4}, \frac{1}{4}, \frac{1}{8}$. Then the probability that the problem is solved correctly by at least one of them is [JEE(Advanced) 2013]

- (1) $\frac{235}{256}$ (2) $\frac{21}{256}$ (3) $\frac{3}{256}$ (4) $\frac{253}{256}$

Sol. Answer (1)

$$P(A) = \frac{1}{2}, P(B) = \frac{3}{4}, P(C) = \frac{1}{4}, P(D) = \frac{1}{8}$$

$$P(A \cup B \cup C \cup D) = 1 - P(\overline{A} \cap \overline{B} \cap \overline{C} \cap \overline{D})$$

$$= 1 - P(\overline{A}) P(\overline{B}) P(\overline{C}) P(\overline{D})$$

$$= 1 - P(\overline{A}) P(\overline{B}) P(\overline{C}) P(\overline{D})$$

$$= 1 - \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{7}{8}$$

$$= 1 - \frac{21}{256} = \frac{235}{256}$$

152. Three boys and two girls stand in a queue. The probability, that the number of boys ahead of every girl is at least one more than the number of girls ahead of her, is [JEE(Advanced)-2014]

- (1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{2}{3}$ (4) $\frac{3}{4}$

Sol. Answer (1)

$$n(S) = 5!$$

Case I : $\overline{\quad} \overline{\quad} \overline{\quad} \overline{\quad} \overline{\quad}$
 G_1
 $2 \times 4! = 48$

Case II : $\overline{\quad} \overline{\quad} \overline{\quad} \overline{\quad} \overline{\quad}$
 G_1
 $2 \times 3! = 12$
 $n(E) = 60$

$$P(E) = \frac{60}{5!} = \frac{1}{2}$$

153. A computer producing factory has only two plants T_1 and T_2 . Plant T_1 produces 20% and plant T_2 produces 80% of the total computers produced. 7% of computers produced in the factory turn out to be defective. It is known that $P(\text{computer turns out to be defective given that it is produced in plant } T_1) = 10P(\text{computer turns out to be defective given that it is produced in plant } T_2)$ where $P(E)$ denotes the probability of an event E . A computer produced in the factory is randomly selected and it does not turn out to be defective. Then the probability that it is produced in plant T_2 is

[JEE(Advanced) 2016]

- (1) $\frac{36}{73}$ (2) $\frac{47}{79}$ (3) $\frac{78}{93}$ (4) $\frac{75}{83}$

Sol. Answer (3)

$$P(T_1) = \frac{1}{5}, P(T_2) = \frac{4}{5}, P(D) = \frac{7}{100}$$

$$P(D) = P\left(\frac{D}{T_1}\right) \cdot P(T_1) + P\left(\frac{D}{T_2}\right) \cdot P(T_2) = 10P\left(\frac{D}{T_2}\right)P(T_1) + P\left(\frac{D}{T_2}\right)P(T_2)$$

On solving $P\left(\frac{D}{T_2}\right) = \frac{1}{40}$

$$P\left(\frac{T_2}{\bar{D}}\right) = \frac{P\left(\frac{\bar{D}}{T_2}\right) \cdot P(T_2)}{P(\bar{D})}$$

$$= \frac{\frac{39}{40} \cdot \frac{4}{5}}{\frac{93}{100}} = \frac{78}{93}$$

154. Three randomly chosen non-negative integers x , y and z are found to satisfy the equation $x + y + z = 10$. Then the probability that z is even, is

[JEE(Advanced)-2017]

- (1) $\frac{6}{11}$ (2) $\frac{36}{55}$ (3) $\frac{1}{2}$ (4) $\frac{5}{11}$

Sol. Answer (1)

$$x + y + z = 10$$

$$n(s) = {}^{10+3-1}C_{3-1} = {}^{12}C_2 = \frac{12 \times 11}{2} = 66$$

Let $z = 2n$, where $n = 0, 1, 2, 3, 4, 5$

$$x + y + 2n = 10$$

$$x + y = 10 - 2n$$

$$\text{Total such solution} = \sum_{n=0}^5 (11 - 2n) = 36$$

$$P(E) = \frac{36}{66} = \frac{6}{11}$$

SECTION - B

Objective Type Questions (More than one options are correct)

1. If $P(A)$ and $P(B)$ are the probabilities of occurrence of two events A and B , then the probability that exactly one of the two events occur, is

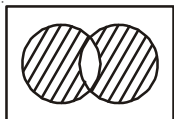
(1) $P(A \cap \bar{B}) + P(\bar{A} \cap B)$

(2) $P(A \cup B) - P(A \cap B)$

(3) $P(A) + P(B)$

(4) $P(A) + P(B) - 2P(A \cap B)$

Sol. Answer (1, 2, 4)



$$\begin{aligned} \text{The probability that exactly one of the two events occur} &= P(A \cup B) - P(A \cap B) \\ &= P(A) + P(B) - 2P(A \cap B) \\ &= P(A \cap \bar{B}) + P(\bar{A} \cap B) \end{aligned}$$

2. Two events A and B have probability 0.25 and 0.5 respectively. The probability that both A and B occur simultaneously is 0.14. Then the probability that neither A nor B occur is

(1) 0.39

(2) 0.61

(3) $P(A' \cap B')$

(4) $P(A' \cup B')$

Sol. Answer (1, 3)

$$P(A) = 0.25$$

$$P(B) = 0.5$$

$$P(A \cap B) = 0.14$$

$$P(A \cup B) = 0.25 + 0.5 - 0.14 = 0.61$$

$$P(A \cup B)' = 1 - 0.61 = 0.39$$

3. Harsha and Ramya throw a dice. The probability that Harsha's throw is not greater than Ramya's is

(1) $\frac{7}{12}$

(2) $\frac{1}{6} \left(\frac{6}{6} + \frac{5}{6} + \frac{4}{6} + \frac{3}{6} + \frac{2}{6} + \frac{1}{6} \right)$

(3) $\frac{{}^7C_2}{{}^4C_2}^2$

(4) $\frac{{}^4C_2}{{}^7C_2}$

Sol. Answer (1, 2, 3)

Output for Harsha	Probability that output of a dice Ramya's is not greater
1	6/6
2	5/6
3	4/6
4	3/6
5	2/6
6	1/6

$$\text{Required probability} = \frac{1}{6} \left(\frac{6}{6} + \frac{5}{6} + \frac{4}{6} + \dots + \frac{1}{6} \right) = \frac{7}{12} = \frac{{}^7C_2}{{}^4C_2}^2$$

4. There are seven seats in a row. Three person sit at random. Probability that middle seat is always occupied and no two persons sit adjacently is

(1) $\frac{8}{35}$

(2) $\frac{4}{35}$

(3) $\frac{4}{7C_3}$

(4) $\frac{{}^4C_1}{{}^7C_3}$

Sol. Answer (2, 3, 4)

$n(S)$ = no. of ways of sitting 3 persons = ${}^7C_3 \times 3!$

Middle sit can be filled by 3 ways

Not to sit together remaining one of two will sit on first two place or last two place

$$n(E) = 3 \times {}^2C_1 \times {}^2C_1 \times 2!$$

$$P(E) = \frac{3 \times {}^2C_1 \times {}^2C_1 \times 2!}{{}^7C_3 \times 3!} = \frac{4}{{}^7C_3} = \frac{{}^4C_1}{{}^7C_3} = \frac{4}{35}$$

5. Box A has 3 white and 2 red balls, box B has 2 white and 4 red balls. If two balls are selected at random without replacements from the box A and 2 more are selected at random from B, the probability that all the four balls are white is

(1) $\frac{1}{10}$

(2) $\frac{1}{50}$

(3) $1 - \left(\frac{{}^2C_2}{{}^5C_2} \cdot \frac{{}^4C_2}{{}^5C_2} \right)$

(4) $\frac{{}^3C_2 \cdot {}^2C_2}{{}^5C_2 \cdot {}^6C_2}$

Sol. Answer (2, 4)

Box A – 3W
 2R

Box B – 2W
 4R

Probability that selected balls from A is while = $\frac{{}^3C_2}{{}^5C_2}$

Probability that selected ball from B is while = $\frac{{}^2C_2}{{}^6C_2}$

$$\text{Required probability} = \frac{{}^3C_2}{{}^5C_2} \cdot \frac{{}^2C_2}{{}^6C_2} = \frac{3}{10} \times \frac{1}{15} = \frac{1}{50}$$

6. If $P(A) = \frac{2}{3}$, $P(B) = \frac{4}{9}$ and $P(A \cap B) = \frac{14}{45}$ then $P(A' \cap B')$ is greater than or equal to

(1) $\frac{37}{45}$

(2) $\frac{1}{5}$

(3) $\frac{4}{5}$

(4) $\frac{41}{45}$

Sol. Answer (1, 2, 3)

$$P(A) = \frac{2}{3}, P(B) = \frac{4}{9}, P(A \cap B) = \frac{14}{45}$$

$$P(A \cup B) = \frac{2}{3} + \frac{4}{9} - \frac{14}{45} = \frac{30 + 20 - 42}{45} = \frac{8}{45}$$

$$P(A' \cap B') = 1 - P(A \cup B) = 1 - \frac{8}{45} = \frac{37}{45}$$

7. A function is selected at random from all the functions defined over a given set A consisting of three distinct elements. The probability that the mapping selected is one to one is

(1) $\frac{2}{9}$

(2) $\frac{1}{9}$

(3) $\frac{3!}{3^3}$

(4) $\frac{{}^3P_3}{27}$

Sol. Answer (1, 3, 4)

$$A = \{x_1, x_2, x_3\}$$

$$\text{Number of function from } A \text{ to } A = 3 \times 3 \times 3 = 27$$

$$\text{Number of one-one function} = 3! = 6$$

$$\text{Required probability} = \frac{6}{27} = \frac{2}{9}$$

8. A matrix is chosen at random from the set of all matrices of order 2 with elements 0 or 1 only. The probability that the determinant value of the matrix chosen has the value non-negative is greater than or equal to

(1) $\frac{3}{8}$

(2) $\frac{13}{16}$

(3) $\frac{1}{4}$

(4) $\frac{15}{16}$

Sol. Answer (1, 2, 3)

$$A = \begin{bmatrix} - & - \\ - & - \end{bmatrix}$$

Each place can be filled by 2 ways

$$n(s) = 2 \times 2 \times 2 \times 2 = 2^4$$

$|A|$ will be negative for following case

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Required probability} = \frac{16 - 3}{16} = \frac{13}{16} = 1 - \frac{3}{16}$$

9. If A and B are two events such that $P(A \cup B) = \frac{3}{4}$, $P(A \cap B) = \frac{1}{4}$, $P(\bar{A}) = \frac{2}{3}$, then

(1) $P(\bar{A} \cap B) = \frac{5}{12}$

(2) $P(B) = \frac{2}{3}$

(3) $P(\bar{A} \cup \bar{B}) = \frac{1}{4}$

(4) $P(\bar{A} \cup \bar{B}) = \frac{3}{4}$

Sol. Answer (1, 2, 4)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

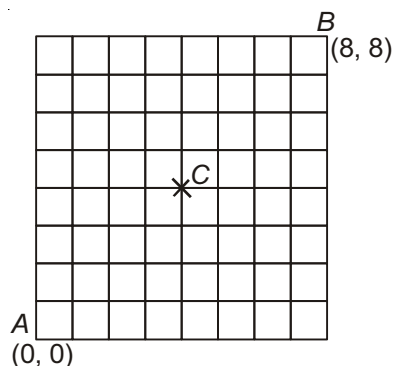
$$\frac{3}{4} = \frac{1}{3} + P(B) - \frac{1}{4}$$

$$\Rightarrow P(B) = \frac{2}{3}$$

$$P(\bar{A} \cap B) = P(B) - P(A \cap B) = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$$

$$P(\bar{A} \cup \bar{B}) = P(A \cap B) = 1 - P(A \cap B) = 1 - \frac{1}{4} = \frac{3}{4}$$

10. Figure shows a jinxed grid in which Shaktimaan has to get to point B from point A . He can either go right or up. The point C is jinxed, upon reaching which Shaktimaan loses all his magical powers. The probability that Shaktimaan will remain a super hero is



(1) $\frac{{}^{16}C_8 - ({}^8C_4)^2}{{}^{16}C_8}$

(2) $\frac{797}{1287}$

(3) $\frac{4}{8}$

(4) $\frac{1}{2}$

Sol. Answer (1, 2)

Total number of ways in which Shaktimaan can reach from A to B is $= \frac{16!}{8! \cdot 8!} = {}^{16}C_8$

Total number of ways in which Shaktimaan can reach from A to C , then C to B is ${}^8C_4 \times {}^8C_4$

$$\begin{aligned} \text{So, } P(E) &= \frac{{}^{16}C_8 - ({}^8C_4 \times {}^8C_4)}{{}^{16}C_8} \\ &= 1 - \frac{4900}{12870} \end{aligned}$$

11. Two dices are rolled and numbers r_1 and r_2 are chosen. Let $1 + \omega^{r_1} + \omega^{r_2} = \lambda$ (where ω is a complex cube root of unity). Then

(1) The probability that λ is zero is $\frac{1}{9}$

(2) The probability that λ is zero is $\frac{2}{9}$

(3) The probability that λ is 3 is $\frac{1}{9}$

(4) The probability that λ is 3 is $\frac{2}{9}$

Sol. Answer (2, 3)

Total number of ways in which r_1 and r_2 can be chosen $= {}^6C_1 \times {}^6C_1 = 36$

Total number of ways in which $1 + \omega^{r_1} + \omega^{r_2} = \lambda$

(i) $\lambda = 0$ if (r_1, r_2) are $\{(1, 5), (5, 1), (1, 2), (2, 1), (4, 2), (2, 4), (5, 4), (4, 5)\}$

$$P(E) = \frac{8}{36} = \frac{2}{9}$$

(ii) $\lambda = 3$ if (r_1, r_2) are $\{(3, 3), (3, 6), (6, 3), (6, 6)\}$

$$\therefore \text{Required probability} = \frac{4}{36} = \frac{1}{9}$$

12. Five letters are put in five identical letter boxes, the probability that no letter box is empty is greater than or equal to

(1) $\frac{1}{53}$

(2) $\frac{1}{52}$

(3) $\frac{1}{42}$

(4) $\frac{1}{40}$

Sol. Answer (1, 2)

Sample space

When there are five letters in one of the box, there is only one way

$$(5, 0, 0, 0, 0) \rightarrow 1 \text{ way}$$

$$(4, 1, 0, 0, 0) \rightarrow {}^5C_4 \times {}^1C_1 \text{ ways} = 5 \text{ ways}$$

$$(3, 2, 0, 0, 0) \rightarrow {}^5C_3 \times {}^2C_2 = 10$$

$$(3, 1, 1, 0, 0) \rightarrow \frac{1}{2!} {}^5C_3 \times {}^2C_1 \times {}^1C_1 = 10$$

$$(2, 2, 1, 0, 0) \rightarrow \frac{1}{2!} {}^5C_2 \times {}^3C_2 \times {}^1C_1 = 15$$

$$(1, 1, 1, 1, 1) \rightarrow 1 \text{ way}$$

$$(1, 1, 1, 2, 0) \rightarrow {}^5C_2 \text{ ways} = 10 \text{ ways}$$

So, total number of ways = 52

$$\text{So, } P(E) = \frac{1}{52}$$

13. Which of the following statements are correct?

(1) The probability of an event can't exceed 1

(2) The probability of an event can be zero

(3) The probability of an event is always greater than zero and less than 1

(4) Probability of an event might exceed one when it is a sure event

Sol. Answer (1, 2)

Probability of an event can never exceed one. It is one when it is sure that event will occur and it is zero when event can never occur.

14. Delegates of the five of the member countries of SAARC decide to hold a round table conference. there are 2 Indians, 2 Bangladeshis, 2 Pakistanis, 2 Srilankans and 2 Nepalis. The probability that people of same nationality sit together is

(1) $\frac{3! \times (2!)^5}{9!}$

(2) $\frac{2}{189}$

(3) $\frac{(4!) \times (2!)^5}{9!}$

(4) $\frac{2}{945}$

Sol. Answer (3, 4)

$$n(S) = 9!$$

$$n(E) = 4! \times (2!)^5$$

$$\text{So, } P(E) = \frac{4! \times 32}{9!} = \frac{32}{9 \times 8 \times 7 \times 6 \times 5} = \frac{2}{945}$$

15. The number 900 is resolved into two factors, x and y the probability that the factors x and y are co-prime is

(1) $\frac{5}{27}$

(2) $\frac{7}{27}$

(3) $\frac{11}{27}$

(4) $\frac{8}{27}$

Sol. Answer (4)

$$\text{Given } N = (2)^2 (3)^2 (5)^2$$

$$\text{Total number of divisors} = 3 \times 3 \times 3 = 27$$

$$\text{Total Number of ways in which it can be resolved into co-prime factors is } 2 \times 2 \times 2 = 8$$

$$\therefore \text{Probability} = \frac{8}{27}$$

16. A coins is tossed thrice. Let events A , B , C be first coin shows heads exactly two heads show and tails on 2nd and 3rd tosses respectively

(1) $P(B \cap C) = 0$

(2) $P(A \cap B) = \frac{1}{4}$

(3) $P(A \cap C) = \frac{1}{8}$

(4) $P(C) = \frac{1}{4}$

Sol. Answer (1, 2, 3, 4)

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$A = \{HHH, HHT, HTH, HTT\}$$

$$B = \{HHT, HTH, THH\}$$

$$C = \{HTT, TTT\}$$

$$B \cap C = \{\} \Rightarrow P(B \cap C) = 0$$

$$A \cap B = \{HHT, HTH\} \Rightarrow P(A \cap B) = \frac{1}{4}$$

$$(A \cap C) = \{HTT\} \Rightarrow P(A \cap C) = \frac{1}{8}$$

$$P(C) = \frac{1}{4}$$

17. If a grandfather along with his two grandsons and four granddaughters are to be seated in a line for a photograph so that he is always in the middle then the probability that his two grandsons are never adjacent to each other is

(1) $\frac{6! - 4! \cdot 4 \cdot 2!}{6!}$

(2) $\frac{192}{720}$

(3) $\frac{528}{720}$

(4) $\frac{4}{15}$

Sol. Answer (1, 3)

$$\text{So, total number of ways} = X \ X \ X \ \text{G.F.} \ X \ X \ X = 6! = 720$$

$$\text{Total number of ways in which two grandsons sit together is} = 4! \cdot 2! \cdot 4 = 192$$

$$\text{So, number of ways in which they do not sit together is} = 6! - 4! \cdot 2! \cdot 4 = 528$$

$$\text{So, } P(E) = \frac{528}{720}$$

18. A real number k is selected from the interval $[0, 20]$, the probability that roots of the equation $x^2 - kx + 9 = 0$, equation will be real is greater than or equal to

- (1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) $\frac{7}{10}$ (4) $\frac{4}{5}$

Sol. Answer (1, 2, 3)

The roots of the equation $x^2 - kx + 9 = 0$ will be real iff.

$$k^2 - 36 \geq 0$$

$$\Rightarrow k \geq 6 \text{ or } k \leq -6$$

$$\text{So, required probability} = \frac{14}{20} = \frac{7}{10}$$

19. A quadratic equation is chosen from the set of all the quadratic equations which are unchanged by squaring their roots. The chance that the equation has unequal roots is

- (1) $\frac{1}{2}$ (2) $\frac{3}{4}$ (3) $\frac{3}{8}$ (4) $\frac{7}{16}$

Sol. Answer (1)

Let α and β be the roots of equation.

Now the condition will be satisfied only if $\alpha = \beta^2$ and $\beta = \alpha^2$

$$\text{Or } \alpha = \beta = \alpha^2 = \beta^2$$

So, clearly there are four such equations

$$(x-1)^2 = 0; \quad \text{roots are } (1, 1)$$

$$x^2 = 0; \quad \text{roots are } (0, 0)$$

$$x(x-1) = 0; \quad \text{roots are } (0, 1)$$

$$x^2 + x + 1 = 0; \quad \text{roots are } (\omega, \omega^2)$$

$$\text{So, the required probability is } \frac{1}{2}$$

20. A sample space consists of 3 sample points with associated probabilities given as $2p$, p^2 , $4p-1$, then

- (1) p is an irrational number (2) p is a rational number
(3) p is greater than $\frac{1}{2}$ (4) p is less than $\frac{1}{2}$

Sol. Answer (1, 4)

As the events are exhaustive

$$p^2 + 2p + 4p - 1 = 1$$

$$\Rightarrow p^2 + 6p - 2 = 0$$

$$\Rightarrow p = \sqrt{11} - 3$$

And hence the answer

21. There are only two women among 20 persons taking part in a pleasure trip. The 20 persons are divided into two groups, each group consisting of 10 persons. Then the probability that two women will be in the same group is

(1) $\frac{9}{19}$

(2) $\frac{18}{19}$

(3) $\frac{18! \times 10! \times 10! \times 2}{10! 8! \times 20!}$

(4) $\frac{18! \times 10! \times 10! \times 2 \times 2}{10! \times 8! \times 20!}$

Sol. Answer (1, 3)

Number of ways in which 20 people can be divided into equal groups = $\frac{20!}{10! 10! 2!}$

18 people can be divided into groups of 10 and 8 is = $\frac{18!}{10! 8!}$

$P(E) = \frac{18!}{10! 8!} \times \frac{10! \times 10! \times 2}{20!} = \frac{10 \cdot 9 \cdot 2}{20 \cdot 19} = \frac{9}{19}$

22. The letter forming the word "Clifton" are placed at random in a row. The probabilities that

(1) Two vowels are together = $\frac{2}{7}$

(2) Two vowels are not together = $\frac{5}{7}$

(3) Two vowels are not together = $\frac{2}{7}$

(4) Two vowels are together = $\frac{5}{7}$

Sol. Answer (1, 2)

Let E be the event that vowels are together

$n(S) = 7!$

$n(E) = 2! 6!$

$P(E) = \frac{n(E)}{n(S)} = \frac{2}{7}$

$\therefore P(\bar{E}) = 1 - \frac{2}{7} = \frac{5}{7}$

23. 6 married couples are standing in a room. If 4 people are chosen at random, then the chance that exactly one married couple is among the 4 is

(1) Greater than $\frac{1}{2}$

(2) Less than $\frac{1}{2}$

(3) Equal to $\frac{16}{33}$

(4) Equal to $\frac{17}{33}$

Sol. Answer (2, 3)

$n(S) = {}^{12}C_4 = 55 \times 9$

$n(E) = {}^6C_1 \cdot {}^5C_2 \times 2^2 = 6 \times 10 \times 4$

$P(E) = \frac{6 \times 10 \times 4}{55 \times 9} = \frac{2 \cdot 2 \cdot 4}{11 \cdot 3} = \frac{16}{33}$

24. If A and B are two independent events such that $P(A' \cap B) = \frac{2}{15}$ and $P(A \cap B') = \frac{1}{6}$, then $P(B)$ is

(1) $\frac{1}{5}$

(2) $\frac{1}{6}$

(3) $\frac{4}{5}$

(4) $\frac{5}{6}$

Sol. Answer (2, 3)

$$P(A' \cap B) = \frac{2}{15} \Rightarrow P(A') \cdot P(B) = \frac{2}{15} \quad \dots (1)$$

$$P(A \cap B') = \frac{1}{6} \Rightarrow P(A) \cdot P(B') = \frac{1}{6} \quad \dots (2)$$

Let, $P(A) = x, P(B) = y$

From (1)

$$(1 - P(A)) (P(B)) = \frac{2}{15}$$

$$\text{Then, } (1 - x)(y) = \frac{2}{15}$$

$$y - xy = \frac{2}{15}$$

$$\text{Also, } x - xy = \frac{1}{6}$$

$$x - y = \frac{1}{30}$$

$$\text{Now, } y - y \left(\frac{1}{30} + y \right) = \frac{2}{15}$$

$$\Rightarrow y - \frac{y}{30} - y^2 = \frac{2}{15} \Rightarrow 29y - 30y^2 = 4$$

$$\Rightarrow 30y^2 - 29y + 4 = 0 \Rightarrow y = \frac{1}{6}, \frac{4}{5}$$

$$\Rightarrow P(B) = \frac{1}{6}, \frac{4}{5}$$

25. For two events A and B , let $P(A) = \frac{3}{5}, P(B) = \frac{2}{3}$, then which of the following is correct?

(1) $P(A \cap B') \leq \frac{1}{3}$

(2) $P(A \cup B) \geq \frac{2}{3}$

(3) $\frac{4}{15} \leq P(A \cap B) \leq \frac{3}{5}$

(4) $\frac{2}{5} \leq P(A/B) \leq \frac{9}{10}$

Sol. Answer (1, 2, 3, 4)

$$P(A) = \frac{3}{5}, P(B) = \frac{2}{3}$$

$$P(A') = \frac{2}{5}, P(B') = \frac{1}{3}$$

$$P(A \cap B') = P\left(\frac{A}{B'}\right) \cdot P(B')$$

$$P\left(\frac{A}{B'}\right)_{\max} = 1$$

$$\Rightarrow P(A \cap B') \leq P(B')$$

$$P(A \cap B') \leq \frac{1}{3}$$

Now, $P(A \cup B)$

$$P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B)_{\max} = \frac{3}{5}$$

$$\therefore P(A \cup B)_{\min} = \frac{3}{5} + \frac{2}{3} - \frac{3}{5} = \frac{2}{3}$$

$$\therefore P(A \cup B) \geq \frac{2}{3}$$

Again $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\Rightarrow 1 = \frac{3}{5} + \frac{2}{3} - P(A \cap B)_{\min}$$

$$\Rightarrow P(A \cap B) = \frac{19}{15} - 1 = \frac{4}{15}$$

$$\therefore \frac{4}{15} \leq P(A \cap B) \leq \frac{3}{5}$$

Also, $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$

$$\frac{4}{15} \leq P(A \cap B) \leq \frac{3}{5}$$

$$\frac{\frac{4}{15}}{P(B)} \leq \frac{P(A \cap B)}{P(B)} \leq \frac{\frac{3}{5}}{P(B)} \Rightarrow \frac{2}{5} \leq P\left(\frac{A}{B}\right) \leq \frac{9}{10}$$

26. A fair coin is tossed n times. Let X be the times the heads occur. If $P(X = 4)$, $P(X = 5)$ and $P(X = 6)$ are in A.P., then the value of n is

(1) 15

(2) 7

(3) 9

(4) 14

Sol. Answer (2, 4)

$$P(x = 4) = {}^nC_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{n-4}$$

$$P(x = 5) = {}^nC_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{n-5}$$

$$P(x=6) = {}^nC_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^{n-6}$$

As, $P(x=4), P(x=5), P(x=6)$ are in A.P.

$$\therefore {}^nC_4 + {}^nC_6 = 2 \cdot {}^nC_5$$

$$\Rightarrow n = 7, 14$$

27. A fair coin is tossed 99 times. If X is the number of times in which head occurred, then $P(X=r)$ is maximum when r is equal to

(1) 49

(2) 50

(3) 51

(4) 48

Answer (1, 2)

$$P(x=r) = {}^{99}C_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{99-r} = {}^{99}C_r \left(\frac{1}{2}\right)^{99}$$

$P(x=r)$ will maximum when ${}^{99}C_r$ will be maximum, ${}^{99}C_r$ is maximum when $r = 49, 50$

28. Consider the following events for a family with children

$A = \{\text{Children of both sexes}\}$

$B = \{\text{At-most one boy}\}$

Then which of the following statements is true?

(1) A and B are independent events if a family has 3 children

(2) A and B are not independent events if a family has 3 children

(3) A and B are not independent events if a family has only 2 children

(4) A and B are independent events if a family has only 2 children

Sol. Answer (1, 3)

If family has 3 children

$$P(A) = \frac{6}{8} = \frac{3}{4}$$

$$P(B) = \frac{4}{8} = \frac{1}{2}$$

$$P(A \cap B) = \frac{3}{8}$$

$$P(A \cap B) = P(A) \cdot P(B) = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$$

$\therefore A$ and B are independent events

If family has 2 children,

$$P(A) = \frac{1}{2}$$

$$P(B) = \frac{3}{4}$$

$$P(A \cap B) = \frac{1}{2}$$

$P(A \cap B) \neq P(A) \cdot P(B)$. So A and B are not independent.

29. From the set $S = \{0, 1, 2, \dots, 53\}$, three distinct numbers are chosen. The probability that the sum of them is 54 is

(1) $\frac{27}{2756}$

(2) $\frac{13}{1378}$

(3) $\frac{29}{2756}$

(4) $\frac{15}{1378}$

Sol. Answer (1)

$$n(S) = {}^{54}C_3$$

$$54 = 18 + 18 + 18$$

When '0' is one selected number, then number of ways such that sum is 54 = 26.

1 is selected, then number of ways such that sum of 54 = 25.

2 is selected, then number of ways such that sum is 54 = 23

.....

.....

17 is selected, then number of ways such that sum is 54 = 1

$$\text{Required probability} = \frac{26 + 25 + 23 + 22 + 20 + \dots + 1}{{}^{54}C_3} = \frac{9 \times 27 \times 6}{54 \times 53 \times 52} = \frac{27}{2756}$$

30. A fair coin is tossed three times. Consider the events

$A = \{\text{First toss is head}\}$

$B = \{\text{Second toss is head}\}$

$C = \{\text{Exactly two heads in a row}\}$

Which of the following statements is true?

(1) A and B are independent

(2) A and B are not independent

(3) A and C are independent

(4) B and C are not independent

Sol. Answer (1, 4)

We have,

$$S = \{(H, H, H), (H, H, T), (H, T, H), (H, T, T), (T, H, H), (T, T, H), (T, H, T), (T, T, T)\}$$

$$A = \{(H, H, H), (H, H, T), (H, T, H), (H, T, T)\}$$

$$B = \{(H, H, H), (H, H, T), (T, H, T), (T, H, H)\}$$

$$C = \{(H, H, T), (T, H, H), (H, T, H)\}$$

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{2}, P(C) = \frac{3}{8}$$

$$P(A \cap B) = \frac{1}{4} = P(A).P(B)$$

$$P(A \cap C) = \frac{1}{4} \neq P(A).P(C)$$

A and B are independent events.

A and C are not independent events

B and C are not independent events.

31. If A and B are two events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$ then which of following is/are true?

- (1) $P(A \cup B) \geq \frac{1}{2}$ (2) $P(A \cap B) \leq \frac{1}{3}$ (3) $P(A \cap \bar{B}) \geq \frac{1}{6}$ (4) $P(\bar{A} \cup \bar{B}) \leq \frac{2}{3}$

Sol. Answer (1, 2, 3, 4)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{2} + \frac{1}{3} - P(A \cap B)$$

$$\geq \frac{1}{2}$$

$$P(A \cap B) \leq \frac{1}{3} \text{ since maximum common is } P(B)$$

$$P(A \cap \bar{B}) = P(A - B)$$

$$= P(A) - P(A \cap B)$$

$$\geq \frac{1}{6}$$

$$P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B)$$

$$\leq \frac{2}{3}$$

32. A, B, C in order roll a dice on the condition that one who gets a six, first wins then which of the following are true

(1) A chances $\frac{36}{91}$

(2) B chances $\frac{5}{6} \cdot \frac{36}{91}$

(3) C chances $\left(\frac{5}{6}\right)^2 \frac{36}{91}$

(4) All three has same chances

Sol. Answer (1, 2, 3)

$$\text{Chances of } A \text{ are } \frac{1}{6} + \left(\frac{5}{6}\right)^3 \frac{1}{6} + \dots \infty = \frac{36}{91}$$

$$\text{Chances of } B \text{ are } \frac{5}{6} \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^4 \frac{1}{6} + \dots \infty = \frac{5}{6} \left(\frac{36}{91}\right)$$

$$\text{Chances of } C \text{ are } \left(\frac{5}{6}\right)^2 \frac{1}{6} + \left(\frac{5}{6}\right)^5 \frac{1}{6} + \dots \infty = \left(\frac{5}{6}\right)^2 \cdot \frac{36}{91}$$

33. There are 3 men, 2 women and 4 children. If 4 persons are selected then which of the following is/are true?

(1) Probability of selecting exactly 2 children = $\frac{10}{21}$

(2) Probability of selecting at most 2 children = $\frac{5}{6}$

(3) Probability of selecting exactly 2 children = $\frac{5}{21}$

(4) Probability of selecting at most 2 children = $\frac{25}{42}$

Sol. Answer (1, 2)

$$\text{Probability of exactly 2 children} = \frac{{}^4C_2 \cdot {}^5C_2}{{}^9C_4} = \frac{10}{21}$$

$$\text{Probability of selecting atmost 2 children} = \frac{{}^5C_4 + {}^4C_1 \cdot {}^5C_3 + {}^4C_2 \cdot {}^5C_2}{{}^9C_4} = \frac{5}{6}$$

34. If the letters of word ASSASSIN are written at random in a row, then which of following are true?

(1) Probability that all 'S' come together is $\frac{1}{14}$

(2) Probability that all 'S' do not come together is $\frac{1}{7}$

(3) Probability that no two 'S' come together is $\frac{1}{14}$

(4) Probability that vowels occupy places of vowel only is $\frac{1}{56}$

Sol. Answer (1, 3, 4)

$$\text{Total arrangements} = \frac{8!}{4!2!} = \frac{8 \cdot 7 \cdot 6 \cdot 5}{2} = 840$$

(1) If all 'S' come together = $\frac{5!}{2!} = 60$, required probability = $\frac{60}{840} = \frac{1}{14}$

(2) Required probability = $1 - \frac{1}{14} = \frac{13}{14}$

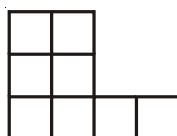
(3) If no two 'S' come together = ${}^5C_4 \cdot \frac{4!}{2!} = 60$

$$\text{Require probability} = \frac{60}{840} = \frac{1}{14}$$

(4) If vowels occupy places of vowel only = $\frac{3!}{2!} \cdot \frac{5!}{4!} = 15$

$$\text{Then required probability} = \frac{15}{840} = \frac{1}{56}$$

35. If alphabets of word RAKESH are written in given boxes, then probability that no row remains empty is



(1) $\frac{13}{14}$

(2) $\frac{{}^8C_6 - 4}{{}^8C_6}$

(3) $\frac{11}{13}$

(4) $\frac{{}^8C_6 - 2}{{}^8C_6}$

Sol. Answer (1, 4)

If 6 boxes are selected from 8 boxes then there are only two ways in which one row remains empty.

$$\text{So, required probability} = \frac{{}^8C_6 - 2}{{}^8C_6} = \frac{13}{14}$$

36. Let E and F be two independent events. The probability that exactly one of them occurs is $\frac{11}{25}$ and the probability of none of them occurring is $\frac{2}{25}$. If $P(T)$ denotes the probability of occurrence of the event T , then

[IIT-JEE 2011]

$$(1) \quad P(E) = \frac{4}{5}, P(F) = \frac{3}{5}$$

$$(2) \quad P(E) = \frac{1}{5}, P(F) = \frac{2}{5}$$

$$(3) \quad P(E) = \frac{2}{5}, P(F) = \frac{1}{5}$$

$$(4) \quad P(E) = \frac{3}{5}, P(F) = \frac{4}{5}$$

Sol. Answer (1, 4)

We have,

$$P(E) + P(F) - 2P(E \cap F) = \frac{11}{25}$$

$$P(E^c \cap F^c) = (1 - P(E))(1 - P(F)) = \frac{2}{25}$$

Solving these equations, we shall get

$$P(E) = \frac{4}{5}, P(F) = \frac{3}{5}$$

$$\text{or } P(E) = \frac{3}{5}, P(F) = \frac{4}{5}$$

37. Let X and Y be two events such that $P(X|Y) = \frac{1}{2}$, $P(Y|X) = \frac{1}{3}$ and $P(X \cap Y) = \frac{1}{6}$. Which of the following is (are) correct?

[IIT-JEE 2012]

$$(1) \quad P(X \cup Y) = \frac{2}{3}$$

$$(2) \quad X \text{ and } Y \text{ are independent}$$

$$(3) \quad X \text{ and } Y \text{ are not independent}$$

$$(4) \quad P(X^c \cap Y) = \frac{1}{3}$$

Sol. Answer (1, 2)

$$P(X) \cdot P(Y|X) = P(Y) \cdot P(X|Y) = P(X \cap Y)$$

$$\Rightarrow P(X) = \frac{1/6}{1/3} = \frac{1}{2}$$

$$P(Y) = \frac{1/6}{1/2} = \frac{1}{3}$$

$$\therefore P(X \cup Y) = \frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{2}{3}$$

$$P(X \cap Y) = \frac{1}{6}$$

$$\text{and } P(X) \cdot P(Y) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

$\Rightarrow$ X and Y are independent events

$$P(X^c \cap Y) = P(X^c) \cdot P(Y) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

38. A ship is fitted with three engines E_1, E_2 and E_3 . The engines function independently of each other with respective probabilities $\frac{1}{2}, \frac{1}{4}$ and $\frac{1}{4}$. For the ship to be operational at least two of its engines must function. Let X denote the event that the ship is operational and let X_1, X_2 and X_3 denote respectively the events that the engines E_1, E_2 and E_3 are functioning. Which of the following is (are) true? **[IIT-JEE 2012]**

(1) $P[X_1^c | X] = \frac{3}{16}$

(2) $P[\text{Exactly two engines of the ship are functioning} | X] = \frac{7}{8}$

(3) $P[X | X_2] = \frac{5}{16}$

(4) $P[X | X_1] = \frac{7}{16}$

Sol. Answer (2, 4)

$$(1) P[X_1^c | X] = \frac{P(\bar{X}_1 \cap X)}{P(X)} = \frac{\binom{1}{2} \binom{1}{4} \binom{1}{4}}{\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4}} = \frac{1}{8}$$

$$(2) \text{ Required probability} = \frac{\frac{3}{2 \cdot 4 \cdot 4} + \frac{3}{2 \cdot 4 \cdot 4} + \frac{1}{2 \cdot 4 \cdot 4}}{\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4}} = \frac{7}{8}$$

$$(3) P[X | X_2] = \frac{P(X \cap X_2)}{P(X_2)} = \frac{\binom{1}{4} \left[\frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{4} + \binom{1}{2} \binom{1}{4} \right]}{\binom{1}{4}} = \frac{\frac{1}{4} \cdot \frac{5}{4}}{\frac{1}{4}} = \frac{5}{8}$$

$$(4) \frac{P(X \cap X_1)}{P(X_1)} = \frac{\binom{1}{2} \left[\frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{3}{4} \right]}{\frac{1}{2}} = \frac{7}{16}$$

39. Let X and Y be two events such that $P(X) = \frac{1}{3}$, $P(X|Y) = \frac{1}{2}$ and $P(Y|X) = \frac{2}{5}$. Then

[JEE (Advanced)-2017]

- (1) $P(Y) = \frac{4}{15}$ (2) $P(X'|Y) = \frac{1}{2}$ (3) $P(X \cap Y) = \frac{1}{5}$ (4) $P(X \cup Y) = \frac{2}{5}$

Sol. Answer (1, 2)

$$P(X) = \frac{1}{3}, P\left(\frac{X}{Y}\right) = \frac{1}{2}, P\left(\frac{Y}{X}\right) = \frac{2}{5}$$

$$P\left(\frac{X}{Y}\right) = \frac{P(X \cap Y)}{P(Y)} = \frac{1}{2}$$

$$P\left(\frac{Y}{X}\right) = \frac{P(X \cap Y)}{P(X)} = \frac{2}{5}$$

$$\Rightarrow P(Y) = \frac{4}{15}, P(X) = \frac{1}{3}, P(X \cap Y) = \frac{2}{15}$$

$$\therefore P\left(\frac{X'}{Y}\right) = \frac{P(X' \cap Y)}{P(Y)} = \frac{P(Y) - P(X \cap Y)}{P(Y)} = \frac{\frac{4}{15} - \frac{2}{15}}{\frac{4}{15}} = \frac{1}{2}$$

$$P(X \cup Y) = \frac{1}{3} + \frac{4}{15} - \frac{2}{15} = \frac{9-2}{15} = \frac{7}{15}$$

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

Three vertices are chosen at random from the vertices of a regular hexagon then

1. The probability that the triangle with these vertices is an equilateral triangle is equal to

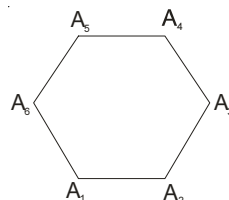
- (1) $\frac{1}{10}$ (2) $\frac{3}{5}$ (3) $\frac{9}{10}$ (4) $\frac{3}{10}$

Sol. Answer (1)

$$n(S) = \text{no. of total triangle} = {}^6C_3 = 20$$

Equilateral triangle is $A_1 A_3 A_5$ and $A_2 A_4 A_6$

$$\therefore P(E) = \frac{2}{20} = \frac{1}{10}$$



2. The probability that the triangle has exactly one side common with the side of the hexagon is

- (1) $\frac{1}{10}$ (2) $\frac{3}{5}$ (3) $\frac{3}{10}$ (4) $\frac{9}{10}$

Sol. Answer (2)Let A_1A_2 is common sideNo. of triangle having $A_1A_2 = 2$ $(\Delta A_1 A_2 A_4 \text{ or } \Delta A_1 A_2 A_5)$ Total no. of triangle having one side common = $6 \times 2 = 12$

$$P(E) = \frac{12}{20} = \frac{3}{5}$$

3. The probability that the triangle has exactly two sides common with the side of the hexagon is

(1) $\frac{1}{10}$

(2) $\frac{3}{5}$

(3) $\frac{3}{10}$

(4) $\frac{9}{10}$

Sol. Answer (3)

Number of triangle having two side common = 6

$$P(E) = \frac{6}{20} = \frac{3}{10}$$

Comprehension-II

5 Indian and 5 Russian couples meet at a party and shake hands.

1. The probability that no wife shakes hands with her husband is

(1) $\frac{3}{5}$

(2) $\frac{18}{19}$

(3) $\frac{23}{38}$

(4) $\frac{27}{38}$

Sol. Answer (2)Total number of possible hand shakes = ${}^{20}C_2 = 190$

No. of hand shakes = hand shakes between Indian male and Russian male + hand shakes between women and men excluding husband of the women.

$$= {}^{10}C_2 + \frac{1}{2} \cdot 10 \times 9 = 90$$

$$\Rightarrow P(\text{no wife shakes hands with her husband}) = \frac{90}{190} = \frac{9}{19}$$

2. The probability that no Indian wife shakes hands with a male is

(1) $\frac{14}{19}$

(2) $\frac{9}{19}$

(3) $\frac{27}{38}$

(4) $\frac{27}{28}$

Sol. Answer (1)

No. of hands shakes when no Indian wife shakes hands with a male

= No. of hand shakes between Indian women and Russian women + hand shakes between Indian men and Russian men

$$= {}^{15}C_2 + {}^5C_2 = 105 + 10 = 115$$

$$\text{Required probability} = \frac{115}{190} = \frac{23}{38}$$

3. The probability that no wife shakes hands with her husband and no Indian wife shakes hands with a male is

(1) $\frac{23}{38}$

(2) $\frac{27}{38}$

(3) $\frac{9}{19}$

(4) $\frac{3}{5}$

Sol. Answer (2)

$$\text{Number of possible shakes hands} = {}^{10}C_2 + {}^{10}C_2 + {}^{10}C_2 = 135$$

$$P(E) = \frac{135}{190} = \frac{27}{38}$$

Comprehension-III

Different words are being formed by arranging the letters of the word "SUCCESS".

1. The probability of choosing a word in which the two C are together but no two S are together is

(1) $\frac{2}{35}$

(2) $\frac{2}{7}$

(3) $\frac{24}{105}$

(4) 1

Sol. Answer (1)

SUCCESS

– U – CC – E –, If 'S' is placed at the place – $n(E) = {}^4C_3 \times 3!$

$$n(S) = \frac{7!}{3!2!}$$

$$P(E) = \frac{{}^4C_3 \times 3!}{\frac{7!}{3! \times 2!}} = \frac{4 \times 3! \times 3! \times 2!}{7!} = \frac{2}{35}$$

2. The probability of word in which no two C and no two S are together is

(1) $\frac{2}{35}$

(2) $\frac{8}{35}$

(3) $\frac{24}{105}$

(4) 1

Sol. Answer (2)

Let us first find the words in which no two S are together.

The remaining letters C, C, U, E can be arranged in $\frac{4!}{2!} = 12$ ways.

These are five favourable places for three SSS $\times C \times C \times U \times E \times$

Thus the total number of ways = $12 \times {}^5C_3 = 120$

Hence the number of words having CC and SSS are separated = $120 - 24 = 96$.

$$\text{Consequently required probability} = \frac{96}{7 \times 6 \times 10} = \frac{8}{35}$$

3. The probability of selecting a word that begins and ends with S is

(1) $\frac{1}{7}$

(2) $\frac{2}{7}$

(3) $\frac{2}{35}$

(4) $\frac{2}{5}$

Sol. Answer (1)

SUCCESS

I, C, C, E, S can be arranged by $\frac{5!}{2!}$

$$n(E) = \frac{5!}{2!}$$

$$n(S) = \frac{\frac{5!}{2!}}{\frac{7!}{3! 2!}} = \frac{5! \times 3!}{7!}$$

$$= \frac{6}{7 \times 6} = \frac{1}{7}$$

Comprehension-IV

There are 20 pairs of shoes in a cupboard. Ten shoes are taken at random.

1. The probability that no complete pair is possible from the selected shoes is

$$(1) \frac{{}^{20}C_3 \times {}^{17}C_4 \times 2^4}{{}^{40}C_{10}} \quad (2) \frac{{}^{20}C_{10} \times 2^{10}}{{}^{40}C_{10}} \quad (3) \frac{{}^{20}C_4 \times {}^{16}C_2 \times 2^2 + {}^{20}C_5}{{}^{40}C_{10}} \quad (4) \frac{{}^{20}C_{10}}{{}^{40}C_{10}}$$

Sol. Answer (2)We have $n(S) = {}^{40}C_{10}$ $n(E_1)$ = no complete pair is possible from the 10 selected shoes = ${}^{20}C_{10} \times 2^{10}$

$$P(E_1) = \frac{{}^{20}C_{10} \times 2^{10}}{{}^{40}C_{10}}$$

2. The probability that exactly three complete pairs are possible from the selected shoes is

$$(1) \frac{{}^{20}C_3 \times {}^{17}C_4 \times 2^4}{{}^{40}C_{10}} \quad (2) \frac{{}^{20}C_{10} \times 2^{10}}{{}^{40}C_{10}} \quad (3) \frac{{}^{20}C_4 \times {}^{16}C_2 \times 2^2 + {}^{20}C_5}{{}^{40}C_{10}} \quad (4) \frac{{}^{20}C_3 \times {}^{17}C_4}{{}^{40}C_{10}}$$

Sol. Answer (1) $n(E_2)$ = exactly three complete pairs are possible from the selected shoes

$$= {}^{20}C_3 \times {}^{17}C_4 \times 2^4$$

$$P(E_2) = \frac{{}^{20}C_3 \times {}^{17}C_4 \times 2^4}{{}^{40}C_{10}}$$

3. The probability that at least four complete pairs are possible from the selected shoes is

$$(1) \frac{{}^{20}C_3 \times {}^{17}C_4 \times 2^4}{{}^{40}C_{10}} \quad (2) \frac{{}^{20}C_{10} \times 2^{10}}{{}^{40}C_{10}} \quad (3) \frac{{}^{20}C_4 \times {}^{16}C_2 \times 2^2 + {}^{20}C_5}{{}^{40}C_{10}} \quad (4) \frac{{}^{20}C_4 + {}^{20}C_5}{{}^{40}C_{10}}$$

Sol. Answer (3)

Number of ways of selecting atleast four complete pairs of shoes = ${}^{20}C_4 \cdot {}^{16}C_2 \times 2^2 + {}^{20}C_5$

$$\text{Required probability} = \frac{{}^{20}C_4 \cdot {}^{16}C_2 \times 2^2 + {}^{20}C_5}{{}^{40}C_{10}}$$

Comprehension-V

A class of problems that requires one to use permutations and combinations for computing probability has at its heart notion of sets and subsets. They are generally an abstract formulation of some concrete situation and require the application of counting techniques.

A is a set containing 10 elements. A subset P_1 of A is chosen and the set A is reconstructed by replacing the elements of P_1 . A subset P_2 of A is chosen and again the set A is reconstructed by replacing the elements of P_2 . This process is continued by choosing subsets $P_1, P_2, \dots, P_{10}$.

1. The number of ways of choosing subsets $P_1, P_2, \dots, P_{10}$ is

- (1) 4^{100} (2) 2^{100} (3) 2^{20} (4) 4^{20}

Sol. Answer (2)

Number of subsets of A containing 10 elements = 2^{10}

For A_1 , we have 2^{10} options, similar for $A_2, \dots, A_{10}$

Hence, the number of ways of choosing subsets $P_1, P_2, \dots, P_{10}$ is

$$= (2^{10})^{10} = 2^{100}$$

2. The probability that $P_i \cap P_j = \phi \forall i \neq j, i, j = 1, 2, \dots, 10$ is

- (1) $\frac{11^{10}}{4^{100}}$ (2) $\frac{10^{11}}{2^{100}}$ (3) $\frac{11^{10}}{2^{100}}$ (4) $\frac{10^{11}}{4^{100}}$

Sol. Answer (3)

Required probability

$$= \frac{{}^{10}C_0 + {}^{10}C_1 10 + {}^{10}C_2 10^2 + \dots + {}^{10}C_{10} 10^{10}}{2^{100}}$$

$$= \frac{(1+10)^{10}}{2^{100}} = \frac{11^{10}}{2^{100}}$$

3. The probability that $P_1 \cap P_2 \cap \dots \cap P_{10} = \phi$ is

- (1) $\frac{(1023)^{10}}{2^{100}}$ (2) $\frac{(1025)^{10}}{4^{100}}$ (3) $\frac{(1023)^{10}}{4^{100}}$ (4) $\frac{(1025)^{10}}{2^{100}}$

Sol. Answer (1)

No. of ways such that $P_1 \cap P_2 \cap P_3 \cap \dots \cap P_{10} = \phi$

is $(2^{10} - 1)^{10} = (1023)^{10}$

$$\text{Required probability} = \frac{(1023)^{10}}{2^{100}}$$

Comprehension-VI

Let A , B and C be three mutually exclusive events such that $P(A) = p_1$, $P(B) = p_2$ and $P(C) = p_3$. Then,

1. Let $p_1 = \frac{1}{2}(1-p)$, $p_2 = \frac{1}{3}(1+2p)$ and $p_3 = \frac{1}{5}(2+3p)$, then p belongs to

(1) ϕ (2) $(-1, 2)$ (3) $\left(-1, \frac{1}{47}\right)$ (4) $\left(-\frac{1}{3}, \frac{2}{3}\right)$

Sol. Answer (1)

We have,

$$0 \leq p_1, p_2, p_3 \leq 1$$

$$\text{and } 0 \leq p_1 + p_2 + p_3 \leq 1$$

$$\Rightarrow -1 \leq p \leq 1, -\frac{1}{2} \leq p \leq 1, -\frac{2}{3} \leq p \leq 1$$

$$\text{And } -\frac{37}{23} \leq p \leq -\frac{7}{23}$$

No such value of p exists.

$$\Rightarrow p \in \phi$$

2. Let p_1 , p_2 and p_3 are roots of $27x^3 - 27x^2 + ax - 1 = 0$, then value of a is

(1) 3 (2) 6 (3) 9 (4) $\frac{1}{3}$

Sol. Answer (3)

$$\text{We have, } p_1 + p_2 + p_3 = 1, p_1p_2 + p_2p_3 + p_3p_1 = a, p_1p_2p_3 = \frac{1}{27}$$

$$\text{Now } \frac{p_1 + p_2 + p_3}{3} \geq (p_1p_2p_3)^{1/3}$$

$$\Rightarrow \frac{1}{3} \geq \left(\frac{1}{27}\right)^{1/3} = \frac{1}{3}$$

$$\text{Equality holds, hence } p_1 = p_2 = p_3 = \frac{1}{3}$$

$$\Rightarrow p_1p_2 + p_2p_3 + p_3p_1 = \frac{9}{27} = \frac{3}{9} = \frac{1}{3}$$

$$\Rightarrow a = 9$$

3. Let $p_1 = \frac{1}{3}(1+3p)$, $p_2 = \frac{1}{4}(1-p)$ and $p_3 = \frac{1}{4}(1-2p)$, then p belongs to

(1) $(-1, 0)$ (2) $\left(-\frac{2}{3}, \frac{2}{3}\right)$ (3) $\left(-\frac{1}{3}, 1\right)$ (4) $\left(-\frac{1}{9}, \frac{1}{4}\right)$

Sol. Answer (4)

Similar as in solution of (1)

$$p \in \left(-\frac{1}{9}, \frac{1}{4}\right)$$

Comprehension-VII

Let bag A contains 3 red balls and 9 black balls. Bag B contains 8 red balls and 4 black balls. Bag C contains 10 red balls and 2 black balls. A card is drawn from a well shuffled pack of 52 playing cards. If a face card is drawn, a ball is selected from Bag A. If an ace is drawn, a ball is selected from Bag B. If any other card is drawn, a ball is selected from Bag C.

1. The probability that a red ball is selected is

(1) $\frac{107}{156}$ (2) $\frac{15}{26}$ (3) $\frac{3}{52}$ (4) $\frac{4}{39}$

2. The conditional probability that Bag A was one from which a ball was selected, given that the ball selected was red

(1) $\frac{156}{107} \times \frac{1}{3}$ (2) $\frac{9}{107}$ (3) $\frac{3}{52}$ (4) $\frac{3}{107}$

3. The conditional probability that Bag B was one from which a ball was selected, given that the ball selected was black is

(1) $\frac{9}{107}$ (2) $\frac{4}{49}$ (3) $\frac{98}{107}$ (4) $\frac{45}{49}$

Solution of Comprehension-VII

1. Answer (1)

2. Answer (2)

3. Answer (2)

Let B_1 be the event that face card is drawn.

B_2 be the ace card is drawn

B_3 be the event that neither face nor ace is drawn.

$$p(\text{red ball is drawn}) = \frac{12}{52} \times \frac{3}{12} + \frac{4}{52} \times \frac{8}{12} + \frac{36}{52} \times \frac{10}{12} = \frac{107}{156}$$

$$p(\text{black ball is drawn}) = \frac{12}{52} \times \frac{9}{12} + \frac{4}{52} \times \frac{4}{12} + \frac{36}{52} \times \frac{2}{12} = \frac{108 + 16 + 72}{52 \times 12} = \frac{196}{52 \times 12} = \frac{49}{156}$$

The conditional probability that Bag-A was one from which a ball was selected, given that the ball selected was red.

$$= P\left(\frac{B_1}{\text{red}}\right) = \frac{\frac{12}{52} \times \frac{3}{12}}{\frac{107}{156}} = \frac{3}{52} \times \frac{156}{107} = \frac{9}{107}$$

The conditional probability that Bag-B was one from which a ball was selected, given that the ball selected was black.

$$= P\left(\frac{B_2}{\text{black}}\right) = \frac{\frac{4}{52} \times \frac{4}{12}}{\frac{49}{156}} = \frac{1}{13 \times 3} \times \frac{156}{49} = \frac{4}{49}$$

Comprehension-VIII

A fair die is tossed repeatedly until a six is obtained. Let X denote the number of tosses required.

[IIT-JEE 2009]

1. The probability that $X = 3$, equals

(1) $\frac{25}{216}$ (2) $\frac{25}{36}$ (3) $\frac{5}{36}$ (4) $\frac{125}{216}$

Sol. Answer (1)

Clearly $p = \frac{1}{6}, q = \frac{5}{6}$

$$p(X = 3) = \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} = \frac{25}{216}$$

2. The probability that $X \geq 3$ equals

(1) $\frac{125}{216}$ (2) $\frac{25}{36}$ (3) $\frac{5}{36}$ (4) $\frac{25}{216}$

Sol. Answer (2)

We have

$$\begin{aligned} p(X \geq 3) &= \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^3 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^4 \cdot \frac{1}{6} + \dots \\ &= \frac{\left(\frac{5}{6}\right)^2 \cdot \frac{1}{6}}{1 - \frac{5}{6}} = \frac{25}{36} \end{aligned}$$

3. The conditional probability that $X \geq 6$ given $X > 3$ equals

(1) $\frac{125}{216}$ (2) $\frac{25}{216}$ (3) $\frac{5}{36}$ (4) $\frac{25}{36}$

Sol. Answer (4)

The conditional probability that $X \geq 6$
given $X > 3$

$$\begin{aligned} &= \frac{\left(\frac{5}{6}\right)^5 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^6 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^7 \cdot \frac{1}{6} + \dots}{\left(\frac{5}{6}\right)^3 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^4 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^5 \cdot \frac{1}{6} + \dots} \\ &= \frac{\left(\frac{5}{6}\right)^5 \cdot \frac{1}{6}}{\left(\frac{5}{6}\right)^3 \cdot \frac{1}{6}} = \left(\frac{5}{6}\right)^2 = \frac{25}{36} \end{aligned}$$

Comprehension-IX

Let U_1 and U_2 be two urns such that U_1 contains 3 white and 2 red balls, and U_2 contains only 1 white ball. A fair coin is tossed. If head appears then 1 ball is drawn at random from U_1 and put into U_2 . However, if tail appears then 2 balls are drawn at random from U_1 and put into U_2 . Now 1 ball is drawn at random from U_2 . [IIT-JEE 2011]

1. The probability of the drawn ball from U_2 being white is

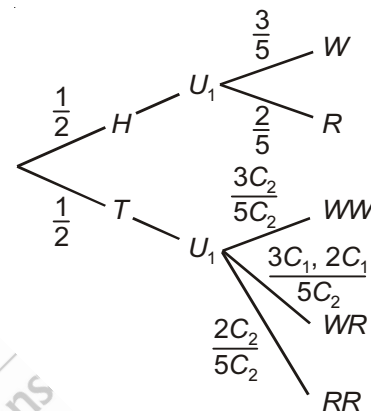
- (1) $\frac{13}{30}$ (2) $\frac{23}{30}$ (3) $\frac{19}{30}$ (4) $\frac{11}{30}$

Sol. Answer (2)

We have the following tree-diagram under given conditions.

Probability of drawing white ball from U_2

$$\begin{aligned}
 &= \frac{1}{2} \cdot \frac{3}{5} \cdot 1 + \frac{1}{2} \cdot \frac{2}{5} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{3}{10} \cdot 1 + \frac{1}{2} \cdot \frac{3}{5} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{10} \cdot \frac{1}{3} \\
 &= \frac{3}{10} + \frac{1}{10} + \frac{3}{20} + \frac{1}{5} + \frac{1}{60} \\
 &= \frac{18+6+9+12+1}{60} \\
 &= \frac{23}{30}
 \end{aligned}$$



2. Given that the drawn ball from U_2 is white, the probability that head appeared on the coin is

- (1) $\frac{17}{23}$ (2) $\frac{11}{23}$ (3) $\frac{15}{23}$ (4) $\frac{12}{23}$

Sol. Answer (4)

Using Baye's theorem,

$$\text{Required probability} = \frac{\frac{1}{2} \times \frac{3}{5} + \frac{1}{2} \times \frac{2}{5} \times \frac{1}{2}}{\frac{23}{30}} = \frac{\frac{4}{10}}{\frac{23}{30}} = \frac{12}{23}$$

Comprehension-X

A box B_1 contains 1 white ball, 3 red balls and 2 black balls. Another box B_2 contains 2 white balls, 3 red balls and 4 black balls. A third box B_3 contains 3 white balls, 4 red balls and 5 black balls. [JEE(Advanced) 2013]

1. If 1 ball is drawn from each of the boxes B_1 , B_2 and B_3 , the probability that all 3 drawn balls are of the same colour is

- (1) $\frac{82}{648}$ (2) $\frac{90}{648}$ (3) $\frac{558}{648}$ (4) $\frac{566}{648}$

Sol. Answer (1)

B_1	B_2	B_3
1W	2W	3W
3R	3R	4R
2B	4B	5B

$$P(WWW + RRR + BBB)$$

$$= \left(\frac{1}{6} \times \frac{2}{9} \times \frac{3}{12} \right) + \left(\frac{3}{6} \times \frac{3}{9} \times \frac{4}{12} \right) + \left(\frac{2}{6} \times \frac{4}{9} \times \frac{5}{12} \right)$$

$$= \frac{6 + 36 + 40}{648}$$

$$= \frac{82}{648}$$

2. If 2 balls are drawn (without replacement) from a randomly selected box and one of the balls is white and the other ball is red, the probability that these 2 balls are drawn from box B_2 is

(1) $\frac{116}{181}$

(2) $\frac{126}{181}$

(3) $\frac{65}{181}$

(4) $\frac{55}{181}$

Sol. Answer (4)

$$P\left(\frac{B_2}{WR}\right) = \frac{P\left(\frac{WR}{B_2}\right) \times P(B_2)}{P\left(\frac{WR}{B_1}\right) \cdot P(B_1) + P\left(\frac{WR}{B_2}\right) \cdot P(B_2) + P\left(\frac{WR}{B_3}\right) \cdot P(B_3)}$$

$$= \frac{\frac{{}^2C_1 \times {}^3C_1}{{}^9C_2} \times \frac{1}{3}}{\left(\frac{{}^1C_1 \times {}^3C_1}{{}^9C_2} \times \frac{1}{3} \right) + \left(\frac{{}^2C_1 \times {}^3C_1}{{}^9C_2} \times \frac{1}{3} \right) + \left(\frac{{}^3C_1 \times {}^4C_1}{{}^{12}C_2} \times \frac{1}{3} \right)}$$

$$= \frac{\frac{2 \times 3}{9 \times 4}}{\frac{3 \times 2}{6 \times 5} + \frac{2 \times 3}{9 \times 4} + \frac{3 \times 4 \times 2}{12 \times 11}}$$

$$= \frac{6 \times 5 \times 6 \times 11}{36 \times 181}$$

$$= \frac{55}{181}$$

Comprehension-XI

Box 1 contains three cards bearing numbers 1, 2, 3; box 2 contains five cards bearing numbers 1, 2, 3, 4, 5; and box 3 contains seven cards bearing numbers 1, 2, 3, 4, 5, 6, 7. A card is drawn from each of the boxes. Let x_i be number on the card drawn from the i^{th} box, $i = 1, 2, 3$. [JEE(Advanced)-2014]

1. The probability that $x_1 + x_2 + x_3$ is odd, is

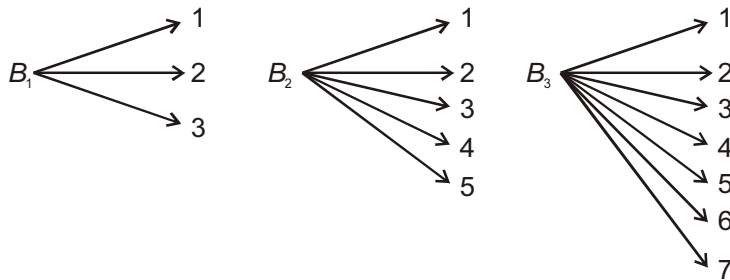
(1) $\frac{29}{105}$

(2) $\frac{53}{105}$

(3) $\frac{57}{105}$

(4) $\frac{1}{2}$

Sol. Answer (2)



$$x_1 + x_2 + x_3 = \text{odd}$$

Case-I :	One odd	$(2 \times 3 \times 2) + (4 \times 1 \times 2) + (3 \times 3 \times 1)$
	Two even	$= 12 + 8 + 9 = 29$
Case-II :	3 odd	$2 \times 4 \times 3 = 24$
		$= 29 + 24 = 53$

$$\text{Total cases} = 3 \times 5 \times 7 = 105$$

$$\text{Probability} = \frac{53}{105}$$

2. The probability that x_1, x_2, x_3 are in an arithmetic progression, is

(1) $\frac{9}{105}$	(2) $\frac{10}{105}$	(3) $\frac{11}{105}$	(4) $\frac{7}{105}$
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Sol. Answer (3)

$$2x_2 = x_1 + x_3$$

 x_1 and x_3 must be even number.

$$\text{both } x_1 \in \{1, 3\}$$

$$x_1 \text{ and } x_3 \text{ odd } x_3 \in \{1, 3, 5, 7\}$$

$$\text{also } x_1 + x_3 \in \{2, 4, 6, 8, 10\}$$

x_1	x_3
1	(1, 3, 5, 7)
3	(1, 3, 5, 7)

8 cases x_1 and x_3 both even

$$x_2 = 2$$

$$x_3 = (2, 4, 6)$$

3 cases

Total = 11 cases

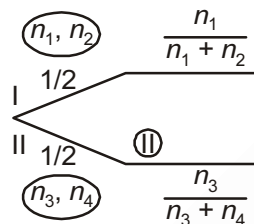
$$\text{Required probability} = \frac{11}{105}$$

Comprehension-XII

Let n_1 and n_2 be the number of red and black balls, respectively, in box I. Let n_3 and n_4 be the number of red and black balls, respectively, in box II. [JEE(Advanced) 2015]

1. One of the two boxes, box I and box II, was selected at random and a ball was drawn randomly out of this box. The ball was found to be red. If the probability that this red ball was drawn from box II is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1, n_2, n_3 and n_4 is(are)
- (1) $n_1 = 3, n_2 = 3, n_3 = 5, n_4 = 15$ (2) $n_1 = 3, n_2 = 6, n_3 = 10, n_4 = 50$
 (3) $n_1 = 8, n_2 = 6, n_3 = 5, n_4 = 20$ (4) $n_1 = 6, n_2 = 12, n_3 = 5, n_4 = 20$

Sol. Answer (1, 2)

$$\frac{1}{3} = \frac{\frac{1}{2} \left(\frac{n_3}{n_3 + n_4} \right)}{\frac{1}{2} \left(\frac{n_1}{n_1 + n_2} + \frac{n_3}{n_3 + n_4} \right)}$$


$$\frac{1}{3} = \frac{\left(\frac{n_3}{n_3 + n_4} \right)}{\left(\frac{n_1}{n_1 + n_2} + \frac{n_3}{n_3 + n_4} \right)}$$

(1) RHS $\frac{\frac{5}{20}}{\frac{5}{20} + \frac{3}{6}} = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{2}} = \frac{\frac{1}{4}}{\frac{1+2}{4}} = \frac{1}{3}$

(2) RHS $\frac{\frac{10}{60}}{\frac{10}{60} + \frac{3}{9}} = \frac{\frac{1}{6}}{\frac{1}{6} + \frac{1}{3}} = \frac{\frac{1}{6}}{\frac{1+2}{6}} = \frac{1}{3}$

(3) $\frac{\frac{5}{25}}{\frac{8}{14} + \frac{5}{25}} = \frac{\frac{1}{5}}{\frac{4}{7} + \frac{1}{5}} = \frac{\frac{1}{5}}{\frac{20+7}{35}} = \frac{7}{27}$

(4) $\frac{\frac{5}{25}}{\frac{6}{18} + \frac{5}{25}} = \frac{\frac{1}{5}}{\frac{5}{15} + \frac{3}{15}} = \frac{\frac{1}{5}}{\frac{8}{15}} = \frac{1}{5} \times \frac{15}{8} = \frac{3}{8}$

2. A ball is drawn at random from box I and transferred to box II. If the probability of drawing a red ball from box I, after this transfer, is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1 and n_2 is(are)
- (1) $n_1 = 4$ and $n_2 = 6$ (2) $n_1 = 2$ and $n_2 = 3$
 (3) $n_1 = 10$ and $n_2 = 20$ (4) $n_1 = 3$ and $n_2 = 6$

Sol. Answer (3, 4)

$$\text{Probability} = \frac{n_1}{n_1 + n_2} \cdot \frac{n_1 - 1}{n_1 + n_2 - 1} + \frac{n_2}{n_1 + n_2} \cdot \frac{n_1}{n_1 + n_2 - 1} = \frac{1}{3}$$

$$\Rightarrow \frac{n_1(n_1 - 1) + n_2 n_1}{(n_1 + n_2)(n_1 + n_2 - 1)} = \frac{1}{3}$$

$$\Rightarrow n_1 = 10, n_2 = 20$$

$$\text{or } n_1 = 3, n_2 = 6$$

Comprehension-XIII

Football teams T_1 and T_2 have to play two games against each other. It is assumed that the outcomes of the two games are independent. The probabilities of T_1 winning, drawing and losing a game against T_2 are $\frac{1}{2}$, $\frac{1}{6}$ and $\frac{1}{3}$, respectively. Each team gets 3 points for a win, 1 point for a draw and 0 point for a loss in a game. Let X and Y denote the total points scored by teams T_1 and T_2 , respectively, after two games.

[JEE(Advanced)-2016]

1. $P(X > Y)$ is

(1) $\frac{1}{4}$

(2) $\frac{5}{12}$

(3) $\frac{1}{2}$

(4) $\frac{7}{12}$

Sol. Answer (2)

Given T_1 and T_2 have to play.

Given independent event

X = total points scored by teams T_1

Y = total points scored by teams T_2

$P(X > Y)$

$$= \frac{1}{2} \times \frac{1}{2} + {}^2C_2 \times \frac{1}{2} \times \frac{1}{6}$$

$$= \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$

2. $P(X = Y)$ is

(1) $\frac{11}{36}$

(2) $\frac{1}{3}$

(3) $\frac{13}{36}$

(4) $\frac{1}{2}$

Sol. Answer (3)

$$P(X = Y) = {}^2C_1 \times \frac{1}{2} \times \frac{1}{3} + \frac{1}{6} \times \frac{1}{6} = \frac{13}{36}$$

Comprehension-XIV

There are five students S_1, S_2, S_3, S_4 and S_5 in a music class and for them there are five seats R_1, R_2, R_3, R_4 and R_5 arranged in a row, where initially the seat R_i is allotted to the student S_i , $i = 1, 2, 3, 4, 5$. But, on the examination day, the five students are randomly allotted the five seats.

[JEE(Advanced) 2018]

1. The probability that, on the examination day, the student S_1 gets the previously allotted seat R_1 , and **NONE** of the remaining students gets the seat previously allotted to him/her is

(1) $\frac{3}{40}$

(2) $\frac{1}{8}$

(3) $\frac{7}{40}$

(4) $\frac{1}{5}$

Sol. Answer (1)

Derangement of S_2, S_3, S_4, S_5 is

$$= 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right) = 9$$

$$\text{Probability} = \frac{9}{5!} = \frac{9}{120} = \frac{3}{40}$$

2. For $i = 1, 2, 3, 4$, let T_i denote the event that the students S_i and S_{i+1} do **NOT** sit adjacent to each other on the day of the examination. Then, the probability of the event $T_1 \cap T_2 \cap T_3 \cap T_4$ is

(1) $\frac{1}{15}$

(2) $\frac{1}{10}$

(3) $\frac{7}{60}$

(4) $\frac{1}{5}$

Sol. Answer (3)

Method (1)

The Possible arrangements are :

(1) $P_1 \quad P_3 \quad P_5 \quad P_2 \quad P_4$

(2) $P_1 \quad P_4 \quad P_2 \quad P_5 \quad P_3$

(3) $P_2 \quad P_4 \quad P_1 \quad P_3 \quad P_5$

(4) $P_2 \quad P_4 \quad P_1 \quad P_5 \quad P_3$

(5) $P_2 \quad P_5 \quad P_3 \quad P_1 \quad P_4$

(6) $P_3 \quad P_1 \quad P_4 \quad P_2 \quad P_5$

(7) $P_3 \quad P_1 \quad P_5 \quad P_2 \quad P_4$

(8) $P_3 \quad P_5 \quad P_2 \quad P_4 \quad P_1$

(9) $P_3 \quad P_5 \quad P_1 \quad P_4 \quad P_2$

(10) $P_4 \quad P_1 \quad P_3 \quad P_5 \quad P_2$

(11) $P_4 \quad P_2 \quad P_5 \quad P_1 \quad P_3$

(12) $P_4 \quad P_2 \quad P_5 \quad P_3 \quad P_1$

(13) $P_5 \quad P_1 \quad P_3 \quad P_2 \quad P_4$

(14) $P_5 \quad P_2 \quad P_4 \quad P_1 \quad P_3$

$$\therefore \text{Required probability} = \frac{14}{120} = \frac{7}{60}$$

Method (2)

$$5! - \underset{\text{at least one pair}}{^4C_1 \times 4! \times 2!} + \underset{\text{at least two pair}}{(^3C_1 \times 3! \times 2! + ^3C_1 \times 3! \times 2! \times 2!)} - \underset{\text{at least 3}}{(^2C_1 \times 2! \times 2! + ^2C_1 \times 2 \times 2! + ^2C_1 \times 2 \times 2!)} + 2 = 14$$

$$\text{Required probability} = \frac{14}{5!} = \frac{7}{60}$$

SECTION - D

Assertion-Reason Type Questions

1. Three distinct numbers are chosen from the set $\{1, 2, 3, \dots, 20\}$.

STATEMENT-1 : The probability that the three numbers are in increasing G.P. is $\frac{2}{285}$.

and

STATEMENT-2 : If a, b, c are in G.P. then $b^2 = ac$.

Sol. Answer (4)

$$n(S) = \text{Number of ways to select 3 numbers from 1 to 20} = {}^{20}C_3 = 1140$$

$$F = \{(1, 2, 4), (1, 3, 9), (1, 4, 16), (2, 4, 8), (2, 6, 18), (3, 6, 12), (4, 6, 9), (4, 8, 16), (5, 10, 20), (8, 12, 18), (9, 12, 16)\}$$

$$n(E) = 11 = \frac{11}{1140}$$

Statement (2) is correct.

2. Three numbers are selected from the set $\{1, 2, 3, \dots, 21\}$.

STATEMENT-1 : The probability that the numbers are in increasing A.P. is $\frac{10}{133}$.

and

STATEMENT-2 : a, b, c are in A.P. if $2b = a + c$.

Sol. Answer (1)

$$\{1, 2, 3, \dots, 21\}$$

$$n(S) = {}^{21}C_3 = 1330$$

Let a, b, c is selected

$$2b = a + c, \text{ to be in A.P.}$$

a and c must be even or a and c must be odd

$$n(E) = {}^{10}C_2 + {}^{11}C_2 = 45 + 55 = 100 \quad \begin{cases} \text{out of 21, 11 number is} \\ \text{odd and 10 number is even} \end{cases}$$

$$P(E) = \frac{100}{1330} = \frac{10}{133}$$

Statement-2 is true and it is a correct explanation of statement-1.

3. STATEMENT-1 : Two cards are drawn at random from a pack of playing cards. The probability that both the cards are of red colour or they are queen is $\frac{55}{221}$.

and

STATEMENT-2 : $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Sol. Answer (1)

A = event that red card is selected

$$P(A) = \frac{{}^{26}C_2}{{}^{52}C_2}$$

B = event that card is queen

$$P(B) = \frac{{}^4C_2}{{}^{52}C_2}$$

$$n(A \cap B) = 2$$

$$P(A \cap B) = \frac{{}^2C_2}{{}^{52}C_2}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{{}^{26}C_2}{{}^{52}C_2} + \frac{{}^4C_2}{{}^{52}C_2} - \frac{{}^2C_2}{{}^{52}C_2} = \frac{325+6-1}{{}^{52}C_2} = \frac{330}{26 \times 51} = \frac{55}{13 \times 17} = \frac{55}{221}$$

Statement-2 is true and it is a correct explanation of statement-1.

4. STATEMENT-1 : Let A and B be two mutually exclusive events such that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$ then

$$P(A' \cap B') = \frac{1}{6}.$$

and

STATEMENT-2 : In case of mutually exclusive events $P(A \cup B) = P(A) + P(B)$.

Sol. Answer (1)

$$P(A \cup B) = P(A) + P(B) = \frac{1}{3} + \frac{1}{2} = \frac{2+3}{6} = \frac{5}{6}$$

$$P(A' \cap B') = 1 - P(A \cup B) = 1 - \frac{5}{6} = \frac{1}{6}$$

Statement-1 is true.

In case of mutually exclusive events $P(A \cap B) = 0$

$$P(A \cup B) = P(A) + P(B)$$

Statement-2 is true and it is a correct explanation of statement-1.

5. STATEMENT-1 : One ticket is drawn at random out of 30 tickets numbered from 1 to 30. The probability that the number on the ticket is a multiple of 5 or 7 is $\frac{1}{3}$.

and

STATEMENT-2 : The set of multiples of 5 and 7 are disjoint sets.

Sol. Answer (3)

A = event that number is divisible by 5

$$A = \{5, 10, 15, 20, 25, 30\}$$

B = event that number is divisible by 7

$$B = \{7, 14, 21, 28\}$$

$$P(A \cup B) = P(A) + P(B) = \frac{6}{30} + \frac{4}{30} = \frac{1}{3}$$

Statement 2 is false

6. STATEMENT-1 : Ten fair coins fall to the floor, the probability that there are five heads and five tails is $\frac{63}{256}$.

and

STATEMENT-2 : Four fair dice are rolled, the probability that the four numbers coming will be all different is $\frac{5}{18}$.

Sol. Answer (2)

Statement 1 : The probability of five heads and five tails is

$$\begin{aligned}
 &= {}^{10}C_5 \cdot \left(\frac{1}{2}\right)^5 \cdot \left(\frac{1}{2}\right)^5 \\
 &= \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2} \cdot \frac{1}{32} \cdot \frac{1}{32} \\
 &= \frac{63}{256}
 \end{aligned}$$

Statement 2 : The probability of four different numbers = $\frac{6 \cdot 5 \cdot 4 \cdot 3}{6 \cdot 6 \cdot 6 \cdot 6} = \frac{5}{18}$

$\therefore$ Statement 2 is also true

$\therefore$ Statement 1 and 2 are both true but statement 2 is not the correct explanation of 1.

7. STATEMENT-1 : If the odds against an event is $\frac{2}{3}$, then probability of occurring of the event is $\frac{3}{5}$.

and

STATEMENT-2 : For two events A and B , $P(A' \cap B') = 1 - P(A \cup B)$

Sol. Answer (2)

Statement (1) odds against an event is 2 : 3

$$\text{Hence } P(E) = \frac{3}{2+3} = \frac{3}{5}$$

Statement 1 is true

Statement 2 is obviously true but it is not the correct explanation of 1.

8. STATEMENT-1 : A lottery agent sells 100 tickets and awards 10 prizes, if Ramesh buys 10 tickets the probability of his winning a prize is $\left(1 - \frac{{}^{90}C_{10}}{{}^{100}C_{10}}\right)$.

and

STATEMENT-2 : If $P(A) = 0.8$, $P(B) = 0.5$, the $P(A \cap B)$ is 0.4.

Sol. Answer (3)

Number of ways of selecting 10 tickets out of 100 is $= {}^{100}C_{10}$

$$\text{Required probability} = (1 - \text{No prize}) = \left(1 - \frac{{}^{90}C_{10}}{{}^{100}C_{10}}\right)$$

Statement 1 is true. But in statement 2 it is not given that A and B are independent. Hence 2 is false.

9. Let for two events A and B , $P(A) = p$ and $P(B) = q$.

STATEMENT-1 : The probability that exactly one of the event A and B occurs is $p + q - 2pq$.

and

STATEMENT-2 : $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Sol. Answer (4)

Probability that exactly one of the events

$$A \text{ and } B \text{ occurs} = P(A \cup B) - P(A \cap B)$$

$$= P(A) + P(B) - 2P(A \cap B)$$

$\therefore$ A and B are not given to be independent, so statement 1 is false.

10. Consider the system of linear equations $ax + by = 0$, $cx + dy = 0$, $a, b, c, d \in \{0, 1\}$

STATEMENT-1 : The probability that the system of equations has a unique solution is $\frac{3}{8}$.

and

STATEMENT-2 : The probability that the system has a solution is 1.

Sol. Answer (2)

The system $ax + by = 0$, $cx + dy = 0$ is homogeneous, so it must have solutions, unique or infinitely many. Therefore, the probability that the system of equations has a solution is 1.

The system $ax + by = 0$, $cx + dy = 0$, has unique solution if $ad - bc \neq 0$.

$a, b, c, d \in \{0, 1\} \Rightarrow$ total ways of selecting a, b, c, d is 16

$ad = bc$ iff $ad = bc = 0$ or 1

If $ad = bc = 0$ or 1, then a, b, c, d can be selected in $9 + 1 = 10$ ways

$$\text{Hence probability} = \frac{16 - 10}{16} = \frac{3}{8}$$

$\therefore$ Statement 1 is true but Statement 2 is not correct explanation of Statement 1.

11. STATEMENT-1 : If $P(B) = \frac{3}{4}$; $P(A \cap B \cap \bar{C}) = \frac{1}{3}$; $P(\bar{A} \cap B \cap \bar{C}) = \frac{1}{3}$, then $P(B \cap C) = \frac{1}{12}$.

and

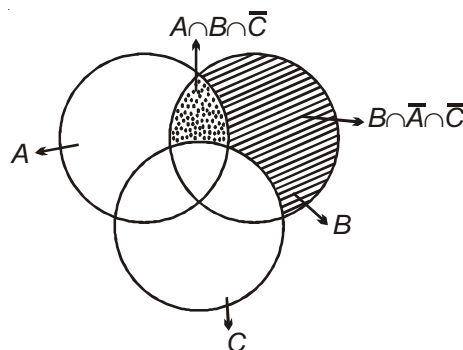
STATEMENT-2 : $P(B \cap C) = P(B) - P(A \cap B \cap \bar{C}) - P(\bar{A} \cap B \cap \bar{C})$.

Sol. Answer (1)

$$P(B \cap C) = P(B) - P(A \cap B \cap \bar{C}) - P(\bar{A} \cap B \cap \bar{C})$$

$$P(B \cap C) = \frac{3}{4} - \frac{1}{3} - \frac{1}{3} = \frac{1}{12}$$

$\therefore$ Statement 1 and Statement 2 both are true and Statement 2 is the correct explanation of Statement 2.



12. STATEMENT-1 : If $P(A) = \frac{1}{2}$, $P(B) = \frac{2}{3}$ and $\frac{1}{6} \leq P(A \cap B) \leq \frac{1}{2}$.

and

STATEMENT-2 : $P(A \cup B) \geq \max. \{P(A), P(B)\}$ and $P(A \cap B) \leq \min. \{P(A), P(B)\}$

Sol. Answer (2)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq 1$$

$$\frac{1}{2} + \frac{2}{3} - P(A \cap B) \leq 1$$

$$\frac{1}{6} \leq P(A \cap B)$$

$$P(A \cap B) \leq \frac{1}{2}$$

Statement 1 and Statement 2 both are correct but statement 2 is not correct explanation of statement 1

13. STATEMENT-1 : Six different balls are put in three different boxes, no box being empty, the probability of putting balls in boxes in equal numbers is $\frac{1}{6}$.

and

STATEMENT-2 : Six letters are posted in 3 letter boxes. The probability that no letter box remains empty is $\frac{20}{27}$.

Sol. Answer (2)

Statement 1 : Number of ways of putting the balls in boxes is $3^6 - {}^3C_1 2^6 + {}^3C_2 1^6 = 540$

Number of ways of putting the balls so that each box gets two balls is $\frac{6!}{2! 2! 2!} = 90$

Hence probability is $\frac{90}{540} = \frac{1}{6}$

Statement 2 : The required probability = $\frac{3^6 - {}^3C_1 2^6 + {}^3C_2 1^6}{3^6} = \frac{20}{27}$

Hence answer (2).

14. A lot contains equal number of defective and non-defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. The events A, B, C are defined as

A : The first bulb is defective

B : The second bulb is non-defective

C : The two bulbs are either both defective or both non-defective.

STATEMENT-1 : A, B, C are pairwise independent.

and

STATEMENT-2 : A, B, C are mutually independent.

Sol. Answer (3)

$P(A)$ = Probability of the first bulb is defective and second bulb is either defective or non-effective

$$= \frac{1}{2} \cdot 1 = \frac{1}{2}$$

$P(B)$ = Probability that second bulb is non-defective

$$= 1 \times \frac{1}{2} = \frac{1}{2}$$

$P(C)$ = Probability that the two bulbs are either defective or non-defective.

$$= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$$

Clearly the events A, B, C are pair-wise and independent but A, B, C are mutually independent.

Statement-1 is true but statement-2 is false.

15. Let $P(A_i)$, the probability of happening of independent events $A_i (i = 1, 2, 3)$ be given by $P(A_i) = \frac{1}{i+1}$.

STATEMENT-1 : The probability that at least one event happens is $\frac{3}{4}$.

and

STATEMENT-2 : $P\left(\bigcup_{i=1}^3 A_i\right) = 1 - \prod_{i=1}^3 P(A'_i)$.

Sol. Answer (1)

Since the events A_1, A_2, A_3 are independent and $P(A_1) = \frac{1}{2}, P(A_2) = \frac{1}{3}, P(A_3) = \frac{1}{4}$, and the probability that neither of the events A_1, A_2, A_3 happen

$$= P(A'_1 \cap A'_2 \cap A'_3) = P(A'_1)P(A'_2)P(A'_3)$$

$$= \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) = \frac{1}{4}$$

$\Rightarrow$ The probability that at least one event happens

$$= 1 - \frac{1}{4} = \frac{3}{4}$$

Statement-1 is true.

Clearly statement-2 is true and it is a correct explanation of statement-1.

16. STATEMENT-1 : If A and B are exhaustive events, then $P(A \cup B) = 1$.

and

STATEMENT-2 : If A and B are independent then $P(A \cap B) = P(A) \cdot P(B)$.

Sol. Answer (2)

A and B are exhaustive events, hence $A \cup B = S$

$$\Rightarrow P(A \cup B) = 1$$

When A and B are independent events, then $P(A \cap B) = P(A)P(B)$.

17. STATEMENT-1 : Let E_1, E_2, E_3 be three events such that $P(E_1) + P(E_2) + P(E_3) = 1$, then E_1, E_2, E_3 are exhaustive events.

and

STATEMENT-2 : If the events E_1, E_2 and E_3 be exhaustive events, then $P(E_1 \cup E_2 \cup E_3) = 1$.

Sol. Answer (4)

Statement-2 is always true but statement-1 is not always true.

For example, let

$$S = \{1, 2, 3, 4\}$$

$$A = \{1, 2\}, B = \{2, 4\}$$

$$P(A) = \frac{2}{4} = \frac{1}{2}, P(B) = \frac{2}{4} = \frac{1}{2}$$

$$P(A) + P(B) = \frac{1}{2} + \frac{1}{2} = 1 \text{ but } A \cup B = \{1, 2, 4\}, \text{ which is not exhaustive.}$$

18. STATEMENT-1 : A fair coin is tossed four times. The probability that heads exceed tails in number is $\frac{5}{16}$.

and

STATEMENT-2 : A fair coin is tossed, then the probability that it lands with heads is $\frac{1}{2}$.

Sol. Answer (2)

Statement-2 is true.

$$n(S) = 2^4 = 16$$

Needs exceeds tails.

$$= 4H + 3H/T = {}^4C_4 + {}^4C_3 = 1 + 4 = 5$$

$$\Rightarrow \text{Probability that heads exceeds tails in number} = \frac{5}{16}$$

Statement-1 is true.

19. STATEMENT-1 : Three fair dice are rolled. The probability that the same number will appear on each of them is $\frac{1}{36}$.

and

STATEMENT-2 : When three dice are rolled, $n(S) = 6^3$.

Sol. Answer (1)

Number of outcomes when three dice are rolled

$$= 6 \times 6 \times 6 = 6^3$$

When the same number will appear on each of then

$$= \{1, 1, 1\}, \{2, 2, 2\}, \{3, 3, 3\}, \dots, \{6, 6, 6\}$$

$$\text{Required probability} = \frac{6}{6^3} = \frac{1}{36}$$

Statement-2 is true and statement-2 is the correct explanation of statement-1.

20. STATEMENT-1 : Let x denotes the number of sixes in 3 consecutive throws of a die, then $P(x = 3) = \frac{1}{216}$.

and

STATEMENT-2 : The probability that sum of faces appeared is 10, when three dice are rolled is $\frac{{}^9C_7}{6^3}$.

Sol. Answer (3)

When three dice are rolled $n(S) = 6^3 = 216$ and event $= \{(6, 6, 6)\}$

$$\Rightarrow P(E) = \frac{1}{216}. \text{ The sum of faces appeared is 10.}$$

$$\Rightarrow x + y + z = 10, n(E) = \text{Coeff. of } x^{10} \text{ in } (x + x^2 + x^3 + \dots + x^6)^3$$

$$\begin{aligned} \Rightarrow n(E) &= \text{Coeff. of } x^{10} \text{ in } x^3(1 - x^6)^3(1 - x)^{-3} \\ &= \text{Coeff. of } x^7 \text{ in } (1 - 3x^6 + 3x^{12} - x^{18})(1 + {}^3C_1x + {}^4C_1x^2 + \dots) \\ &= {}^9C_2 - 3 \cdot {}^5C_3 \neq {}^9C_2 = {}^9C_7 \end{aligned}$$

$\Rightarrow$ Statement-2 is false.

21. STATEMENT-1 : Mutually exclusive events, none of which being an impossible event, are not independent.

and

STATEMENT-2 : Non-impossible independent events are not mutually exclusive.

Sol. Answer (1)

Explanation is clear from the definition of independent and exclusive events.

22. One card is lost from a pack of 52 cards, then

STATEMENT-1 : If two cards are drawn then probability that lost card is Ace if both are Ace cards is $\frac{1}{25}$.

and

STATEMENT-2 : If two cards are drawn then probability that lost card is Spade if both cards are Spade is $\frac{11}{50}$.

Sol. Answer (2)

Statement -1 :

A – Ace card is lost

$\bar{A}$ – Ace card is not lost

A_1 – Both cards are ace.

$$P(A) = \frac{1}{13} \quad P(\bar{A}) = \frac{12}{13}$$

$$P(A / A_1) = \frac{\frac{1}{13} \frac{{}^3C_2}{{}^{51}C_2}}{\frac{1}{13} \frac{{}^3C_2}{{}^{51}C_2} + \frac{12}{13} \frac{{}^4C_2}{{}^{51}C_2}} = \frac{3}{3 + 12(6)} = \frac{1}{25}$$

Statement-2 :

S – Spade card is lost

$\bar{S}$ – Spade card is not lost

S_1 – Both cards are spade

$$P(S) = \frac{1}{4} \quad P(\bar{S}) = \frac{3}{4}$$

$$P(S / S_1) = \frac{\frac{1}{4} \frac{{}^{12}C_2}{{}^{51}C_2}}{\frac{1}{4} \frac{{}^{12}C_2}{{}^{51}C_2} + \frac{3}{4} \frac{{}^{13}C_2}{{}^{51}C_2}} = \frac{12 \times 11}{12 \times 11 + 3 \times 13 \times 12} = \frac{11}{50}$$

23. Let $H_1, H_2, \dots, H_n$ be mutually exclusive and exhaustive events with $P(H_i) > 0, i = 1, 2, \dots, n$. Let E be any other event with $0 < P(E) < 1$.

STATEMENT-1 : $P(H_i | E) > P(E | H_i) \cdot P(H_i)$ for $i = 1, 2, \dots, n$.

and

STATEMENT-2 : $\sum_{i=1}^n P(H_i) = 1$

[IIT-JEE 2007]

Sol. Answer (2)

$$P(B) P(A/B) = P(A) P(B/A)$$

$$\Rightarrow P(A/B) = \frac{P(A) P(B/A)}{P(B)}$$

as $P(B) < 1$, we have $P(A/B) > P(A) P(B/A)$

Identifying A and B with H_i and E we get the result.

24. Consider the system of equations $ax + by = 0$, $cx + dy = 0$, where $a, b, c, d \in \{0, 1\}$.

[IIT-JEE 2008]

STATEMENT-1 : The probability that the system of equations has a unique solution is $\frac{3}{8}$.

and

STATEMENT-2 : The probability that the system of equations has a solution is 1.

Sol. Answer (2)

The number of all possible determinants of the form $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 2^4 = 16$

Out of which only 10 determinants given by

$$\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}, \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix}, \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix}, \begin{vmatrix} 0 & 1 \\ 0 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}, \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix}, \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix}, \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix}, \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix}$$

vanish and remaining six determinants have non-zero value.

Hence the required probability = $\frac{6}{16} = \frac{3}{8}$

$\Rightarrow$ Statement-1 is correct.

Statement-2 is also correct as the homogeneous equations have always a solution and Statement-2 does not explain the explanation of Statement-1.

SECTION - E

Matrix-Match Type Questions

1. Two fair dice are thrown. The probability that the difference between the numbers is $K(0 \leq K \leq 5)$, is denoted by $P(K)$. Match the following.

Column I

(A) $K = 2$

(B) $K = 3$

(C) $K = 0$

(D) $K = 5$

Column II

(p) $P(K) = \frac{1}{18}$

(q) $P(K) = \frac{1}{6}$

(r) $P(K) = \frac{2}{9}$

(s) $P(K) = \frac{4}{9}$

Sol. Answer A(r), B(q), C(q), D(p)

(A) $K = 2$, $E = \{(1, 3), (3, 1), (2, 4), (4, 2), (3, 5), (5, 3), (4, 6), (6, 4)\}$

$$n(S) = 36$$

$$P(E) = \frac{8}{36} = \frac{2}{9}$$

$$(B) K = 3, E = \{(1, 4), (4, 1), (2, 5), (5, 2), (3, 6), (6, 3)\}$$

$$P(E) = \frac{6}{36} = \frac{1}{6}$$

$$(C) K = 0, E = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

$$P(E) = \frac{6}{36} = \frac{1}{6}$$

$$(D) K = 5, E = \{(1, 6), (6, 1)\}$$

$$P(E) = \frac{2}{36} = \frac{1}{18}$$

2. Let us consider $n = 2^8 \times 3^6 \times 5^{10} \times 6^8 \times 10^5$. A factor (positive) of n is chosen at random.

Then match the statement given in column I with the corresponding probability given in column II.

Column I

Column II

(A) The probability of a factor chosen at random is a perfect square, is

(p) $\frac{19 \times 7 \times 13}{22 \times 15 \times 8}$

(B) The probability of a factor chosen at random is a perfect cube, is

(q) $\frac{1}{22}$

(C) The probability of a factor chosen at random is a perfect fourth power is

(r) $\frac{1}{55}$

(D) The probability of a factor chosen at random is divisible by 1800

(s) $\frac{2}{15}$

Sol. Answer A(s), B(q), C(r), D(p)

We have,

$$\begin{aligned} n &= 2^8 \times 3^6 \times 5^{10} \times 6^8 \times 10^5 \\ &= 2^8 \times 3^6 \times 5^{10} \times 2^8 \times 3^8 \times 2^5 \times 5^5 \\ &= 2^{21} \times 3^{14} \times 5^{15} \end{aligned}$$

$\Rightarrow$ The number of possible positive factor = $22 \times 15 \times 16$.

(A) Number of factors which are perfect squares = $11 \times 8 \times 8$

$$\Rightarrow P(A) = \frac{11 \times 8 \times 8}{22 \times 15 \times 16} = \frac{2}{15}$$

(B) Number of factors which are perfect cubes = $8 \times 5 \times 6$

$$\text{Required probability} = \frac{8 \times 5 \times 6}{22 \times 15 \times 16} = \frac{1}{22}$$

(C) Number of factors which are perfect fourth powers = $6 \times 4 \times 4$

$$\Rightarrow P(C) = \frac{6 \times 4 \times 4}{22 \times 15 \times 16} = \frac{1}{55}$$

(D) Number of factors which are divisible by 1800 = $19 \times 14 \times 13$

$$\Rightarrow P(D) = \frac{19 \times 14 \times 13}{22 \times 15 \times 16}$$

3. Three dice are rolled and sum of the number of appeared is x . Match the following.

Column I

Column II

(A) $x = 8$, then $P(x)$

(p) $\frac{10}{216}$

(B) $x = 9$, then $P(x)$

(q) $\frac{21}{216}$

(C) $x = 10$, then $P(x)$

(r) $\frac{27}{216}$

(D) $x = 15$, then $P(x)$

(s) $\frac{25}{216}$

Sol. Answer A(q), B(s), C(r), D(p)

We have,

$$P(S) = 6^3 = 216$$

$$\begin{aligned} \text{(A) } n(x=8) &= \text{Coeff. of } x^8 \text{ in } (x + x^2 + \dots + x^6)^3 \\ &= \text{Coeff. of } x^5 \text{ in } (1 - x^6)^3 (1 - x)^{-3} \\ &= \text{Coeff. of } x^5 \text{ in } (1 - 3x^6 + 3x^{12} - x^{18}) (1 + 3x + {}^4C_2 x^2 + {}^5C_3 x^3 + \dots) \\ &= {}^7C_5 = 21 \end{aligned}$$

$$\Rightarrow P(x=8) = \frac{21}{216}$$

(B) The number of ways that the sum of faces is 9 when three dice are rolled = 25

$$\Rightarrow \text{Required probability} = \frac{25}{216}$$

(C) The number of ways that the sum of faces is 10 when three dice are rolled = 27

$$\Rightarrow \text{Required probability} = \frac{27}{216}$$

(D) The number of ways that the sum of faces is 15 when three dice are rolled = 10

$$\Rightarrow \text{Required probability} = \frac{10}{216}$$

4. Match the following.

Column I

Column II

(A) The probability of throwing largest number with a single die in two trials is

(p) $\frac{1120}{3003}$

(B) A person draws a card from a pack, replaces it and shuffles the pack till he draws a spade. The probability that he will have to make at least three trials is

(q) $\frac{11}{36}$

(C) In question (B), the probability that he will have to make exactly three trials

(r) $\frac{9}{16}$

(D) 6 balls are drawn one by one from a bag containing 8 white and 6 black balls. The probability that 3 white and 3 black balls are drawn is

(s) $\frac{9}{64}$

(t) $\frac{{}^6C_3 \times {}^8C_3}{{}^{14}C_6}$

Sol. Answer A(q), B(r), C(s), D(p, t)

(A) The probability of not throwing six (the largest number on die) in one trial = $\frac{5}{6}$

$$\text{The probability of not throwing ace in 2nd trial} = \left(\frac{5}{6}\right)\left(\frac{5}{6}\right)$$

$$\text{Required probability} = 1 - \left(\frac{5}{6}\right)^2 = \frac{11}{36}$$

$$(B) \text{ Required probability} = \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} + \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} + \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} + \dots = \frac{\frac{9}{64}}{1 - \frac{3}{4}} = \frac{9}{16}$$

$$(C) \text{ Required probability} = \left(\frac{3}{4}\right)\left(\frac{3}{4}\right)\left(\frac{1}{4}\right) = \frac{9}{64}$$

$$(D) \text{ Required probability} = \frac{{}^6C_3 \times {}^8C_3}{{}^{14}C_6} = \frac{1120}{3003}$$

5. A box contains 3 red, 4 black and 5 white balls. Three balls are drawn at random. Then the probability that

Column I

(A) 2 are black and 1 is white

(B) 1 is red, 1 is white and 1 is black

(C) No ball is white

(D) All are of same colour

Column II(p) $\frac{3}{11}$ (q) $\frac{7}{44}$ (r) $\frac{3}{44}$ (s) $\frac{3}{22}$

Sol. Answer A(s), B(p), C(q), D(r)

$$n(S) = {}^{12}C_3$$

$$(A) \quad n(E) = {}^4C_2 \times {}^5C_1$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{{}^4C_2 \times {}^5C_1}{{}^{12}C_3} = \frac{30}{220} = \frac{3}{22}$$

$$(B) \quad n(E) = {}^3C_1 \times {}^4C_1 \times {}^5C_1$$

$$P(E) = \frac{n(E)}{n(S)} = \frac{{}^3C_1 \times {}^4C_1 \times {}^5C_1}{{}^{12}C_3} = \frac{60}{220} = \frac{3}{11}$$

(C) Method I :

$$n(E) = {}^3C_3 + {}^3C_2 \times {}^4C_1 + {}^3C_1 \times {}^4C_2 + {}^4C_3 = 1 + 12 + 18 + 4 = 35$$

$$P(E) = \frac{35}{220} = \frac{7}{44}$$

OR Method II :

$$n(E) = {}^{12}C_3 - {}^5C_1 \times {}^7C_2 - {}^5C_2 \times {}^7C_1 - {}^5C_3$$

$$= {}^{12}C_3 - 105 - 70 - 10$$

$$= 220 - 185 = 35$$

$$P(E) = \frac{35}{220} = \frac{7}{44}$$

$$(D) \quad n(E) = {}^3C_3 + {}^4C_3 + {}^5C_3$$

$$= 1 + 4 + 10 = 15$$

$$P(E) = \frac{15}{220} = \frac{3}{44}$$

6. S is a set containing n elements. A subset X of S is chosen at random and the set S is reconstructed by replacing the elements of X . A subset Y of S is now chosen at random. Match the events with their probabilities. ($|A|$ denotes the cardinality of set A)

Column-I
(Events)

Column-II
(Probability)

(A) $X = Y$

(p) $\left(\frac{3}{4}\right)^n$

(B) $X \cup Y = S$

(q) $\frac{{}^{2n}C_{n-1}}{4^n}$

(C) $|X| = |Y|$

(r) $\frac{1}{2^n}$

(D) $|Y| = |X| + 1$

(s) $\frac{{}^{2n}C_n}{4^n}$

Sol. Answer : A(r), B(p), C(s), D(q)

(A) number of subsets of ' S ' = 2^n

$$P(X = Y) = \frac{1}{2^n}$$

(B) $X \cup Y = S$

Let number of elements in $X = r$.

R	x no. of ways to select	Y	Corresponding probability
0	1	$1 = 2^0$	$\frac{{}^nC_0 \times 2^0}{2^n}$
1	${}^nC_{n-1}$	$2 = 2^1$	$\frac{{}^nC_1 \times 2^1}{2^n}$
2	${}^nC_{n-2}$	${}^2C_0 + {}^2C_1 + {}^2C_2 = 2^2$	$\frac{{}^nC_2 \times 2^2}{2^n}$
3	${}^nC_{n-3}$	${}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3 = 2^3$	
.....			
N	nC_0	2^n	$\frac{{}^nC_n \cdot 2^n}{2^n}$

$$\begin{aligned}
 \text{Required probability} &= \frac{\frac{{}^nC_0 \times 2^0}{2^n} + \frac{{}^nC_1 \times 2^1}{2^n} + \dots + \frac{{}^nC_n 2^n}{2^n}}{2^n} \\
 &= \frac{{}^nC_0 + {}^nC_1 2 + \dots + {}^nC_n 2^n}{2^n \cdot 2^n} \\
 &= \frac{3^n}{4^n} = \left(\frac{3}{4}\right)^n
 \end{aligned}$$

(C) $|X| = |Y|$. Let r be the number, of the element in X .

R	X	Y
0	nC_0	nC_0
1	nC_1	nC_1
N	nC_n	nC_n

$$\begin{aligned}
 \text{Required probability} &= \frac{\frac{{}^nC_0 \cdot {}^nC_0}{2^n} + \frac{{}^nC_1 \cdot {}^nC_1}{2^n} + \dots + \frac{{}^nC_n \cdot {}^nC_n}{2^n}}{2^n} \\
 &= \frac{({}^nC_0)^2 + ({}^nC_1)^2 + \dots + ({}^nC_n)^2}{4^n} \\
 &= \frac{{}^{2n}C_n}{4^n}
 \end{aligned}$$

(D) r = number of element in X .

R	No. of ways to select X	No. of ways to select Y
0	nC_0	nC_1
1	nC_0	nC_2
...	...	...
...	...	...
...	...	...
$n-1$	${}^nC_{n-1}$	nC_n

$$\begin{aligned}
 \therefore \text{ Required probability} &= \frac{{}^nC_0 \cdot {}^nC_1}{2^n} + \frac{{}^nC_1 \cdot {}^nC_2 + \dots + {}^nC_{n-1} \cdot {}^nC_n}{2^n} \\
 &= \frac{{}^nC_0 \cdot {}^nC_1 + {}^nC_1 \cdot {}^nC_2 + \dots + {}^nC_{n-1} \cdot {}^nC_n}{4^n} \\
 &= \frac{{}^{2n}C_{n-1}}{4^n}
 \end{aligned}$$

7. In a monthly test, 25% students failed in mathematics, 15% failed in physics and 10% failed in both mathematics and physics. A student is selected at random.

Column-I

Column-II

(A) The probability that the student failed mathematics or physics is

(p) $\frac{3}{10}$

(B) The probability that the student failed neither mathematics nor physics is

(q) $\frac{7}{10}$

(C) If the student failed mathematics, the probability that he or she failed physics is

(r) $\frac{2}{5}$

(D) If the student failed physics, the probability that he or she failed mathematics is

(s) $\frac{2}{3}$

Sol. Answer : A(p), B(q), C(r), D(s)

Let M be the set of students who failed in mathematics and P be the set of students who failed in physics.

Then $P(M) = 0.25$, $P(\text{Phy}) = 0.15$, $P(M \cap P) = 0.10$

(A) The probability that the student failed in mathematics or physics

$$= P(M \cup P) = P(M) + P(P) - P(M \cap P)$$

$$= 0.25 + 0.15 - 0.10 = 0.30 = \frac{3}{10}$$

(B) The probability that the student failed neither in mathematics nor in Physics

$$= P(M \cup P)' = 1 - \frac{3}{10} = \frac{7}{10}$$

$$(C) \quad P\left(\frac{M \cap P}{M}\right) = \frac{P(M \cap P)}{P(M)} = \frac{0.10}{0.25} = \frac{2}{5}$$

$$(D) \quad P\left(\frac{M \cap P}{P}\right) = \frac{P(M \cap P)}{P(P)} = \frac{0.10}{0.15} = \frac{2}{3}$$

8. Match the following

Column-I

Column-II

(A) Let E_1 and E_2 be two events such that

(p) $\frac{1}{4}$

 $E_2 \subset E_1$. Then $P\left(\frac{E_1}{E_2}\right)$ is(B) If two events E_1 and E_2 are such that

(q) $\frac{5}{6}$

$P(E_1^c) = 0.3$, $P(E_2^c) = 0.4$ and $P(E_1 E_2^c) = 0.5$

then $P\left(\frac{E_2}{E_1 \cup E_2^c}\right)$ is(C) If $P(E_1 \cup E_2) = 0.9$, $P(E_2) = 0.4$ and

(r) 1

 E_1 and E_2 are independent events then $P(E_1)$ equals(D) Let E_1 and E_2 be two independent events

(s) Odds in favour of getting a multiple of 5 when a ticket is drawn from a bag containing 25

and $P(E_1 \cap E_2) = \frac{1}{12}$ and the probabilitythat neither E_1 nor E_2 happens is $\frac{1}{2}$,

tickets numbered 1 to 25.

then $P(E_1)$ or $P(E_2)$ is

(t) $\frac{1}{3}$

Sol. Answer : A(r), B(p, s), C(q), D(p, t)

(A) $P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{P(E_2)}{P(E_2)} = 1$ as $E_2 \subset E_1$

(B) $P(E_1) = 0.3$, $P(E_2^c) = 0.4$

$\Rightarrow P(E_1) = 0.7$

$P(E_1 E_2^c) = P(E_1) - P(E_1 E_2)$

$\Rightarrow P(E_1 E_2) = 0.7 - 0.5 = 0.2$

$$P\left(\frac{E_2}{E_1 \cup E_2^c}\right) = \frac{P(E_2 \cap (E_1 \cup E_2^c))}{P(E_1 \cup E_2^c)} = \frac{P(E_1 \cap E_2)}{P(E_1) + P(E_2^c) - P(E_1 E_2^c)} = \frac{0.2}{0.8} = \frac{1}{4}$$

$$(C) P(E_1 \cup E_2) = 0.9 = P(E_1) + 0.4 - 0.4P(E_1)$$

$$\Rightarrow P(E_1) = \frac{0.5}{0.6} = \frac{5}{6}$$

$$(D) P(E_1 \cap E_2) = P(E_1)P(E_2) = \frac{1}{12} \text{ and } (1 - P(E_1))(1 - P(E_2)) = \frac{1}{2}$$

$$\Rightarrow P(E_1) = \frac{1}{4} \text{ or } \frac{1}{3} \text{ and } P(E_2) = \frac{1}{3} \text{ or } \frac{1}{4}$$

9. Words are made using all letters of the word INDEPENDENCE and a word is chosen at random.

Column-I

(A) Probability that vowels occupy even places is not less than

(B) Probability that all vowels do not occupy even places is not less than

(C) Probability that vowels are together is not less than

(D) Probability that all vowels are not together is not less than

Column-II

(p) $\frac{1}{2}$

(q) $\frac{1}{4}$

(r) $\frac{3}{4}$

(s) 0

(t) 1

Sol. Answer : A(s), B(p, q, r, s), C(s), D(p, q, r, s)

I 1 N 3 D 2 E 5 P 1 C 1

$$\text{Total words} = \frac{13!}{5!3!2!} = 4324320$$

$$P(\text{All vowels occupy even places}) = \frac{6 \times 7!}{3!2! \times 4324320} = \frac{1}{1716}$$

$$P(\text{All vowels do not occupy even places}) = \frac{1715}{1716}$$

$$P(\text{vowels are together}) = \frac{8 \times 6 \times 7!}{3! \times 2! \times 4324320} = \frac{2}{429}$$

$$P(\text{All vowels are not together}) = \frac{427}{429}$$

10. In a binomial distribution $\left(\frac{3}{5} + \frac{2}{5}\right)^6$ probability of success of any event is $\frac{2}{5}$ then

Column-I

Column-II

(A) Mean is

(p) $\frac{864}{3125}$

(B) Variance is

(q) $\frac{12}{5}$

(C) Standard deviation is

(r) $\frac{6}{5}$ (D) $P(x = 3)$ is(s) $\frac{36}{25}$ (t) $\frac{6}{25}$

Sol. Answer : A(q), B(s), C(r), D(p)

The binomial distribution is $(q + p)^n$

Here $q = \frac{3}{5}$; $p = \frac{2}{5}$; $n = 6$

(A) Hence mean = $np = 6 \times \frac{2}{5} = \frac{12}{5}$

(B) Variance = $npq = 6 \times \frac{2}{5} \times \frac{3}{5} = \frac{36}{25}$

(C) Standard deviation = $\sqrt{npq} = \sqrt{\frac{36}{25}} = \frac{6}{5}$

(D) $P(x = 3) = {}^6C_3 \left(\frac{2}{5}\right)^3 \cdot \left(\frac{3}{5}\right)^3$

$$= \frac{6 \times 5 \times 4}{3 \times 2 \times 1} \times \frac{8}{5 \times 5 \times 5} \times \frac{27}{5 \times 5 \times 5}$$

$$= \frac{864}{3125}$$

SECTION - F

Integer Answer Type Questions

1. All possible triangles are formed by joining the vertices of decagon. The probability that all the three sides of triangle are also diagonal of decagon is a/b (no factor of a and b is common) then a is _____.

Sol. Answer (5)

$$n(S) = {}^{10}C_3 = 120$$

Number of triangle having two side common = 10

Number of triangle having one side common = $10 \times 6 = 60$

$$n(E) = {}^{10}C_3 - 10 - 60 = 50$$

$$p(E) = 50/120 = 5/12$$

2. Letters are arranged in all possible ways of the word 'DISCRETE MATHEMATICS'. One word is selected at random then the probability that all T are at maximum distance is a/b (a and b has no common factor) then a is _____.

Sol. Answer (1)

DISCRETE MATHEMATICS

D	–	1
I	–	2
S	–	2
C	–	2
R	–	1
E	–	3
T	–	3
M	–	2
A	–	2
H	–	1

$$n(E) = \frac{16!}{(2!)^5 \cdot (3!)}$$

$$n(S) = \frac{19!}{(2!)^5 \cdot (3!)^2}$$

$$p(E) = \frac{3}{19 \times 18 \times 17} = \frac{1}{1938}$$

3. Two different numbers are selected at random from the set $S = \{1, 2, 3, \dots, 10\}$, then the probability that sum of selected numbers is divisible by 2 is a/b , where a and b are co-prime then b is equal to _____.

Sol. Answer (9)

For $x + y$ to be divisible, either both x and y are odd or even

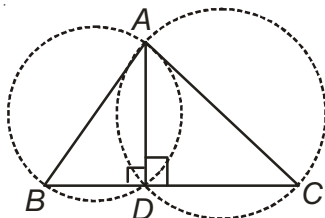
$$\therefore n(E) = {}^5C_2 + {}^5C_2$$

$$n(S) = {}^{10}C_2$$

$$P(E) = \frac{{}^5C_2 + {}^5C_2}{{}^{10}C_2} = \frac{20}{45} = \frac{4}{9}$$

4. Consider a triangle ABC . Circles are drawn with AB and AC as diameters, if p be the probability that two circles intersect on BC or BC produced, then $4p$ is equal to _____.

Sol. Answer (4)



Let ABC be an acute angle triangle.

Here D is the foot of perpendicular from A upon BC .

Clearly, the two circles intersect at D .

Similarly it can be shown that the circles meet on the sides or sides produced so, $p = 1$

$$\Rightarrow 4p = 4$$

5. A rack has 5 different pairs of shoes. The probability that 4 shoes are chosen from it and there is no complete pair is p , then $\left\lceil \frac{1}{p} \right\rceil$ is equal to, where $[\cdot]$ denotes the greatest integer function _____.

Sol. Answer (2)

$$n(S) = {}^{10}C_4 = 210$$

$$n(E) = \frac{10 \times 8 \times 6 \times 4}{4!} = 80$$

$$p = \frac{80}{210}; \quad \frac{1}{p} = \frac{210}{80} \Rightarrow \left\lceil \frac{1}{p} \right\rceil = 2$$

6. Ten equally spaced points are lying on a circle, if p is the probability that a chosen chord is diameter of the circle, then $18p + 2$ is equal to _____.

Sol. Answer (4)

As the ten points are equally spaced.

Any two points will give us a chord

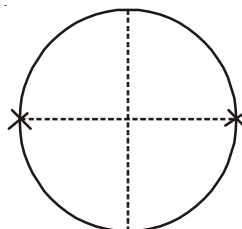
$$\text{So, total number of chord(s)} = {}^{10}C_2 = 45$$

$$\text{Total number of chords which are diameter} = 5$$

$$\text{So, } p = \frac{5}{45} = \frac{1}{9}$$

$$\Rightarrow 9p = 1$$

$$\Rightarrow 18p + 2 = 4$$



7. Let p be the probability that sum of digits of a five digit number is even then $4p$ is equal to _____.

Sol. Answer (2)

Total number of 5 digit number = 9×10^4

Total number of 5 digit number whose sum is even is = $\frac{9 \times 10^4}{2}$

$$p = \frac{1}{2} \Rightarrow 4p = 2$$

8. Six persons go to a birthday party. They leave their top-coats in the lounge and pick them while returning back.

The probability that exactly one person picks up his own top-coat, is p , then $\left[\frac{1}{p}\right]$ is equal to _____.

Sol. Answer (2)

Total number of ways in which they can pick up their top-coat is $6!$ ways.

$n(E)$ = Total number of ways in which exactly one gets his own top-coat, while others do not get theirs.

$$= {}^6C_1 D_5; \text{ where } D \text{ refers to Derray. i.e., } D_5 = 5! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right]$$

$$= 264$$

$$p = \frac{264}{6!}$$

$$\Rightarrow \frac{1}{p} = \frac{6!}{264} \Rightarrow \left[\frac{1}{p} \right] = 2$$

9. A and B are said to be independent if $P(A \cap B) = P(A) \cdot P(B)$. If B is independent with every subset of sample space then the value of $6 P(B)$ is _____

Sol. Answer (6)

B is independent with every element only if B is sample space

$$\therefore P(B) = 1$$

$$\therefore 6 \cdot P(B) = 6$$

10. If the probability that two queens, placed at random on a chess-board, do not take on each other, is $\frac{23K}{6}$, then $12K$ equals _____

Sol. Answer (2)

Queen will be facing each other in case both are placed on same column, same row or same diagonal of chess-board. There are eight rows, eight columns, two principal diagonals and four sets each containing diagonal lines of 7, 6, 5, 4, 3, 2, 1 squares.

Hence, probability of queens taking on each other is

$$= \frac{18^8 C_2 + 4 \{ {}^7 C_2 + {}^6 C_2 + \dots + {}^2 C_2 \}}{{}^{64} C_2} = \frac{13}{36}$$

Hence, probability that two queens do not take on each other, is

$$P(A) = 1 - \frac{13}{36} = \frac{23}{36} = \frac{1}{6} \left[\frac{23}{6} \right]$$

$$\Rightarrow K = \frac{1}{6}, \text{ hence, } 12K = \boxed{2}$$

11. Each of n urns contains 4 white and 6 black balls. The $(n + 1)^{\text{th}}$ urn contains 5 white and 5 black balls. One of the $(n + 1)$ urns is chosen at random and two balls are drawn from it without replacement. Both balls turn out to be black. If probability that $(n + 1)^{\text{th}}$ urn was chosen to draw the balls is $\frac{1}{16}$, then value of $(n - 5)$ equals _____.

Sol. Answer (5)

Let E_1 denotes event that one of first n urns is chosen and E_2 denotes event $(n + 1)^{\text{th}}$ urn is selected. A denotes event that two balls drawn are black. Then,

$$P(E_1) = \frac{n}{n+1}, \quad P(E_2) = \frac{1}{n+1}$$

$$P(A/E_1) = \frac{{}^6C_2}{{}^{10}C_2} = \frac{1}{3}; \quad P(A/E_2) = \frac{{}^5C_2}{{}^{10}C_2} = \frac{2}{9}$$

Using Baye's Theorem, required probability,

$$\begin{aligned} &= P(E_2/A) = \frac{P(E_2) \cdot P(A/E_2)}{P(E_1) P(A/E_1) + P(E_2) P(A/E_2)} \\ \Rightarrow \frac{1}{16} &= \frac{\frac{1}{n+1} \cdot \frac{2}{9}}{\frac{1}{n+1} \cdot \frac{1}{3} + \frac{1}{n+1} \cdot \frac{2}{9}} \Rightarrow n = 10 \Rightarrow n - 5 = \boxed{5} \end{aligned}$$

12. Of the three independent events E_1 , E_2 and E_3 , the probability that only E_1 occurs is α , only E_2 occurs is β and only E_3 occurs is γ . Let the probability p that none of events E_1 , E_2 or E_3 occurs satisfy the equations $(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$. All the given probabilities are assumed to lie in the interval $(0, 1)$.

Then $\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3} =$

[JEE(Advanced) 2013]

Sol. Answer (6)

$$P(E_1) P(\bar{E}_2) P(\bar{E}_3) = \alpha \quad \dots(i)$$

$$P(\bar{E}_1) P(E_2) P(\bar{E}_3) = \beta \quad \dots(ii)$$

$$P(\bar{E}_1) P(\bar{E}_2) P(E_3) = \gamma \quad \dots(iii)$$

$$P(\bar{E}_1) P(\bar{E}_2) P(\bar{E}_3) = p \quad \dots(iv)$$

Divide (i) by (iv),

$$\frac{P(E_1)}{P(\bar{E}_1)} = \frac{\alpha}{\rho} = \frac{\alpha}{\frac{\alpha\beta}{\alpha-2\beta}} = \frac{\alpha-2\beta}{\beta}$$

$$\Rightarrow \frac{P(\bar{E}_1)}{P(E_1)} = \frac{\beta}{\alpha-2\beta}$$

$$\Rightarrow \frac{1-P(E_1)}{P(E_1)} = \frac{\beta}{\alpha-2\beta}$$

$$\Rightarrow \frac{1}{P(E_1)} - 1 = \frac{\beta}{\alpha-2\beta}$$

$$\Rightarrow \frac{1}{P(E_1)} = \frac{\alpha-\beta}{\alpha-2\beta}$$

$$\Rightarrow P(E_1) = \frac{\alpha-2\beta}{\alpha-\beta} \quad \dots(v)$$

Now, $\frac{\alpha\beta}{\alpha-2\beta} = \frac{2\beta\gamma}{\beta-3\gamma}$

$$\Rightarrow \frac{\alpha}{\alpha-2\beta} = \frac{2\gamma}{\beta-3\gamma}$$

$$\Rightarrow \alpha\beta - 3\gamma\alpha = 2\gamma\alpha - 4\beta\gamma$$

$$\Rightarrow \alpha\beta = 5\gamma\alpha - 4\beta\gamma$$

$$\gamma = \frac{\alpha\beta}{5\alpha - 4\beta} \quad \dots(vi)$$

Divide (iii) by (iv)

$$\frac{P(E_3)}{P(\bar{E}_3)} = \frac{\gamma}{\rho} = \frac{\gamma}{\frac{\alpha\beta}{\alpha-2\beta}} = \frac{\gamma(\alpha-2\beta)}{\alpha\beta}$$

$$\frac{P(E_3)}{P(\bar{E}_3)} = \frac{\alpha-2\beta}{5\alpha-4\beta}$$

$$\frac{P(\bar{E}_3)}{P(E_3)} = \frac{5\alpha-4\beta}{\alpha-2\beta}$$

$$\frac{1-P(E_3)}{P(E_3)} = \frac{5\alpha-4\beta}{\alpha-2\beta}$$

$$\frac{1}{P(E_3)} = \frac{6\alpha-6\beta}{\alpha-2\beta} = \frac{6(\alpha-\beta)}{\alpha-2\beta}$$

$$P(E_3) = \frac{\alpha - 2\beta}{6(\alpha - \beta)}$$

$$\frac{P(E_1)}{P(E_3)} = \frac{\frac{\alpha - \beta}{6(\alpha - \beta)}}{\frac{\alpha - 2\beta}{6(\alpha - \beta)}} = 6$$

13. The minimum number of times a fair coin needs to be tossed, so that the probability of getting at least two heads is at least 0.96, is **[JEE(Advanced) 2015]**

Sol. Answer (8)

$$P(2) + P(3) + \dots \geq 0.96$$

$$1 - P(0) - P(1) \geq 0.96$$

$$1 - 0.96 \geq {}^nC_0 \left(\frac{1}{2}\right)^n + {}^nC_1 \left(\frac{1}{2}\right)^n$$

$$\Rightarrow (0.04)2^n \geq 1 + n$$

$$\Rightarrow 4 \cdot 2^n \geq (n + 1)100$$

$$\Rightarrow 2^{n+2} \geq (n + 1)100$$

$$\text{as } 2^{10} \geq 900$$

$$\Rightarrow n = 8$$

SECTION - G

Multiple True-False Type Questions

1. From set $S = \{1, 2, 3, 4, 5, 6\}$ three numbers are drawn without replacement.

STATEMENT-1 : The probability that second number is greater than first is $\frac{1}{2}$.

STATEMENT-2 : The probability that numbers are in increasing A.P. is $\frac{7}{60}$.

STATEMENT-3 : The probability that numbers are in ascending order is $\frac{1}{6}$.

(1) F F F

(2) T T F

(3) T F T

(4) F F T

Sol. Answer (3)

Statement-1: Total number of ways of selecting number in such a way that second number is greater than first

$$= 4 + 8 + 12 + 16 + 20 = 60$$

$$\text{Probability} = \frac{60}{6 \times 5 \times 4} = \frac{1}{2} \quad (\text{True})$$

Statement-2: Required number of ways = 6

$$\text{Required probability} = \frac{6}{6 \times 5 \times 4} = \frac{1}{20} \quad (\text{False})$$

$$\text{Statement-3 : Required probability} = \frac{{}^6C_3}{{}^6C_3 \times 3!} = \frac{1}{6} \quad (\text{True})$$

2. STATEMENT-1 : A coin is tossed 16 times. For the first 15 consecutive times, it shows heads. The probability that it will show heads in the 16th toss is less than $\frac{1}{2}$.

STATEMENT-2 : The probability that a heart will be drawn from a pack of well shuffled cards is $\frac{1}{4}$.

STATEMENT-3 : The probability that a king will be drawn from a pack of well shuffled pack of cards is $\frac{1}{13}$.

(1) T T F

(2) T F T

(3) F F T

(4) F T T

Sol. Answer (4)

Statement 1 : Since the tosses are independent, so the probability of a head is $\frac{1}{2}$.

Statement 2, 3 : They are obvious

3. STATEMENT-1 : The probability that a natural number chosen from the set $\{1, 2, 3, \dots, 100\}$ will satisfy the

$$\text{equation } \left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x}{3} \right\rfloor = \frac{5x}{6} \text{ is } \frac{4}{25}.$$

STATEMENT-2 : The probability that a chosen number from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ will be prime is $\frac{2}{5}$.

STATEMENT-3 : In a game of dice which A and B play alternately and A starts first, the probability that A gets 5 with a throw of two dice before B gets 7 is $\frac{3}{7}$.

(1) T T T

(2) F T T

(3) F T F

(4) F F F

Sol. Answer (1)

Statement 1 : Clearly x must be a multiple of 6

$$\text{So, } n(E) = 16; n(S) = 100$$

$$P(E) = \frac{16}{100} = \frac{4}{25}$$

Statement 2 : Clearly $P(E) = \frac{4}{10} = \frac{2}{5}$

Statement 3 : Required probability $= \frac{1}{9} + \frac{8}{9} \times \frac{5}{6} \times \frac{1}{9} + \frac{8}{9} \times \frac{5}{6} \times \frac{8}{9} \times \frac{5}{6} \times \frac{1}{9} + \dots \infty$

$$= \frac{1}{9} \left(1 + \left(\frac{8 \times 5}{54} \right) + \left(\frac{8 \times 5}{54} \right)^2 + \dots \infty \right)$$

$$= \frac{1}{9} \times \frac{1}{1 - \frac{40}{54}} = \frac{1}{9} \times \frac{54}{14} = \frac{3}{7}$$

4. STATEMENT-1 : In a double decker bus consisting of 5 seats on the ground floor and 5 on the first floor, the chance that two old person will take the ground floor and three children will take first floor while the five

youth may take any floor is $\frac{{}^5C_2 \times 2! \times {}^5C_3 \times 3! \times 5!}{10!}$.

STATEMENT-2 : Two person each make a single throw with a pair of dice. The chance that the throws are

equal is coefficient of x^{10} in $\frac{\left(\frac{1-x^6}{1-x} \right)^4}{(36)^2}$.

STATEMENT-3 : Given that $x + y = 2a$, where a is constant and that all values of x between 0 and $2a$ are equally likely, then the chance that $xy > \frac{3}{4}a^2$ is $\frac{1}{2}$.

(1) T T T

(2) F T T

(3) F F T

(4) F T F

Sol. Answer (1)

Statement 1 : $n(S) = 10!$

$$n(E) = {}^5C_2 \times 2! \times {}^5C_3 \times 3! \times 5!$$

$$\text{Hence } P(E) = \frac{{}^5C_2 \times 2! \times {}^5C_3 \times 3! \times 5!}{10!}$$

Statement 2 : $x_1 + x_2 = k$

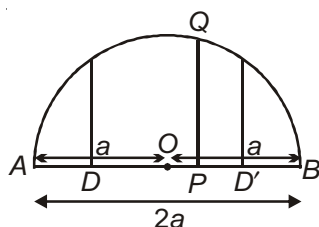
Now k varies from 2 to 12

$$= \text{coefficient of } x^{10} \text{ in } \frac{\left(\frac{1-x^6}{1-x} \right)^4}{(36)^2}$$

Statement 3 : Clearly $AP \cdot PB = PQ^2$

Clearly $DD' = a$

$$P(E) = \frac{a}{2a} = \frac{1}{2}$$



5. STATEMENT-1 : The chance of casting 5 with a throw of two dice is $\frac{2}{9}$.

STATEMENT-2 : The chance that the face of a die will show an odd number given that the number that was observed on the face of the die was prime is $\frac{1}{2}$.

STATEMENT-3 : Five balls are drawn from a bag containing 6 white and 4 black balls. The chance that 3 white and 2 black balls are drawn is $\frac{10}{21}$.

(1) T F F

(2) T T F

(3) F F T

(4) F F F

Sol. Answer (3)

Statement 1 : The total number of ways of getting 5 from the throw of two dice is
 $= \{(1, 4), (2, 3), (3, 2), (4, 1)\}$

$$\text{So, } P(E) = \frac{4}{36} = \frac{1}{9}$$

Statement 2 : $n(S) = 3$

$$n(E) = 2$$

$$P(E) = \frac{2}{3}$$

Statement 3 : $P(E) = \frac{n(E)}{n(S)} = \frac{{}^6C_3 \cdot {}^4C_2}{{}^{10}C_5} = \frac{10}{21}$

6. STATEMENT-1 : In a single cast with three dice, the chance of throwing four-five-six is $\frac{1}{36}$.

STATEMENT-2 : In a single cast with three dice, the chance that the sum of numbers will be 11 is $\frac{1}{8}$.

STATEMENT-3 : In a single cast with three dice, the chance that the sum of numbers appearing is more than 10 is $\frac{1}{2}$.

(1) T T T

(2) T F F

(3) F T T

(4) F F T

Sol. Answer (1)

Statement 1 : $P(E) = \frac{n(E)}{n(S)} = \frac{3!}{6^3} = \frac{1}{36}$

Statement 2 : $n(E)$ = Coefficient of x^{11} in $(x + x^2 + x^3 + \dots + x^6)^3 = 27$

$$P(E) = \frac{27}{6^3} = \frac{1}{8}$$

Statement 3 : Total possibilities of sum are 3 to 18.

$$n(S) = 6 \times 6 \times 6 = 216$$

Out of which 108 will give sum from 3 to 10 and 108 will give sum 11 to 18.

e.g., sum 3 $\Rightarrow$ (1, 1, 1)

Sum 18 (6, 6, 6) and so on.

$$\therefore \text{Required probability} = \frac{1}{2}$$

7. STATEMENT-1 : An urn contains m white and n black balls. All the balls except for one ball, are drawn from it. The probability that the last ball remaining in the urn is white is equal to $\frac{m}{m+n}$.

STATEMENT-2 : An urn contains m white balls and n black balls. Balls are drawn one by one till all the balls are drawn. The probability that the second drawn ball is white is $\frac{n}{m+n}$.

STATEMENT-3 : An urn contains 6 balls, 3 white and 3 black ones, a person selects at random an even number of balls (all the different ways of drawing an even number of balls are considered equally probable, irrespective of their number). Then the probability that there will be the same number of black and white balls among them is $\frac{11}{25}$.

(1) T T T

(2) F T T

(3) T F T

(4) F F F

Sol. Answer (3)

STATEMENT-1

$$n(S) = {}^{m+n}C_{m+n-1}; n(A) = {}^mC_{m-1} \cdot {}^nC_n$$

$$\Rightarrow P = \frac{m}{m+n}$$

STATEMENT-2

E = Event that 2nd drawn ball is white

$$P(E) = P(BW \text{ or } WW)$$

$$= \frac{n}{m+n} \cdot \frac{m}{m+n-1} + \frac{m}{m+n} \cdot \frac{m-1}{m+n-1}$$

$$= \frac{m\{n+m-1\}}{(m+n)(m+n-1)}$$

$$= \frac{m}{m+n}$$

STATEMENT-3

Total number of possible cases = 3 (either 2 or 4 or 6 are drawn)

$$\begin{aligned} \text{Hence required probability} &= \frac{1}{3} \left(\frac{{}^3C_1 \times {}^3C_1}{{}^6C_2} + \frac{{}^3C_2 \times {}^3C_2}{{}^6C_4} + \frac{{}^3C_3 \times {}^3C_3}{{}^6C_6} \right) \\ &= \frac{11}{15} \end{aligned}$$

8. STATEMENT-1 : Three natural number are taken at random from the set $A = \{x : 1 < x \leq 100, x \in \mathbb{N}\}$ the probability that the A.M of the numbers taken is 25 is equal to $\frac{{}^{74}C_2}{{}^{100}C_3}$.

STATEMENT-2 : Let $A = \{2, 3, 4 \dots 20\}$. A number is chosen at random from set A and it is bound to be a prime number. The probability that it is more than 10 is $\frac{1}{5}$.

STATEMENT-3 : The probability of three person having the same date and month for the birthday is $\frac{1}{(365)^2}$.

(1) T T T

(2) F F T

(3) T F T

(4) F F F

Sol. Answer (1)

STATEMENT-1

Since the A.M. of three numbers is 25

$\therefore$ Their sum = 25

$$x_1 + x_2 + x_3 = 75 ; x_1 \geq 1, x_2 \geq 1, x_3 \geq 1$$

$$\Rightarrow {}^{75-1}C_{3-1} = {}^{74}C_2$$

Hence probability is $\frac{{}^{74}C_2}{{}^{100}C_3}$

STATEMENT-2

The required probability = (probability of drawing a prime number) $\times$ (probability of drawing more than 10 from prime number)

$$= \frac{8}{20} \times \frac{4}{8} = \frac{1}{5}$$

STATEMENT-3

Total number of cases = $(365)^3$ and number of favourable cases are 365

$$\text{Hence, probability} = \frac{365}{(365)^3} = \frac{1}{(365)^2}$$

9. If the squares of a 8×8 chess board are painted either white or black at random then

STATEMENT-1 : The probability that not all squares are in any column, are alternating in colour is $\left(1 - \frac{1}{2^7}\right)^8$.

STATEMENT-2 : The probability that all the squares in any column of same colour and alternating colour in any row is equal to $\frac{1}{2}$.

STATEMENT-3 : The probability that the chess board contains equal number of white and black squares is

$$\frac{64!}{2^{64} \cdot 32!}$$

(1) T F T

(2) F T T

(3) T T T

(4) T F F

Sol. Answer (4)

STATEMENT-1

The total number of ways of painting first column when colours are not alternating is $28 - 2$.

Hence, the probability when no column has alternating colours is $\left(\frac{2^8 - 2}{2^8}\right)^8 = \left(1 - \frac{1}{2^7}\right)^8$.

STATEMENT-2

This is possible only when the columns are alternating white and black colour.

$$\text{Probability} = \frac{2}{2^{64}} = \frac{1}{2^{63}}$$

STATEMENT-3

The number of ways the square has equal number of white and black square is ${}^{64}C_{32}$.

Hence, probability is equal to $\frac{{}^{64}C_{32}}{2^{64}}$

10. Let A and B be two independent even such that $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{4}$, then

STATEMENT-1 : $P\left(\frac{A}{A \cup B}\right)$ is equal to $\frac{1}{12}$.

STATEMENT-2 : $P\left(\frac{A'}{B}\right)$ is equal to 0.

STATEMENT-3 : $P\left(\frac{B}{A' \cap B'}\right)$ is equal to $\frac{2}{3}$.

(1) T F T

(2) F T T

(3) F F F

(4) T F F

Sol. Answer (3)

STATEMENT-1

$$P\left(\frac{A}{A \cup B}\right) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A \cup B)} = \frac{1/3}{1/2} = \frac{2}{3}$$

$$\text{Here, } P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{3} + \frac{1}{4} - \frac{1}{12} = \frac{1}{2}$$

STATEMENT-2

$$P\left(\frac{A'}{B}\right) = \frac{P(A' \cap B)}{P(B)} = \frac{P(A')P(B)}{P(B)} = \frac{2}{3}$$

STATEMENT-3

$$P\left(\frac{B}{A' \cap B'}\right) = \frac{P(B \cap (A' \cap B'))}{P(A' \cap B')} = \frac{P(\emptyset)}{P(A' \cap B')} = 0$$

SECTION - H

Aakash Challengers Questions

1. Two points P, Q are taken at random on a straight line OA of length a . The chance that $PQ > b$, where $b < a$ is

(1) $\frac{|a-b|}{a}$

(2) $\frac{|b-a|}{a}$

(3) $\frac{(a-b)^2}{a^2}$

(4) $\frac{(b-a)^2}{b^2}$

Sol. Answer (3)



Let the given line segment of OA is of length a . Let $PQ = x$ and $OP + QA = y$

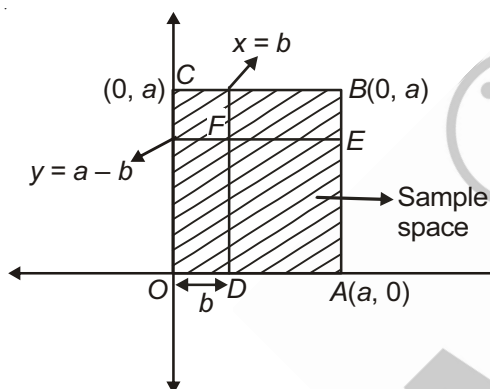
Clearly $x + y = a$

Now let us calculate the sample space

Clearly $0 \leq x, y \leq a$

But $x > b$

$\Rightarrow y < a - b$



Sample space = $OABC$

Favourable case = $\text{ar.}(\triangle AEF)$

$$P(E) = \frac{(a-b)^2}{a^2}$$

2. The sides of a rectangle are chosen at random, each less than a given length a , all such lengths being equally likely. The chance that the diagonal is less than a is

(1) $\frac{1}{2}$

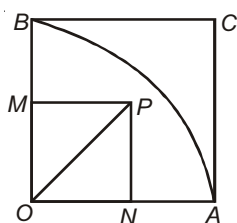
(2) $\frac{2}{3}$

(3) $\frac{3}{4}$

(4) $\frac{\pi}{4}$

Sol. Answer (4)

Draw a square, $OACB$, whose sides is a , and the arc AB .



If $ONPM$ is one the rectangle, the favourable cases are when P lies in the quadrant OAB .

$$\text{So, } P(E) = \frac{\frac{\pi a^2}{4}}{a^2} = \frac{\pi}{4}$$

3. A floor is paved with rectangular bricks, each of length a and breadth b . A circular disc of diameter c is thrown on the floor. The chance that it falls entirely on one brick is

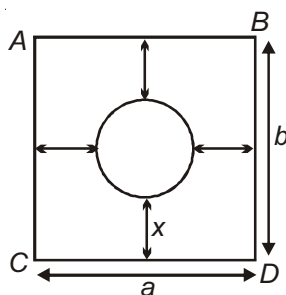
(1) $\frac{c}{ab}$

(2) $\frac{c}{\sqrt{ab}}$

(3) $\frac{(a-c)(b-c)}{c^2}$

(4) $\frac{(a-c)(b-c)}{ab}$

Sol. Answer (4)



Sample space

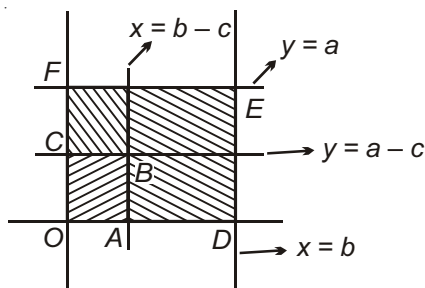
$$0 \leq x \leq b$$

$$0 \leq y \leq a$$

Favourable cases

$$0 \leq x \leq b - c$$

$$0 \leq y \leq a - c$$



$$\text{So, required probability } P(E) = \frac{\text{ar}(OABC)}{\text{ar}(ODEF)}$$

$$= \frac{(b-c)(a-c)}{ab}$$

4. A point P is taken at random on a straight line AB . The chance that the greater of the parts AP and PB is at least k times the smaller is

(1) $\frac{2}{k+1}$

(2) $\frac{2}{k+2}$

(3) $\frac{2}{k^2+1}$

(4) $\frac{2}{k^2+2}$

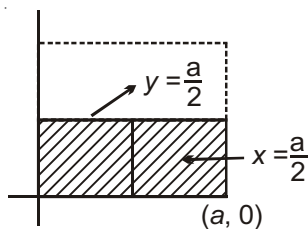
Sol. Answer (1)

Let AB be a line segment of length " a " and P be a point on it, such that

$$AP = x \text{ and } PB = y$$

Now clearly $x + y = a$

$$\text{Since } x > y \Rightarrow x > \frac{a}{2} \text{ and } y < \frac{a}{2}$$



Clearly $x > ky$

$$\text{So, the required probability is } = \frac{a \cdot \frac{a}{k+1}}{a^2} = \frac{1}{k+1}$$

But since another case $y > kx$ may also arise, so the total probability is $\frac{2}{k+1}$

5. If two points are taken at random on the circumference of a circle, the chance that their distance apart is greater than the radius of the circle is

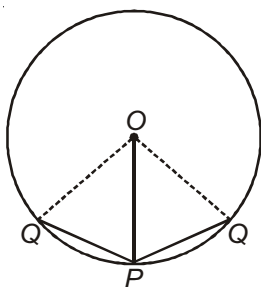
(1) $\frac{1}{4}$

(2) $\frac{1}{3}$

(3) $\frac{2}{3}$

(4) $\frac{3}{4}$

Sol. Answer (3)



Choose any point P on the circumference of the circle. The next chosen point Q should be such that

$$PQ > r$$

Now, clearly Δ 's OQP is equilateral if $PQ = r$

$$\text{So, the favourable cases } = \frac{\pi - \frac{\pi}{3}}{\pi} = \frac{2}{3}$$

6. If three points P, Q, R are taken at random on the circumference of a circle, the chance that do not lie on the same semicircle is

(1) $\frac{1}{4}$

(2) $\frac{1}{3}$

(3) $\frac{1}{2}$

(4) $\frac{3}{4}$

Sol. Answer (1)

Choose P . Choose Q on one semicircle on PP' ; R must lie on $P'Q'$.

So, If $PQ = s$,

$$\text{Number of favourable cases} = \int_0^{\frac{1}{2}C} s ds = \frac{1}{8}C^2$$

$$\text{Total number of cases} = \frac{1}{2} \cdot C \cdot C$$

$$\therefore P(E) = \frac{\frac{1}{8}C^2}{\frac{1}{2}C^2} = \frac{1}{4}$$

7. 3 six faced dice are thrown. One die bears numbers 0 to 5, second bears numbers 1 to 6 and third has numbers 2 to 7. The probability that sum of numbers shown is six is

(1) $\frac{7}{108}$

(2) $\frac{5}{108}$

(3) $\frac{5}{216}$

(4) $\frac{7}{216}$

Sol. Answer (2)

Favourable cases = coefficient of x^6 in $(x^0 + x^1 + \dots + x^5)(x^1 + x^2 + \dots + x^6)(x^2 + x^3 + \dots + x^7)$

$$= \text{Co-efficient of } x^3 \text{ in } \left(\frac{1-x^6}{1-x} \right)^3$$

$$= \text{Co-efficient of } x^3 \text{ in } (1 - 3x^6 + \dots)(1 + {}^3C_1x + {}^4C_2x^2 + \dots)$$

$$= {}^5C_2 = 10$$

$$\therefore \text{Required probability} = \frac{10}{216} = \frac{5}{108}$$

8. Two natural numbers r, s are drawn one at a time, without replacement from the set $S = \{1, 2, 3, \dots, n\}$, then

$P\left[r \leq \frac{p}{s} \leq p\right]$, is equal to (where $p \in S$).

(1) $\frac{p-1}{n-1}$

(2) $\frac{p-1}{n(n-1)}$

(3) $\frac{(p-1)^2}{(n-1)^2}$

(4) $\frac{2(p-1)}{(n-1)}$

Sol. Answer (1)

Clearly the given condition is $sr \leq p \leq ps$

No for any value of p ; $ps \geq p$, as the minimum value of s can be 1

So, the only condition which is to be satisfied is $p \geq sr$

So, that p should be greater than or equal to the product of two chosen numbers.

Case I Favourable cases

$$p = 1 \quad 0$$

$$p = 2 \quad (1, 2), (2, 1)$$

$$p = 3 \quad (1, 2), (1, 3), (2, 1), (3, 1)$$

$$p = 4 \quad (1, 2), (1, 3), (1, 4), (2, 1), (3, 1), (4, 1)$$

... ..

... ..

... ..

So, $(p - 1)$ choices

$$\Rightarrow P(E) = \frac{p-1}{n-1}$$

9. An urn contains m white and n black balls. A ball is drawn at random and is put back into the urn along with k additional balls of the same colour as that of the ball drawn. A ball is again drawn at random. Show that the probability of drawing 4 white ball now does not depend on k .

Sol. Either a white ball or a black ball will be drawn.

Case I : The probability of drawing a white ball in the first attempt is $\frac{m}{m+n}$

The probability of occurrence of white ball if the drawn ball was white $\frac{m+k}{m+n+k}$

$$\text{So, probability in this case} = \frac{m}{m+n} \times \frac{(m+k)}{(m+n+k)}$$

Case II : The probability of drawing a black ball in the first attempt is $\frac{n}{m+n}$

The probability of occurrence of white ball if the drawn ball was black is $\frac{m}{m+n+k}$

$$\text{So, probability in this case} = \frac{n}{m+n} \times \frac{m}{m+n+k}$$

$$\text{So, total probability} = \frac{m}{m+n} \times \frac{(m+k)}{(m+n+k)} + \frac{n}{m+n} \times \frac{m}{m+n+k}$$

$$= \frac{m(m+n+k)}{(m+n)(m+n+k)}$$

$$= \frac{m}{m+n}; \text{ which is independent of } k.$$

10. A bag contains $(2n + 1)$ coins. It is known that n of these coins have a head on both sides where as rest of the coins are fair. A coin is picked up at random from the bag and is tossed. If the probability that the toss result in a head is $\frac{31}{42}$, the total number of coins is

- (1) 10 (2) 21 (3) 31 (4) 41

Sol. Answer (2)

Total numbers of coins = $(2n + 1)$

Total number of ways to chose a biased coin = ${}^nC_1 = n$ ways

If the coin is biased, then probability of occurrence of head is $\frac{{}^nC_1}{2n+1} = \frac{n}{2n+1}$

Total number of ways to chose an unbiased coin = ${}^{n+1}C_1 = (n + 1)$ ways

If the coin is unbiased, then probability of occurrence of head is $\frac{n+1}{2(2n+1)}$

$$\text{Total probability} = \frac{n}{2n+1} + \frac{n+1}{2(2n+1)} = \frac{3n+1}{2(2n+1)}$$

$$\Rightarrow \frac{3n+1}{2(2n+1)} = \frac{31}{42}$$

$$\Rightarrow \frac{3n+1}{2n+1} = \frac{31}{21}$$

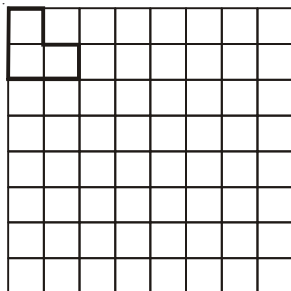
$$n = 10$$

So, total number of coins = 21

11. Three squares of size 1×1 are chosen at random on a chessboard, the probability that they will form a letter "L" on the chessboard is

- (1) $\frac{49}{64C_3}$ (2) $\frac{98}{64C_3}$ (3) $\frac{147}{64C_3}$ (4) $\frac{196}{64C_3}$

Sol. Answer (4)



We can vary them in ${}^7C_1 \times {}^7C_1$ ways on one side.

So, in total there will be $4 \times {}^7C_1 \times {}^7C_1$ ways

$$P(E) = \frac{4 \times 7 \times 7}{64C_3} = \frac{196}{64C_3}$$

12. There are 50 black balls and 50 white balls in a box. Now any number of these two types of balls are distributed among two different boxes (whose selection is equally likely) in such a way that probability of getting a white ball is maximum, then this maximum probability is equal to

- (1) $\frac{1}{2}$ (2) $\frac{3}{4}$ (3) $\frac{5}{6}$ (4) $\frac{74}{99}$

Sol. Answer (4)

Clearly the distribution should be like one white ball in one box and 49 white and 50 black in the other

$$\text{So, required probability is equal to} = \frac{1}{2} \left(1 + \frac{49}{99} \right) = \frac{1}{2} \times \frac{148}{99} = \frac{74}{99}$$

13. Every person in the room has shake hands with a certain number of other persons. The probability that count of the number of people who have shaken hands an odd number of times is even is

- (1) 0 (2) $\frac{1}{2}$ (3) $\frac{2}{3}$ (4) 1

Sol. Answer (4)

Consider the total number of handshakes which have been completed at any moment. This must be an even number, since every handshake is participated in by two people, thus the total number is increased by two. The number of handshakes, however, is also the sum of the handshakes made by each individual person. Since, this sum is an even number, the count of the people who have shaken hands and odd number of times must be even (otherwise, odd times odd would give an odd contribution to the total).

Hence the required probability is 1.

14. The probability that any number which is not a power of 2 can be represented as the sum of at least two consecutive positive integers is

- (1) 0 (2) $\frac{1}{2}$ (3) $\frac{3}{4}$ (4) 1

Sol. Answer (4)

Let N be an integer, which is not a power of 2. Then following equation can be written

$$N = 2^k (2l + 1)$$

where 2^k is the greatest power of 2 appearing as a factor of N , $k \geq 0$, $l \geq 1$ and $2l + 1$ is the greatest odd divisor of N . consider the A.P.

$$(2^k - l) + (2^k - l + 1) + \dots + (2^k - l + 2l - 1) + (2^k - l + 2l)$$

$$= \frac{(2l + 1)(2^k - l + 2^k - l + 2l)}{2} = 2^k (2l + 1) = N$$

If some of the $2l + 1$ consecutive integers which form the progression are negative (that $l > 2^k$), then it is possible to cancel them with first appearing positive integers. It is readily shown that atleast final two terms of the progression must remain uncanceled.

So, required probability = 1

15. Three horses participate in a race, the probability that it will be a tie (all three reach the winning point together) is

(1) $\frac{1}{4}$

(2) $\frac{1}{6}$

(3) $\frac{1}{13}$

(4) $\frac{1}{7}$

Sol. Answer (3)

Consider the case when they all end up differently

Total number of cases = $3! = 6$

Suppose there is a tie between the two = ${}^3C_2 \times 2! = 6$

If there is tie between all = 1

So, total number of cases = 13

Favourable ones = 1

$$P(E) = \frac{1}{13}$$

16. Seven digits from the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 are written in random order. Let the probability that this number is divisible by 9 be $p = \frac{a}{b}$, where a, b are relatively prime positive integers, then the value of $a + b$ is

Sol. 1, 2, 3, 4, 5, 6, 7, 8, 9

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45$$

Out of 9 numbers, those two number should be removed whose sum is 9

i.e., (1, 8) or (2, 7) or (3, 6) or (4, 5)

$$n(S) = {}^9P_7$$

$$n(E) = 4 \times 7!$$

$$P(E) = \frac{4 \times 7! \times 2!}{9!} = \frac{4 \times 2}{9 \times 8} = \frac{1}{9} = \frac{a}{b}$$

$$\therefore a + b = 10$$

17. The number of ways to choose 3 distinct numbers in increasing GP from the set $\{1, 2, 3, \dots, 100\}$ is ... and corresponding probability is

Sol. Let a, b, c are in G.P.

a	c	a	c
1	4	2	8
9		18	
16		32	
25		50	
36		72	
49		98	
64			
81			
100			

Similarly we can count for other groups

$\therefore$ Number of such groups = 53

$$P(E) = \frac{53}{{}^{100}C_3}$$

18. A person throws four standard six sided distinguishable dice. The probability that the product of four numbers on the upper faces is 144 is p then $81p$ is equal to

Sol. $n(S) = 6 \times 6 \times 6 \times 6 = 6^4$

$$a.b.c.d. = 144, 1 \leq a, b, c, d \leq 6.$$

Factors are (6, 6, 2, 2), (6, 6, 4, 1), (6, 4, 3, 2) and (4, 4, 3, 3)

$$\text{Number of ways} = \frac{4!}{2!2!} + \frac{4!}{2!} + 4! + \frac{4!}{2!2!} = 48$$

$$P(E) = \frac{48}{6^4} = \frac{1}{27} = p \Rightarrow 81p = 3$$

19. The sum of the digits of a seven digit number is 59. If the probability that the number is divisible by 11 is p , then $441p$ is equal to

Sol. We observe that $7 \times 9 = 63$

Then $6 \times 9 = 54$

Thus in order that the sum of the digits to be 59 atleast three of them should be 9 and no digits should be less than 5.

Thus for the sum of 7 digits to be 59 the 7 digits must be

(i) 9, 9, 9, 8, 8, 8, 8

(ii) 9, 9, 9, 9, 8, 8, 7

(iii) 9, 9, 9, 9, 9, 7, 7

(iv) 9, 9, 9, 9, 9, 8, 6

(v) 9, 9, 9, 9, 9, 9, 5

$$\therefore n(S) = \frac{7!}{3!4!} + \frac{7!}{4!2!} + \frac{7!}{5!2!} + \frac{7!}{5!} + \frac{7!}{6!} = 35 + 105 + 21 + 42 + 7 = 210$$

In order that the number is to be divisible by 11 the difference of the sum of the digits at odd places and the sum of the digits at even places should be divisible by 11.

Let a = the sum of the digits at odd places.

$$\Rightarrow a \leq 9 \times 4 = 36$$

b = the sum of the digits at even places.

The $a - b = 0, 11, 22, 33, 44, 55$

And $a + b$ = the sum of all the 7 digits = 59

The following table will illustrate the possible outcomes :

Value of $a - b$	Value of $a + b$	Value of a	Value of b	Remark
$a - b = 0$	$a + b = 59$	$a = 29.5$	$b = 29.5$	Not possible
$a - b = 11$	$a + b = 59$	$a = 35$	$b = 24$	Possible
$a - b = 22$	$a + b = 59$	$a = 40.5$	$b = 18.5$	Not Possible
$a - b = 33$	$a + b = 59$	$a = 46$	$b = 13$	Not Possible
$a - b = 44$	$a + b = 59$	$a = 51.5$	$b = 7.5$	Not Possible
$a - b = 55$	$a + b = 59$	$a = 57$	$b = 2$	Not Possible

Thus the only possibility is $a = 35$, $b = 24$.

In Case (i), 4 odd places should be occupied by 9, 9, 9, 8 and 3 even places by 8, 8, 8.

In Case (ii), 4 odd places should be occupied by 9, 9, 9, 8 and 3 even places by 9, 8, 7.

In Case (iii), number divisible by 11 will not be formed as $a = 35$.

In Case (iv), 4 odd places should be occupied by 9, 9, 9, 8 and 3 even places by 9, 9, 6.

In Case (v), no number divisible by 11 is formed.

$$\therefore n(E) = \frac{4! \cdot 3!}{3! \cdot 3!} + \frac{4! \cdot 3!}{3! \cdot 3!} + 0 + \frac{4! \cdot 3!}{3! \cdot 2!} + 0 = 4 + 24 + 12 = 40$$

$$\Rightarrow P(E) = p = \frac{40}{210} = \frac{4}{21}$$

$$\Rightarrow 441p = 441 \times \frac{4}{21} = 21 \times 4 = 84$$

20. The probability of the number of permutations of the letters AAAABBBC in which the A's appear together in a block of four letters or the B's appear in a block of three letters is p then $770p$ is equal to

Sol. The number of permutations of the letters AAAABBBC = $\frac{8!}{4!3!} = 280$

Number of ways in which 4 A's are together in a block of four letters = $\frac{5!}{3!} = 20$; eg AAAA BBBC

Number of ways in which B's appear together in a block of three letters = $\frac{6!}{4!} = 30$; eg BBB AAAAC

Number of ways in which A's appear together in a block of 4 and B appear in a block of three

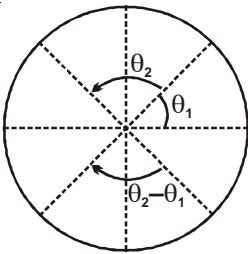
$$= 3! \cdot \boxed{\text{AAAA}} \cdot \boxed{\text{BBB}} \cdot \text{C}$$

$$\therefore n(E) = 20 + 30 - 6 = 44, P(E) = \frac{44}{280} \Rightarrow p = \frac{11}{70}$$

$$770p = 770 \times \frac{11}{70} = 121$$

21. Three arbitrary tangents are drawn on a fixed circle, then what is the probability that the triangle formed by the tangents will contain the circle.

Sol.



Let the eccentric angles of the points be θ_1 and θ_2 .

So, the third point must lie in $\theta_2 - \theta_1$ region. Suppose $\theta_2 - \theta_1 = \theta$

The probability that angle between them will be θ is $\frac{\theta}{2\pi}$.

The total probability $\frac{\theta}{2\pi} \times \frac{\theta}{2\pi} = \frac{\theta^2}{4\pi^2}$

Let the angle be now $(\theta + d\theta)$

$$\text{So, } P_{\text{new}} = \frac{(\theta + d\theta)^2}{4\pi^2}$$

$$dp = \frac{(\theta + d\theta)^2 - \theta^2}{4\pi^2}$$

$$= \frac{2\theta d\theta}{4\pi^2}; \text{ neglecting infinitesimal of higher order}$$

$$P = \frac{2 \int_0^\pi \theta d\theta}{4\pi^2} = \frac{1}{4}$$

22. In a mixed doubles tennis match 5 couples participate, what is the probability that no-one is with or in opposition to their spouses in a match?

Sol. As it is a mixed doubles, so total number of ways in which team can be formed $= {}^5C_2 \cdot {}^5C_2 \cdot 2!$

Total number of ways in which no one gets their spouses $= {}^5C_2 \cdot {}^3C_2 \cdot 2!$

$$\text{So, } P(E) = \frac{{}^5C_2 \cdot {}^3C_2 \cdot 2!}{{}^5C_2 \cdot {}^5C_2 \cdot 2!} = \frac{3}{10}$$

23. A triangle is formed with vertices of an n -sided regular polygon. What is the probability that triangle have exactly one side common with the sides of the regular polygon.

Sol. Number of sides of polygon $= n$

$$\therefore n(S) = \text{Number of triangles so formed} = {}^nC_3 = \frac{n(n-1)(n-2)}{6}$$

Let E_1 be the event that triangle formed have one side common with given polygon.

So, number of favourable ways = if on side AD of n sided polygon be selected then to form triangle, we need to select on vertex from $(n - 4)$ vertices.

$$\text{So, } n(E) = n(n - 4)$$

$$P(E) = \frac{6(n - 4)}{(n - 1)(n - 2)}$$

24. A triangle is formed with vertices of an n -sided regular polygon. What is the probability that triangle formed have exactly two sides common with the sides of the polygon.

Sol. Let E_4 be the event that formed triangle have at most one side common with the sides of a polygon.

$\therefore n(E_4) = \text{number of triangles with one side common} + \text{Number of triangle having no common side}$

$$= \frac{n(n - 1)(n - 2)}{6} - \frac{n}{1} = \frac{n(n + 1)(n - 4)}{6}$$

$$\therefore P(E) = \frac{(n + 1)(n - 4)}{(n - 1)(n - 2)}$$

25. Let $A = \{1, 2, \dots, 100\}$, a number x is selected at random from A , the probability that $\frac{x^2 - 60x + 800}{x - 30} < 0$ will be

Sol. For given $\frac{x^2 - 60x + 800}{x - 30} < 0$ and $n(S) = 100$

$$\frac{(x - 20)(x - 40)}{(x - 30)} < 0$$

$$\Rightarrow x \in \{1, 2, \dots, 19\} \cup \{31, 32, \dots, 39\}$$

$$n(x) = 19 + 9 = 28$$

$$\therefore P(x) = \frac{n(x)}{n(S)} = \frac{28}{100} = \frac{7}{25}$$

26. The digits 1, 2, 3, ..., 9 are arranged at random to write 9 digit number. What is the chance that the difference of the digits at equal distances from both the ends is always one?

Sol. $n(S) = 9!$

The 9 digits can be split into pairs (1, 2), (2, 3), (3, 4), ..., (8, 9) such that the difference of digit is 1. The total number of ways of filling up of first place is 9 and its pair should be placed in the last place. The second place from the beginning can be filled in 7 ways. The second place from last can be filled in only one way. And similarly for the other cases.

$$\therefore P(E) = \frac{9 \times 7 \times 5 \times 3}{9!} = \frac{1}{384}$$

27. Six couples are arranged at random for a photograph. What is the chance that two members of each couple are always together?

Sol. $n(S) = 12!$

$$n(E) = 6! (2!)^6$$

$$P(E) = \frac{6! (2!)^6}{12!}$$

28. 3 identical green balls, 2 identical red balls and 5 identical blue balls are arranged in a row at random. What is the chance that first green ball comes before first red ball?

Sol. $n(S) = \frac{10!}{3! \cdot 2! \cdot 5!} = 2520$

Let E be the event that first green ball comes before first red ball.

Position	No. of ways of arrangement
G	$\frac{9!}{2! \cdot 2! \cdot 5!}$
BG	$\frac{8!}{2! \cdot 2! \cdot 4!}$
BBG	$\frac{7!}{2! \cdot 2! \cdot 3!}$
BBBG	$\frac{6!}{2! \cdot 2! \cdot 2!}$
BBBBG	$\frac{5!}{2! \cdot 2!}$
BBBBBG	$\frac{4!}{2! \cdot 2!}$

Required Chance = $\frac{1512}{2520}$

29. A bag contains $n + 1$ coins, one of which is a two-headed coin and the rest are fair. A coin is selected at random and tossed. If the probability. That the toss results in a head is $\frac{7}{12}$ then the value of n is

Sol. A be the event that selected coin is fair

$$P(A) = \frac{n}{n+1}$$

$$P(A') = \frac{1}{n+1}$$

$$P(H) = P(A) \cdot P\left(\frac{H}{A}\right) + P(A') \cdot P\left(\frac{H}{A'}\right)$$

$$\Rightarrow \frac{7}{12} = \frac{n}{n+1} \cdot \frac{1}{2} + \frac{1}{n+1} \cdot 1 \quad \Rightarrow \quad \frac{7}{12} = \frac{n+2}{2(n+1)}$$

$$\Rightarrow 14n + 14 = 12n + 24 \quad \Rightarrow \quad 2n = 10$$

$$\Rightarrow \boxed{n = 5}$$

30. A bag contains m white and n black marbles. Two players A and B alternately draw a marble from the bag, replacing the marble each time after the draw till one of them draws a white marble and wins the game. If A begins the game and the probability of winning of A be three times as much as that of B , the value of $123k$, k being the ratio of m to n in simplest form, is

Sol. Let E_1 be the event that white ball is selected

$$P(E_1) = \frac{m}{m+n}, \quad P(E_1') = \frac{n}{m+n}$$

The probability that A wins the game

$$\begin{aligned} &= \frac{m}{m+n} + \frac{n}{m+n} \cdot \frac{n}{m+n} \cdot \frac{m}{m+n} + \frac{n}{m+n} \cdot \frac{n}{m+n} \cdot \frac{n}{m+n} \cdot \frac{m}{m+n} + \dots \\ &= \frac{m}{m+n} \left[1 + \left(\frac{n}{m+n} \right)^2 + \left(\frac{n}{m+n} \right)^4 + \dots \right] \\ &= \frac{m}{m+n} \cdot \frac{1}{1 - \left(\frac{n}{m+n} \right)^2} = \frac{m(m+n)}{m^2 + 2mn} = \frac{m(m+n)}{m(m+2n)} = \frac{m+n}{m+2n} \end{aligned}$$

$$\text{Probability of winning } B = 1 - \frac{m+n}{m+2n} = \frac{n}{m+2n}$$

$$\frac{m+n}{m+2n} = 3 \frac{n}{m+2n}$$

$$\Rightarrow m = 2n$$

$$\text{Now, } \frac{m}{n} = 2$$

$$\text{Now, } 123k = 246.$$

31. An ellipse of eccentricity $\frac{2\sqrt{2}}{3}$ is inscribed in a circle and a point within the circle is chosen at random. Let the probability that this point lies outside the ellipse be p . Then the value of $105p$ is

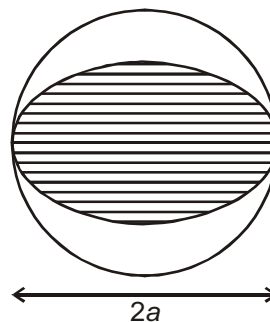
Sol. Let a be the radius of circle.

$$\text{Length of semi-minor axis of ellipse} = a \sqrt{1 - \left(\frac{2\sqrt{2}}{3} \right)^2} = \frac{a}{3}$$

Probability that point lies outside the ellipse

$$p = \frac{\pi a^2 - \pi a \cdot \frac{a}{3}}{\pi a^2} = \frac{\pi a^2 \left(1 - \frac{1}{3} \right)}{\pi a^2} = \frac{2}{3}$$

$$\Rightarrow 105p = 105 \times \frac{2}{3} = 70$$



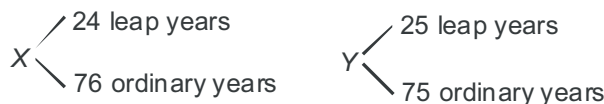
32. A number is chosen randomly from one of the two sets $X = \{2001, 2002, 2003, \dots, 2100\}$, $Y = \{1901, 1902, 1903, \dots, 2000\}$. If the number chosen represents a calendar year and p is the probability that selected year has 53 Sundays, then $2800p$ is equal to

Sol. We have,

$$X = \{2001, 2002, 2003, \dots, 2100\}$$

$$Y = \{1901, 1902, 1903, \dots, 2000\}$$

Then



Let E be the event that randomly chosen year has 53 Sundays

$$P(E) = P(E \cap L) + P(E \cap O)$$

$$= P(L) \cdot P\left(\frac{E}{L}\right) + P(O)P\left(\frac{E}{O}\right)$$

$$= \frac{1}{2} \left[\frac{24}{100} \cdot \frac{2}{7} + \frac{76}{100} \cdot \frac{1}{7} \right] + \frac{1}{2} \left[\frac{25}{100} \cdot \frac{2}{7} + \frac{75}{100} \cdot \frac{1}{7} \right]$$

$$= \frac{249}{1400} = p$$

$$\Rightarrow 2800p = 2 \times 249 = 498$$

33. Consider a town with n people. A person spreads a rumour to a second, who in turn repeats it to a third and so on. Suppose that at each stage, the recipient of the rumour is chosen at random from the remaining $(n-1)$ people. The probability that the rumour will be repeated n times without being repeated to the originator is

(1) $\frac{n-1}{n}$

(2) $\frac{n-1}{(n-2)(n-3)}$

(3) $\left(1 - \frac{1}{n-1}\right)^{(n-1)}$

(4) $\left(\frac{n-2}{n-1}\right)$

Sol. Answer (3)

Let there are n people $A_1, A_2, A_3 \dots A_n$

So, Let us assume that A_1 starts the rumors so, A_1 can has $(n-1)$ options. The next person has $(n-1)$ options and similarly they all have $(n-1)$ options.

So, total number of ways in which rumors can spread is $(n-1)^n$.

$$\Rightarrow n(S) = (n-1)^n$$

Now let us consider the case in which the rumors does not get back to the originator.

Now, A_1 has $(n-1)$ choices

The next person has $(n-2)$ choices

The next person again has $(n-2)$ choices

The last person has $(n-2)$ choices

So, total number of ways is $(n-1)(n-2)^{n-1}$

$$P(E) = \frac{(n-1)(n-2)^{n-1}}{(n-1)^n} = \left(\frac{n-2}{n-1}\right)^{n-1}$$

$$P(E) = \left(1 - \frac{1}{n-1}\right)^{(n-1)}$$

34. Two players P_1 and P_2 , are playing the final of a chess championship, which consists of series of matches.

Probability of P_1 winning a match is $\frac{2}{3}$ and for P_2 is $\frac{1}{3}$. The winner will be the one who is ahead by two games as compared to the other player and wins at least 6 games. Now if the player P_2 wins first four matches, then the probability that P_1 will win the championship is

(1) $\frac{1077}{3645}$

(2) $\frac{1088}{3645}$

(3) $\frac{1099}{3645}$

(4) $\frac{1}{5}$

Sol. Answer (2)

P_1 can win in following mutually exclusive ways

(a) P_1 wins the next six matches

(b) P_2 wins five out of next six matches, so that after next six matches scores of P_1 and P_2 are tied up. This tie continues up to next " $2n$ " matches ($n \geq 0$) and finally P_1 wins 2 consecutive matches.

Now, probability of first case = $\left(\frac{2}{3}\right)^6$

And probability of ties after 6 matches = ${}^6C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right) = \frac{2^6}{3^5}$

Now probability that scores are still tied up after another next two matches = $\frac{2}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{9}$

[1st match won by P_1 and 2nd by P_2 or 1st by P_2 , & 2nd by P_1]

Similarly probability that scores are still tied up after $2n$ matches = $\left(\frac{4}{9}\right)^n$

$\Rightarrow$ Total probability of winning is $\left(\frac{2}{3}\right)^6 + \frac{2^6}{3^5} \cdot \left(\sum_{n=0}^{\infty} \left(\frac{4}{9}\right)^n \left(\frac{2}{3}\right)^2\right)$

$= \left(\frac{2}{3}\right)^6 + \frac{2^6}{3^5} \cdot \left(\frac{2}{3}\right)^2 \left(\frac{1}{1 - \frac{4}{9}}\right)$

$= \frac{17}{5} \left(\frac{2}{3}\right)^6 = \frac{1088}{3645}$

35. If a and b are chosen randomly from the set consisting of numbers 1, 2, 3, 4, 5, 6 with replacement. The

probability that $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2}\right)^{2/x} = 6$ is

(1) $\frac{1}{2}$

(2) $\frac{1}{4}$

(3) $\frac{1}{8}$

(4) $\frac{1}{9}$

Sol. Answer (4)

$\lim_{x \rightarrow 0} \left(1 + \frac{a^x + b^x - 2}{2}\right)^{\frac{2(a^x + b^x - 2)}{(a^x + b^x - 2)x}} = 6$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{a^x + b^x - 2}{x} = e^{\ln ab} = 6$$

$$\Rightarrow ab = 6$$

Total number of ways in which we can choose b is $6 \times 6 = 36$.

Total number of possible ways = 4.

$$P(E) = \frac{4}{36} = \frac{1}{9}$$

36. Two dice are thrown simultaneously to get the coordinates of a point on x-y plane. The probability that this point lies on or inside the region bounded by $|x| + |y| = 3$.

(1) $\frac{1}{3}$

(2) $\frac{1}{4}$

(3) $\frac{1}{12}$

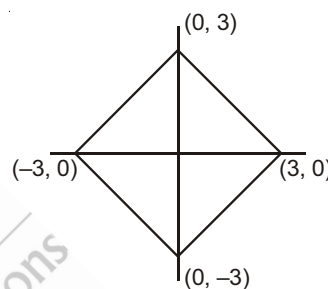
(4) $\frac{1}{36}$

Sol. Answer (3)

Now any point on or inside the region bounded by $|x| + |y| = 3$ will have x coordinates lying between -3 and $+3$.

Total such integral points are 3 which will turn up as a throw of dice i.e., $(1, 1)$, $(1, 2)$, $(2, 1)$.

Hence the probability is $\frac{3}{36} = \frac{1}{12}$



37. A function f is such that $f(x) = x^3 + ax^2 + bx + c$, where a, b, c are chosen by throwing a die three times. The probability that f is an increasing function is

(1) $\frac{1}{9}$

(2) $\frac{2}{9}$

(3) $\frac{1}{3}$

(4) $\frac{4}{9}$

Sol. Answer (4)

$$f(x) = x^3 + ax^2 + bx + c$$

$$f'(x) = 3x^2 + 2ax + b$$

Now for f to be increasing

$$f'(x) \geq 0 \quad \forall x$$

$$\Rightarrow (2a)^2 - 12b \leq 0$$

$$\Rightarrow a^2 \leq 3b$$

This is true for exactly 16 ordered pairs.

So, $P(E) = \frac{16}{36} = \frac{4}{9}$

